The U(1) Higgs Model

II. The Infinite Volume Limit

Abstract. We construct the infinite volume limit of the $U(1)$ Higgs model in three dimensions, and verify reflection positivity, translation invariance, analyticity, regularity and rotation invariance for the Schwinger functions of the field strength, the last by a comparison with periodic models on tori which carry a mass term away from the sources. The proof uses a combination of renormalization group techniques and correlation inequalities.

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1. Introduction

In a previous paper [I], we considered the lattice $U(1)$ Higgs model in a finite volume with periodic boundary conditions; throughout this paper $d = 3$. We proved the convergence of this model to a finite limit as the lattice spacing approaches zero. The method of proof was based

on the renormalization transformations introduced in [Ba 1–4]. In this paper, we continue our analysis of the model by constructing the infinite volume theory and verifying, for the field strength, reflection positivity, translation invariance, regularity and a suitable analyticity property, and rotation invariance (Sect. 6.7).

This model was previously constructed in two dimensions in the papers [BFS 1–3]. In particular, a collection of useful correlation inequalities was established in [BFS 1]. We use these inequalities to define the infinite volume limit of the continuum theory derived in [I]. The limit is shown to satisfy reflection positivity, translation invariance, and a suitable analyticity property.

Since we approach the continuum limit through a sequence of models defined on lattices with a fixed orientation, it is not obvious that the continuum theory satisfies rotation invariance.

Rotation invariance rests on two estimates. The first, Theorem 2.4, states that two lattices with the same spacing whose orientations differ by a special angle θ_0 (incommensurate with 2π) give the same continuum limit for the field strength on the tori of the class described in Sect. 4, which carry a mass term away from the sources, for sources which are 2-forms evaluated by each lattice at the centers of its own plaquettes. The second, Theorem 6.24, compares the infinite volume limit with the limits of these periodic models.

The first is proved in Sect. 4 and the second in Sect. 6.7; the estimates of the renormalization group analysis which these proofs require are taken from [M]. Sect. 6.7 derives the rotation invariance of the infinite volume limit from the two estimates.

For periodic boundary conditions the correlation inequalities of [BFS 1] give no monotonicity in the volume, since a torus is not obtained from a larger one by removing couplings, and we do not use a cluster expansion; the comparison of Sect. 6.7 takes its place.

The paper is organized as follows. Section 2 contains a statement of the results proved in this paper. In Sect. 3, we construct the infinite volume limit and verify the properties mentioned before. Section 4 contains the comparison of the two lattices on which rotation invariance rests. In Sect. 5 we extend the ultra-violet stability results of [Ba 1–4] to include the different boundary conditions used in this paper and the rotated lattice of Sect. 4. Finally, in Sect. 6 we prove the regularity bounds for the field strength, construct the limit measure, treat the scalar sources, and derive the rotation invariance of the infinite volume limit from Theorems 2.4 and 6.24 (Sect. 6.7), using the continuity of the limit in the source and the cubic symmetry of the limit (Sect. 6.3). The questions left open are collected in Sect. 6.10.

We recommend that the reader study [I] before reading this paper. This paper is a corrected and extended edition of [K 2]; the corrected edition of [K 1] is cited as [I] (Sect. 6.10).

2. Statement of Results

The lattice $U(1)$ Higgs model was defined in [I]. In order to apply the results obtained in [BFS 1–3] concerning correlation inequalities, we restrict attention to the case of a two-component scalar field, though the ultra-violet estimates do not depend on this restriction. For convenience, the field is written

$$\phi(x) = r(x)e^{i\theta(x)}\,, \tag{2.1}$$

where $r \in [0,\infty)$ and $\theta \in [-\pi,\pi)$. The model is defined on a finite lattice $\Omega \subset \varepsilon\mathbb{Z}^d$, where the scalar field is a map $\phi : \Omega \to \mathbb{C}$. The vector field is considered either as a map $A : \Omega \to \mathbb{R}^d$, or as a map $A : \Omega^* \to \mathbb{R}$, where Ω^* is the set of oriented bonds on Ω. These meanings are identified by

$$A_\mu(x) = A_{\langle x,\, x+\varepsilon e_\mu\rangle}\,, \tag{2.2}$$

where e_μ is the unit lattice vector in the μth direction. The covariant derivative of the scalar field is

$$(D^\varepsilon_\mu\phi)(x) = \varepsilon^{-1}\big(\exp[ie\varepsilon A_\mu(x)]\phi(x+\varepsilon e_\mu) - \phi(x)\big)\,, \tag{2.3}$$

where e is the electric charge. Under a local gauge transformation χ, these fields transform as

$$\begin{aligned} \phi(x) &\to \exp[ie\chi(x)]\phi(x)\,, \\ A_\mu(x) &\to A_\mu(x) - \partial_\mu\chi(x)\,, \\ (D^\varepsilon_\mu\phi)(x) &\to \exp[ie\chi(x)](D^\varepsilon_\mu\phi)(x)\,. \end{aligned} \tag{2.4}$$

Denoting by ∂^ε the usual lattice derivative on Ω, our lattice action is

$$\begin{aligned} &S^\varepsilon(\Omega, A, \phi) \\ &= \frac{1}{2}\sum_{\mu=1}^{d}\sum_{b\in\Omega^*}\varepsilon^d|\partial^\varepsilon A_\mu(b)|^2 + \frac{1}{2}\,\mu_0^2\sum_{\mu=1}^{d}\sum_{x\in\Omega}\varepsilon^d|A_\mu(x)|^2 \\ &\quad + \frac{1}{2}\sum_{\langle x,y\rangle\in\Omega^*}\varepsilon^{d-2}\big[r(x)^2 + r(y)^2 - 2r(x)r(y)\cos\big(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y)\big)\big] \\ &\quad + \sum_{x\in\Omega}\varepsilon^d\left[\frac{1}{2}m^2 r(x)^2 + \lambda r(x)^4 + \frac{1}{2}\,\delta m^2\, r(x)^2\right] + E_0 + E_1\,. \end{aligned} \tag{2.5}$$

The quadratic part of the action for the scalar field comes from expanding the gauge-invariant term

$$\frac{1}{2}\sum_{\mu=1}^{d}\sum_{x\in\Omega}\varepsilon^d|(D^\varepsilon_\mu\phi)(x)|^2\,.$$

We have introduced a vector field mass μ_0^2 for technical reasons connected with ultra-violet stability. The constant E_0 is a normalization for the Gaussian parts of the action (2.5):

$$\begin{aligned} \exp(E_0) = \int (dA)(d\phi)\exp\Bigg[&-\frac{1}{2}\sum_{\mu=1}^{d}\sum_{b\in\Omega^*}\varepsilon^d|\partial^\varepsilon A_\mu(b)|^2 - \frac{1}{2}\,\mu_0^2\sum_{\mu=1}^{d}\sum_{x\in\Omega}\varepsilon^d|A_\mu(x)|^2 \\ &- \frac{1}{2}\sum_{b\in\Omega^*}\varepsilon^d|\partial^\varepsilon\phi(b)|^2 - \frac{1}{2}\,m^2\sum_{x\in\Omega}\varepsilon^d|\phi(x)|^2\Bigg]. \end{aligned} \tag{2.6}$$

The mass and vacuum-energy counterterms δm^2 and E_1 are introduced to cancel the usual divergences of perturbation theory, and they are defined in [I]; δm^2 is defined in Sect. 3.1 with free boundary conditions and independently of the site, so that it does not depend on the region Ω. For definiteness, we take Ω to be a rectangular parallelepiped with equal sides. Define for integer J,

$$C_J = \{x \in \mathbb{R}^d :\ -MJ \leqq x_\mu < MJ,\ \mu = 1, \dots, d\}\,, \tag{2.7}$$

where M is the large integer used in Sect. 2.3 of [I], a multiple of 50 (this is used for the tori of Sect. 4). As before, we define lattice spacings $\{\varepsilon_K\}$, $K = 0, \dots, \infty$, by

$$\varepsilon_K = L^{-K}\,, \tag{2.8}$$

where L is the small integer which determines the size of the blocks used in the renormalization transformation; $L = 5$ throughout, as in [I].

In the action (2.5), we will use Neumann boundary conditions (N b.c.) for the vector field A_μ on ∂C_J. We will use Dirichlet boundary conditions (D b.c.) for the scalar field ϕ on ∂C_J (this choice of boundary conditions for ϕ is gauge-invariant).

More precisely, the scalar field vanishes on the layers of sites $x_\mu = -MJ-\varepsilon$ and $x_\mu = MJ-\varepsilon$, $\mu = 1, \dots, d$, so that the two Dirichlet planes in each direction are a distance $2MJ$ apart, a multiple of every block side. The scalar bond sum in (2.5) runs over the bonds joining two sites of the slab $-MJ - \varepsilon \leqq x_\mu \leqq MJ - \varepsilon$, and the vector field terms are unchanged.

This is the cube on which the renormalization transformation of Sect. 3.1 is carried out. The cube in which every site of $C_J \cap \varepsilon\mathbb{Z}^d$ carries a scalar variable, which is the cube of the correlation inequalities of Sect. 3.2 and of the action (3.38), has the same infinite volume limit for the field strength, in the sense that every limit of its cube expectations as $K \to \infty$ along a subsequence, followed by $J \to \infty$, is the limit (2.18) (Sect. 6.1, Lemma 6.4).

With these choices, the partition function in C_J is written

$$Z^{\varepsilon}_{N,D}(C_J) = \int (dA)(d\phi) \exp[-S^{\varepsilon}(C_J, A, \phi)] \,, \tag{2.9}$$

and the expectation of an operator $\mathcal{O}$ is defined by

$$\langle \mathcal{O} \rangle^{\varepsilon, C_J}_{N,D} = Z^{\varepsilon}_{N,D}(C_J)^{-1} \int (dA)(d\phi)\, \mathcal{O}(A,\phi) \exp[-S^{\varepsilon}(C_J, A, \phi)] \,. \tag{2.10}$$

We are primarily interested in gauge invariant operators $\mathcal{O}$. Denote by Ω^{**} the set of plaquettes on Ω, and define the field strength tensor $F : \Omega^{**} \to \mathbb{R}$ by

$$F(p) = \sum_{b \in \partial p} \varepsilon^{-1} A_b \,. \tag{2.11}$$

Then for a function $g : \Omega^{**} \to \mathbb{R}$, we have

$$F(g) = \sum_{p \in \Omega^{**}} \varepsilon^d F(p) g(p) \,. \tag{2.12}$$

As in [I], $g(p) = g_{\mu\nu}(x_p)$ is the value at the center x_p of the plaquette p of a real 2-form g; this is the form in which the comparison of two lattices in Sect. 4 holds.

For a function $h : \Omega \to \mathbb{R}$, the gauge-invariant operator for the scalar field is

$$:\phi^2: h = \sum_{x \in \Omega} \varepsilon^d h(x) \big[r(x)^2 - 2(-\Delta^{\varepsilon} + m^2)^{-1}(x,x) - a_{\varepsilon} \big] \,, \tag{2.13}$$

where $-\Delta^{\varepsilon}$ is the Laplacian on $\varepsilon\mathbb{Z}^d$, so that the subtraction is the same for every Ω (see the proof of Theorem 3.2), and a_{ε} is the constant of (2.8) of [I]. The dependence of the results on the choice of a_{ε} is discussed in Sects. 6.6 and 6.10. The general operator we consider is

$$\mathcal{O}(zg, wh) = \exp\big[zF(g) + w :\phi^2:(h) \big] \,, \tag{2.14}$$

where $g, h \in C_0^{\infty}(\mathbb{R}^d)$ and $z, w \in \mathbb{C}$. For the source h the results below hold on a tube in w. We say that a real function k on $\mathbb{R}^d$ lies in the tube if

$$\sup_x k(x)_+ < \sigma_+/2 \qquad \text{and} \qquad \sup_x k(x)_- < H/2 \,, \tag{2.15}$$

where $k_{\pm}$ are the positive and negative parts of k and $\sigma_+, H > 0$ are the constants of Sect. 6.6, which depend on the coupling constants and on the auxiliary choices made there; we write $(w,h) \in \mathcal{D}$ if $\operatorname{Re} w\, h$ lies in the tube. Every (w,h) with w purely imaginary is in $\mathcal{D}$, and for each h so is every w with $|\operatorname{Re} w|$ small enough. The results below with $h \neq 0$ are proved under a condition on the coupling constants, stated in Sect. 6.8, which holds on the explicit regions of couplings described there (for instance, when $\mu_0 = m$, for e^2 and λ small compared with m); for $h = 0$ there is no condition on the couplings.

It will be understood below that the results stated hold for dimension $d = 3$. The first result extends Theorem 2.1 of [I] to include the boundary conditions used in this paper, and the second result gives the existence of the infinite volume limit.

Theorem 2.1. *Let C_J be defined by* (2.7)*, and let g be a real 2-form and h a real function, both C^∞ and supported at distance greater than* 1 *from the complement of C_J. Then for all $z \in \mathbb{C}$ and all $w \in \mathbb{C}$ with*

$$2 \sup_x (\operatorname{Re} w\, h(x))_+ < \sigma_+ \,, \tag{2.16}$$

where $\sigma_+ > 0$ is the constant of Sect. 6.6,

$$\exists \lim_{K\to\infty} \langle \mathcal{O}(zg, wh)\rangle_{N,D}^{\varepsilon_K, C_J} = \langle \mathcal{O}(zg, wh)\rangle_{N,D}^{C_J} \,. \tag{2.17}$$

The set of w in (2.16) is an open vertical strip containing the imaginary axis; it depends on h only through $\sup h_+$ and $\sup h_-$, and it contains the half-plane $\operatorname{Re} w \leqq 0$ when $h \geqq 0$.

Theorem 2.2. *Let g, h be C^∞ functions with compact support, $z \in \mathbb{C}$ and $(w, h) \in \mathcal{D}$. Then*

$$\exists \lim_{J\to\infty} \langle \mathcal{O}(zg, wh)\rangle_{N,D}^{C_J} = \langle \mathcal{O}(zg, wh)\rangle \,. \tag{2.18}$$

We also establish some properties of the limit (2.18). We denote by Tf the function obtained from f by some translation T in the Euclidean group on $\mathbb{R}^d$. Also, if $\operatorname{supp} f \subset \{x \in \mathbb{R}^d : x_\nu > c\}$ for some real c, we denote by $\theta_\nu f$ the function obtained from f by reflection in the plane $x_\nu = c$.

Theorem 2.3. *The infinite volume limit* (2.18) *satisfies the following properties:*

(i)

$$\langle \mathcal{O}(Tg, Th)\rangle = \langle \mathcal{O}(g, h)\rangle \,; \tag{2.19}$$

(ii) *if $\operatorname{supp} g \cup \operatorname{supp} h \subset \{x \in \mathbb{R}^d : \ x_\nu > c\}$, some ν and some real c, and, when $h \neq 0$, $\sup_x |\operatorname{Re} w\, h(x)| < \sigma_+/8$, then*

$$\langle \overline{\mathcal{O}}(\theta_\nu g, \theta_\nu h)\, \mathcal{O}(g, h)\rangle \geqq 0 \,, \tag{2.20}$$

where the bar denotes complex conjugation (the limit (2.18) *exists at these points also when $(w, h) \notin \mathcal{D}$, see Sect. 6.6);*

(iii) *if the $2N$ functions $\{g_i\}$, $\{h_j\}$, $i, j = 1, \ldots, N$, are in $C_0^\infty(\mathbb{R}^d)$, and $\{z_i\}$, $\{w_j\}$ are in $\mathbb{C}$, the function*

$$\left\langle \mathcal{O}\left(\sum_{i=1}^N z_i g_i, \ \sum_{j=1}^N w_j h_j \right) \right\rangle \tag{2.21}$$

is entire in $(z_i) \in \mathbb{C}^N$, and holomorphic in (w_j) on the open set where $\operatorname{Re} \sum_j w_j h_j$ lies in the tube.

The properties we have listed in Theorem 2.3 are a subset of the Osterwalder-Schrader axioms presented in [GJ 1]. To complete the list, we would need to prove ergodicity, rotation invariance and regularity. Ergodicity corresponds to uniqueness of the vacuum, and has been proved for other models using cluster expansion techniques. To establish regularity we would need to keep more careful track of the source functions g, h when proving ultra-violet stability using the renormalization transformation method of [Ba 1–2]. We follow a different route: regularity is proved in Sect. 6.2 by bounds independent of the renormalization transformation, invariance under the cubic group in Sect. 6.3 and, for the field strength, under all rotations in Sect. 6.7; in place of ergodicity we obtain in Sect. 6.5 the decomposition of the limit measure into ergodic components. We do not prove rotation invariance for the scalar sources.

In order to prove rotation invariance, it is sufficient to show that the limit $\langle \mathcal{O}(zg, 0)\rangle$ is the same when constructed from two sequences of lattices whose orientations differ by an angle θ_0 incommensurate with 2π. The two lattices we use are illustrated in Fig. 1. Both have spacing ε; the second is the image of the first under the rotation R through the angle $\theta_0 = \tan^{-1}(3/4)$ in a coordinate plane. Since $5R$ is an integer matrix, both lattices contain the sublattice $R(5\varepsilon\mathbb{Z}^d)$;

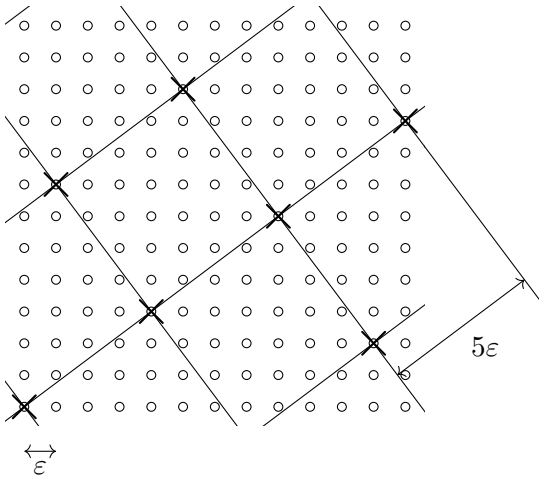


Fig. 1: Two lattices with different orientations. The circles are the sites of $\varepsilon\mathbb{Z}^2$; the lines are those of the sublattice $R(5\varepsilon\mathbb{Z}^2)$ of the rotated lattice $R(\varepsilon\mathbb{Z}^2)$, and the crosses mark its sites, which lie on both lattices

in Fig. 1 the rotated lattice is represented by the lines of this sublattice. For $d > 2$, the lattices have the same orientation in the other $d-2$ coordinate directions not shown in Fig. 1.

The methods we use to construct the continuum limit require quite explicit expressions for lattice operators with periodic, Neumann and Dirichlet boundary conditions. These hold only when the boundary is a lattice hyperplane orthogonal to a lattice coordinate direction. Clearly the two lattices in Fig. 1 never share such a boundary. Our solution of this problem is to use periodic boundary conditions. We define a torus D_J by (2.7) with M replaced by $25M$. For $L = 5$, we can "fit" D_J by lattices with the same orientation as the sides of D_J and with lattice spacings $\varepsilon_K = L^{-K}$. The expectation defined by this lattice is written

$$\langle \mathcal{O} \rangle_p^{\varepsilon_K, D_J} . \tag{2.22}$$

From Fig. 1, and since the length of each side of D_J is a multiple of 25 and $5R^{-1}$ is an integer matrix, we see that D_J is also fitted by the rotated lattice $R(\varepsilon_K\mathbb{Z}^d)$, $\varepsilon_K = L^{-K}$. The expectation defined by this rotated lattice is written

$$\langle \mathcal{O} \rangle_p^{R(\varepsilon_K), D_J} . \tag{2.23}$$

We can now state our result concerning rotation invariance.

Let ω be a smooth non-negative function on D_J, which we call a mass term. The periodic model with the term $\sum_x \varepsilon^d \omega(x) r(x)^2$ added to the action (2.5) has the expectation $\langle\cdot\rangle_p^{\varepsilon_K, D_J, \omega}$, and similarly for the rotated lattice. The class of mass terms and of tori for which the following theorem is proved is described in Sect. 4; it does not contain $\omega = 0$.

Theorem 2.4. *Let ω be a mass term of the class of Sect. 4 on D_J, and let g be a real 2-form supported in D_J at distance greater than 1 from the support of ω, sampled by each lattice at the centers of its own plaquettes, $g(p) = g_{x_p}(u_p, v_p)$ with u_p, v_p the unit vectors spanning p in that lattice's own frame. Then for $z \in \mathbb{C}$ the limits*

$$\lim_{K\to\infty} \langle \mathcal{O}(zg, 0) \rangle_p^{\varepsilon_K, D_J, \omega} \qquad \text{and} \qquad \lim_{K\to\infty} \langle \mathcal{O}(zg, 0) \rangle_p^{R(\varepsilon_K), D_J, \omega} \tag{2.24}$$

exist and are equal.

$$\textit{Also, } \theta_0/2\pi \textit{ is irrational.} \tag{2.25}$$

The theorem holds with R replaced by R^{-1}, since $R^{-1} = SRS$ with S a coordinate reflection, which maps the torus D_J to itself and the class of Sect. 4 to itself. The mass term enters the

lower bounds used there; the comparison in the presence of the mass term is all that the proof of rotation invariance requires. Theorem 2.4 is proved in Sect. 4 by comparing the renormalization transformations of the two lattices at equal spacing; the estimates of this comparison which are taken from [M] are stated there.

The theorem does not hold for a scalar function g sampled on all the plaquettes of both lattices: already at $e = 0$, where both sides of (2.24) are Gaussian, the two variances differ for suitable g. Nor do we state it for $h \neq 0$. Without the constant a_ε in (2.13) the expectation of $:\phi^2:(h)$ would contain, in perturbation theory, a term of order e^2 proportional to $\ln \varepsilon^{-1} \int h$; the comparison of Sect. 4 is made at equal spacing, where both lattices carry the same constant, and the statement with h is not needed.

Because the expectations in (2.24) are defined with P b.c., we cannot use correlation inequalities to take the infinite volume limit along boxes $\Lambda \nearrow \mathbb{R}^d$. We proceed differently.

For $b > 0$ denote by Ω_b the set of pairs $(D_{J'}, \omega)$ admitted in Theorem 2.4 such that the cube Q_{b+2}, where $Q_b = \{x : \ |x_\mu| \leqq b\}$, lies inside $D_{J'}$ and ω vanishes at distance less than 2 from Q_b. For a real 2-form g supported in Q_b and z purely imaginary, Theorem 2.4 gives the limit

$$W(zg; D_{J'}, \omega) = \lim_{K\to\infty} \langle \mathcal{O}(zg, 0)\rangle_p^{\varepsilon_K, D_{J'}, \omega}\,, \qquad (D_{J'}, \omega) \in \Omega_b\,, \tag{2.26}$$

which is real and lies in $[-1, 1]$, since $\mathcal{O}(zg, 0) = \cos(tF(g)) + i \sin(tF(g))$ for $z = it$ and the expectation of $\sin(tF(g))$ vanishes by the symmetry $(A, \theta) \to (-A, -\theta)$; and the correlation inequalities (3.24)–(3.26) give

$$\langle \mathcal{O}(zg, 0)\rangle_{N,D}^{\varepsilon_K, C_J} \leqq \langle \mathcal{O}(zg, 0)\rangle_p^{\varepsilon_K, D_{J'}, \omega} \tag{2.27}$$

whenever $\operatorname{supp} g$ lies at distance greater than 1 from the complement of C_J and $MJ \leqq b$. These inequalities hold for every measure of the class of Sect. 6.1 (Lemma 6.1), in particular on the torus $D_{J'}$ with the term $\sum_x \varepsilon^d \omega(x) r(x)^2$ and on the lattice $R^{-1}(\varepsilon_K \mathbb{Z}^d)$; what fails with P b.c. is the monotonicity in the volume, since no coupling can be removed to pass from one torus to a smaller one.

The cube C_J with its boundary conditions is obtained from the torus by removing the gradient couplings of the vector field and the cosine couplings of the scalar field on the bonds crossing ∂C_J, the diagonal parts $\frac{1}{2}\varepsilon^{d-2} r^2$ remaining as the Dirichlet terms, and by setting the scalar field to zero on the top layers of C_J, neither of which increases the expectation (Lemmas 6.2, 6.3); and $\omega = 0$ on C_J.

The class of Sect. 4 contains, for every b, pairs $(D_{J'}, \omega)$ with ω vanishing at distance less than 2 from Q_b: it suffices to take $\omega = 2c_0$ outside Q_{b+3}, c_0 the constant of (4.2), with a smooth transition on $Q_{b+3} \setminus Q_{b+2}$, and J' so large that $|Q_{b+3}| \leqq \frac{1}{2}|D_{J'}|$. So Ω_b is not empty. Therefore the infimum $I_b(zg)$ of (2.26) over Ω_b, which is nondecreasing in b, satisfies

$$\langle \mathcal{O}(zg, 0)\rangle \leqq \lim_{b\to\infty} I_b(zg)\,. \tag{2.28}$$

Equality holds in (2.28): this is Theorem 6.24 below, whose proof in Sect. 6.7 rests on estimates taken from [M]. Let R be the rotation through the angle θ_0 in the $\mu\nu$-coordinate plane and Rg the transform of the 2-form g. The expectation of $\mathcal{O}(zRg, 0)$ in the cube C_J on the lattice $\varepsilon_K \mathbb{Z}^d$ equals the expectation of $\mathcal{O}(zg, 0)$ in the rotated cube $R^{-1}C_J$ on the rotated lattice $R^{-1}(\varepsilon_K \mathbb{Z}^d)$, whose faces lie on lattice hyperplanes of that lattice. By the correlation inequalities on the rotated lattice this is at most the expectation on any torus $D_{J'}$ carrying a mass term which vanishes near $R^{-1}C_J$, that is on every $(D_{J'}, \omega) \in \Omega_{7MJ/5}$, and by Theorem 2.4, applied to the rotation R^{-1}, the limits $K \to \infty$ of the latter on the two lattices agree:

$$\lim_{K\to\infty} \langle \mathcal{O}(zRg, 0)\rangle_{N,D}^{\varepsilon_K, C_J} = \lim_{K\to\infty} \langle \mathcal{O}(zg, 0)\rangle_{N,D}^{R^{-1}(\varepsilon_K), R^{-1}C_J} \leqq W(zg; D_{J'}, \omega)\,, \tag{2.29}$$

so the limit of the left side is at most $I_{7MJ/5}(zg)$. Since $R^{-1}C_J$ contains the interior of $Q_{5MJ/7}$ and is contained in $Q_{7MJ/5}$, letting $J \to \infty$ gives

$$\langle \mathcal{O}(zRg, 0)\rangle \leqq \lim_{b\to\infty} I_b(zg) = \langle \mathcal{O}(zg, 0)\rangle \,, \tag{2.30}$$

and the reverse inequality follows by applying (2.30) to R^{-1} and Rg.

By iterating the rotation R and using the fact that $\theta_0/2\pi$ is irrational, we generate a dense set of rotations in the $\mu\nu$-plane; all rotations in that plane follow from the continuity of (2.18) in g (Sect. 6.3). Repeating the procedure for each μ, ν, and using the cubic symmetry of Sect. 6.3, gives invariance under the full orthogonal group for z purely imaginary, and then for all z, since both sides are entire in z by Theorem 2.3(iii).

Theorem 2.5. *For g a real 2-form in $C_0^\infty(\mathbb{R}^d)$, $z \in \mathbb{C}$ and $R \in O(d)$,*

$$\langle \mathcal{O}(z\, Rg, 0)\rangle = \langle \mathcal{O}(zg, 0)\rangle \,, \tag{2.31}$$

where Rg is the transform of the 2-form g under R.

Theorem 2.5 is proved in Sect. 6.7 from Theorem 2.4 and the equality in (2.28). Together with Theorems 2.1–2.3 and the results of Sects. 6.2–6.6, it gives the following theorem, whose complete statement is given in Sect. 6.9.

Theorem 2.6. *The Schwinger functions $S_n(g_1, \dots, g_n) = \partial_{z_1} \cdots \partial_{z_n} \langle \mathcal{O}(\sum_i z_i g_i, 0)\rangle|_{z=0}$ of the limit* (2.18) *have the following properties.*

(i) *They are tempered distributions, symmetric in their arguments, satisfying the linear growth condition (E0′) of* [OS 2]*, and for real g_i*

$$|S_n(g_1, \dots, g_n)| \leqq C^n (n!)^{1/2} \prod_{i=1}^n \|g_i\|_2 \,, \tag{2.32}$$

where C is a numerical constant.

(ii) *They are invariant under the Euclidean group, F transforming as a 2-form, and reflection positive about every plane; so they satisfy all the axioms of Osterwalder and Schrader* [OS 1–2] *except the cluster property.*

(iii) *They are the moments of a unique probability measure on the tempered distributions with values in 2-forms, which is an integral of translation-ergodic, reflection positive measures, almost every one of which satisfies the cluster property.*

(iv) *When the coupling constants satisfy the condition of Sect. 6.8, the joint Schwinger functions of $\langle \mathcal{O}(zg, wh)\rangle$ at $(w, h) \in \mathcal{D}$ are tempered distributions, invariant under translations and under the cubic group, and reflection positive about the coordinate planes.*

3. The Infinite Volume Limit

3.1. The Continuum Limit

The renormalization transformation method was introduced in Sect. 2.2 of [I]. The space of field configurations is divided into cells, each corresponding to fluctuations of the field on a certain length scale. Beginning with the shortest length scales, the integral over fields in the partition function is successively analyzed in each cell. This method of phase space localization was first introduced in [GJ 2]. It was developed for the $U(1)$ Higgs model in [Ba 1–4], using a block spin transformation to separate fields into average and fluctuation parts.

In order to apply the renormalization transformation to the case when the scalar field ϕ has D b.c., we modify the averaging operators defined in [I]. Given a block $B(y)$ containing L^d sites, we consider a bounded function $f : B(y) \to \mathbb{R}$ such that

$$\sum_{x\in B(y)} L^{-d} f(x) = 1 \,. \tag{3.1}$$

For each $y \in L\varepsilon\mathbb{Z}^d$, we introduce f on the block $B(y)$, giving a function f on $\varepsilon\mathbb{Z}^d$. We now define the averaging operator for the scalar field by

$$(Q_k(A)\phi)(y) = L^{-kd} \sum_{x \in B^k(y)} f(x) \exp[ieA(\Gamma^{(k)}_{y,x})]\phi(x)\,, \tag{3.2}$$

where $y \in L^k\varepsilon\mathbb{Z}^d$ and $A(\Gamma^{(k)}_{y,x})$ is defined in [I]. The operators and notation used in [I] will be taken over unchanged, with the understanding that $Q_k(A)$ is replaced by (3.2). We leave unchanged the averaging operator for the vector field, its blocks being the blocks $B^k(y)$ placed as below. The results stated in Sect. 2.2 of [I] still hold, in particular the representation of the propagator, (2.18) of [I].

The blocks $B^k(y)$ are placed so that the planes $x_\mu = 0$ and $x_\mu = L_\mu$ of (3.3) below are top layers of blocks, the plane $x_\mu = 0$ being the top layer of the block below Ω: in the coordinates of (3.3), in which the sites of Ω are $\varepsilon \leqq x_\mu \leqq L_\mu$, the blocks are $B(y) = \{x :\ y_\mu \leqq x_\mu < y_\mu + L\varepsilon\}$ with $y \in (\varepsilon + L\varepsilon\mathbb{Z})^d$, and $B^k(y)$ with $y \in (\varepsilon + L^k\varepsilon\mathbb{Z})^d$, while in the coordinates of (2.7), in which (3.2) and (3.11) are written, the corners are the points of $L^k\varepsilon\mathbb{Z}^d$. So the top layers $x_\mu = y_\mu + (L^k - 1)\varepsilon$ of the blocks lie on the planes $x_\mu \in L^k\varepsilon\mathbb{Z}$, in particular on the Dirichlet planes $x_\mu = 0$ and $x_\mu = L_\mu$, on which f vanishes, as (3.9) requires, and Ω is a union of blocks $B^k(y)$ for every k satisfying (3.4), its faces lying midway between blocks as (3.7) requires.

With corners $y \in L\varepsilon\mathbb{Z}^d$ in these coordinates, reflection in $x_\mu = 0$ would not carry the support of f in a block onto the support of f in a block, $P_k(0)$ would not be reflection invariant, and the derivation of (3.9) by odd images would break down.

For the cube C_J of (2.7), whose Dirichlet planes are $x_\mu = -MJ - \varepsilon$ and $x_\mu = MJ - \varepsilon$, these are the blocks $B^k(y)$ with corners $y \in L^k\varepsilon\mathbb{Z}^d$, the sites of C_J being those of (3.3) with $L_\mu = 2MJ$ after a translation by $MJ + \varepsilon$ in each direction; the lattices $T^{(k)}$ of (3.11) are the sets of these corners in C_J.

Using the method of multiple reflections, we can write an operator with P, N, or D b.c. on a lattice hyperplane as a sum of operators with free boundary conditions. Suppose $\Omega \subset \varepsilon\mathbb{Z}^d$ is a rectangular parallelepiped given by

$$\Omega = \{x \in \varepsilon\mathbb{Z}^d :\ 0 < x_\mu \leqq L_\mu,\ \mu = 1, \dots, d\}\,, \tag{3.3}$$

and the dimensions L_μ satisfy

$$(L^k\varepsilon)^{-1} L_\mu \in \mathbb{N}\,, \quad \text{each} \quad \mu = 1, \dots, d\,. \tag{3.4}$$

Recall that the renormalization transformation produces a sequence of covariances defined by

$$G^\varepsilon_k(\Omega) = \left(-\Delta^{\varepsilon,\Omega} + m^2 + a_k(L^k\varepsilon)^{-2}P_k(0)\right)^{-1}, \tag{3.5}$$

where $P_k(0) = Q^*_k(0)Q_k(0)$ and $-\Delta^{\varepsilon,\Omega}$ is the Laplacian on Ω with some boundary conditions on $\partial\Omega$. The operator G^ε_k with free boundary conditions on $\varepsilon\mathbb{Z}^d$ is given by (3.5) with $-\Delta^{\varepsilon,\Omega}$ replaced by $-\Delta^\varepsilon$. We have an explicit Fourier representation for G^ε_k and so we want to write covariances with other boundary conditions in terms of G^ε_k. For P b.c. on Ω, we take $f = 1$ in (3.1) and then

$$G^\varepsilon_k(\Omega)(x, y) = \sum_{n \in \mathbb{Z}^d} G^\varepsilon_k \left(x,\ y + \sum_{\mu=1}^d n_\mu L_\mu e_\mu\right). \tag{3.6}$$

For N b.c. we also take $f = 1$ and then

$$G^\varepsilon_k(\Omega)(x, y) = \sum_{n \in \mathbb{Z}^d} \sum_{m \in \{0,1\}^d} G^\varepsilon_k \left(x,\ y + \sum_{\mu=1}^d 2n_\mu L_\mu e_\mu - \sum_{\nu=1}^d m_\nu(2y_\nu - \varepsilon)e_\nu\right). \tag{3.7}$$

Finally, for D b.c. we use a different function f. For $x \in B(y)$, define

$$f(x) = \begin{cases} \left(\dfrac{L}{L-1}\right)^d & \text{if} \quad x_\mu - y_\mu \leqq (L-2)\varepsilon \quad \text{each} \quad \mu = 1, \dots, d\,, \\ 0 & \text{otherwise}\,. \end{cases} \tag{3.8}$$

We take $-\Delta^{\varepsilon,\Omega}$ to have D b.c. on $\{x \in \varepsilon\mathbb{Z}^d : \ x_\mu = 0 \text{ or } x_\mu = L_\mu, \text{ some } \mu = 1, \dots, d\}$. Then we can write

$$G_k^\varepsilon(\Omega)(x,y) = \sum_{n \in \mathbb{Z}^d} \sum_{m \in \{0,1\}^d} (-1)^{\Sigma_\nu m_\nu}\, G_k^\varepsilon \left(x,\ y + \sum_{\mu=1}^d 2n_\mu L_\mu e_\mu - \sum_{\nu=1}^d 2m_\nu y_\nu e_\nu \right). \tag{3.9}$$

If we had taken $f = 1$, the representation (3.9) would not hold.

We now wish to prove Theorem 2.1, giving the continuum limit in a region C_J. In Sect. 5, Theorems 3.1 and 3.3 of [I] are extended to include the case when the vector field has N b.c. and the scalar field has D b.c. Therefore, the existence of the continuum limit in C_J will follow from the convergence of the field-dependent part of the effective action, which was proved for P b.c. in [I].

Recall that Theorem 3.1 of [I] states that the generating functional $Z_{N,D}^{\varepsilon}(C_J)\langle \mathcal{O}(zg, wh)\rangle_{N,D}^{\varepsilon,C_J}$ is well approximated by the integral of an effective action $S^{(k),L^k\varepsilon}$ on the lattice with spacing $L^k\varepsilon$ in C_J, when z, w are real. Since the error in that approximation is at most $e^{-p(L^k\varepsilon)^2 + C|C_J|}$, while $Z_{N,D}^{\varepsilon}(C_J) \geqq e^{-C|C_J|}$ by the lower bound of Sect. 5.2 and $\langle \mathcal{O}(zg, wh)\rangle_{N,D}^{\varepsilon,C_J} \leqq e^{C|C_J|}$ for real z, w by the upper bound of Sect. 5, the error relative to the functional at zero source is at most $e^{-p(L^k\varepsilon)^2 + C|C_J|}$, which tends to zero as $L^k\varepsilon \to 0$ with J fixed; this is the form we use, the limit $\varepsilon \to 0$ being taken first at fixed $L^k\varepsilon$. By proving convergence of the field-dependent part of $S^{(k),L^k\varepsilon}$ as $\varepsilon \to 0$, with constants uniform in the number of steps, we deduce convergence of the ratio of the generating functional to its value at zero source, by the argument of Sect. 3.2 of [I], and hence of $\langle \mathcal{O}(zg, wh)\rangle_{N,D}^{\varepsilon,C_J}$ for z, w real, provided that the part of $S^{(k),L^k\varepsilon}$ which is independent of the fields but depends on the sources converges.

For $h = 0$ this part consists of the graphs with source vertices and no field. Under the charge conjugation $A \to -A$, $\phi \to \bar{\phi}$ the action, the boundary conditions and the renormalization transformation (3.2) are invariant, since $Q_k(-A)\bar{\phi} = \overline{Q_k(A)\phi}$, while $F(g)$ of (2.12) changes sign; the field-independent part of $S^{(k)}$ is therefore even in g, the terms with an odd number of curvature vertices vanish, and those with two or more are covered by Theorem 3.4 of [I], extended to C_J in Sect. 6.1. Furthermore,

$$\left|\langle \mathcal{O}(zg, wh)\rangle_{N,D}^{\varepsilon,C_J}\right| \leqq \langle \mathcal{O}(\operatorname{Re} zg, \operatorname{Re} wh)\rangle_{N,D}^{\varepsilon,C_J}\,. \tag{3.10}$$

Therefore, using Vitali's theorem, since $\ln\langle \mathcal{O}(tg, 0)\rangle_{N,D}^{\varepsilon,C_J}$ is convex in real t, so that, by (3.10), the family is bounded on $|\operatorname{Re} z| \leqq T$ by its values at $z = \pm T$, which converge, it follows that $\langle \mathcal{O}(zg, 0)\rangle_{N,D}^{\varepsilon,C_J}$ converges to an entire function of z as $\varepsilon \to 0$. When $h \neq 0$ the same argument applies for w in the set of Theorem 2.1, which is open and convex and contains with each w the real point $\operatorname{Re} w$, using (3.10), the joint convexity of $\ln\langle \mathcal{O}(sg, th)\rangle_{N,D}^{\varepsilon,C_J}$ in real s, t and the convergence for real w, which is established in Sect. 6.6.

In the remainder of this section we will prove convergence of the effective action $S^{(k),L^k\varepsilon}$ in C_J for z, w real. For convenience, we adopt the following notation:

$$T_{L^k\varepsilon_K}^{(k)} = C_J \cap L^k\varepsilon_K\mathbb{Z}^d\,, \qquad k = 0, 1, \dots, K\,. \tag{3.11}$$

When $k = 0$, we write (3.11) as T_ε. Under a re-scaling $\varepsilon \to \eta$, (3.11) becomes $T_{L^k\eta}^{(k)}$. We will henceforth reserve η to mean L^{-k}. Recall that for an integer $n \geqq 1$, $\eta' = L^{-n}\eta = L^{-(k+n)}$.

The effective action $S^{(k),L^k\varepsilon}$ is defined on the lattice $T^{(k)}_{L^k\varepsilon}$. For convenience, we re-scale $L^k\varepsilon \to 1$ and then we have the following formula for $S^{(k),1}$ (see Sect. 3.6 of [I]):

$$\begin{aligned} S^{(k),1}(T^{(k)}, A_k, \phi_k, g, h) &= \tfrac{1}{2}\langle A_k, \Delta^{(k)} A_k\rangle + \ln N_k \\ &\quad + \tfrac{1}{2}\langle \phi_k, \Delta^{(k)}(A^{(k)})\phi_k\rangle + \ln N_k(0) \\ &\quad - \ln\left[\frac{Z_k(A^{(k)})}{Z_k(0)}\right] + P^{(k),1}(T^{(k)}, A_k, \phi_k, g, h)\,. \end{aligned} \tag{3.12}$$

The operators $\Delta^{(k)}$, $\Delta^{(k)}(A^{(k)})$ are effective Laplacians for the fields A_k and ϕ_k. The interaction term $P^{(k),1}$ is represented by a sum of connected graphs with vertices on the lattice T_η. These graphs were described in detail in [I] and in [Ba 3]. The only way that the boundary conditions enter this description is through the propagators $G^\eta_k(A^{(k)})$ and G^η_k connecting the vertices. In our case, $G^\eta_k(A^{(k)})$ has D b.c. and G^η_k has N b.c. on ∂T_η. We shall specify below how the proof of convergence in Sect. 3.4 of [I] must be modified to take account of this.

When we localize the vertices of a graph in a cube $\square$, it may happen that $\partial\square \cap \partial T^{(k)}$ is non-empty. In this case the vertices will see a "sharp boundary;" this is the main difference between periodic boundary conditions and those used here. The following remarks refer to Sect. 3.4 of [I].

Having localized the vertices of a graph H in a block $\square$, we obtained the expression $E^{(k)}(H, A^{(k)}; \{y_i\}, \{z_q\})$. We now replace each propagator $G_k(A^{(k)})$ by the difference $G_k(\square', A^{(k)}) - \delta G_k(\square', A^{(k)})$, where we take $G_k(\square', A^{(k)})$ to have D b.c. on $\partial\square'$; here $\square'$ is a rectangular parallelepiped which is a union of large blocks of the unit lattice $T^{(k)}$, of the form (3.3)–(3.4), so that Proposition 5.1 and (3.9) apply on $\partial\square'$.

Here, as in (3.9), D b.c. on $\partial\square'$ means that the field vanishes on the top layer of sites of $\square'$ and on the layer of sites immediately below $\square'$ in each coordinate direction; the propagators $G_k(\Omega, A)$ of Sect. 5 and Appendix A are defined with this convention, the only one for which (3.9) holds with the blocks of this section. In the expansions (3.65)–(3.71) of [I], we take $B = 0$ and $A' = \tilde{A}^{(k)}$. We omit the replacement (3.72) of [I], so every scalar field propagator in our graph is $G_k(\square', 0)$. We shall prove Propositions 3.8 and 3.9 of [I] for $G_k(\square', 0)$ and G_k in Appendix A. There is a minor technical point connected with the operator $Q_{k+n}(0)$. After all propagators and sources have been replaced in a graph in $E^{(k+n)}(H)$, we have (for some function K)

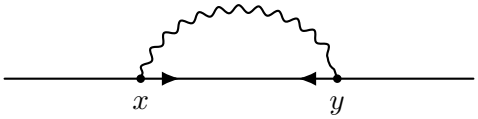
$$\begin{aligned} (\eta')^d \sum_{z\in B^k(y)} \sum_{x\in B^n(z)} f(x)K(z) &= \eta^d \sum_{z\in B^k(y)} K(z) \\ &= (Q_k K)(y) + \eta^d \sum_{z\in B^{k-1}(y)} \sum_{x\in B(z)} (1 - f(x))|x - z|^\alpha \\ &\quad \cdot \left[\frac{K(x) - K(z)}{|x - z|^\alpha}\right]. \end{aligned} \tag{3.13}$$

The second term in (3.13) gives a factor $L^{-\alpha k}$, since $|x - z| \leqq \sqrt{d}L^{-k+1}$ for all x, z.

There are other modifications required when we examine subgraphs with non-positive degree. First, note that it is sufficient to assume that $\partial\square' \cap \partial T_\eta$ is non-empty; otherwise, the vertices of the graph never see a sharp boundary, and the methods of [I] go straight through.

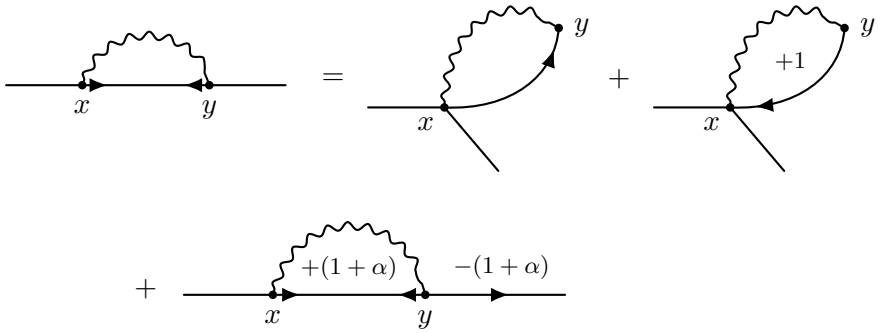
We shall define the mass counterterm δm^2 using free boundary conditions; that is the propagators used to define the graphical expansion of δm^2 (see Sect. 3.3 of [I]) have free boundary conditions on ∂T_η. This is necessary in order to take the infinite volume limit (see the next section). At first glance, this appears to cause problems since the divergent diagrams produced by perturbation theory have propagators with N and D b.c. on ∂T_η. However, because we use D b.c. for the scalar field, the propagators entering a mass vertex go to zero near the boundary, and this is sufficient to keep the graph finite, once the most divergent part has been removed.

As a specific example, consider the following linearly divergent graph for the self-energy of the scalar field:



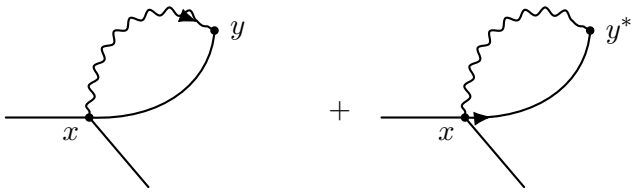
(3.14)

As explained in Sect. 3.5 of [I], this graph may be written as



(3.15)

The third graph on the right-hand side of (3.15) is convergent. In the first graph on the right-hand side, we integrate by parts at y to give



(3.16)

where the asterisk denotes differentiation of the localization function $\chi(y)$. Using the representations (3.7) and (3.9) we now write (3.16) and the second graph on the right-hand side of (3.15) as a sum of graphs with free propagators. The graph in δm^2 corresponding to (3.14) has an expansion similar to (3.16), and we may cancel some of the terms produced. The remaining graphs have one or more propagators connecting x to some reflection of y in ∂T_η, or they have y summed over $\eta\mathbb{Z}^d \setminus T_\eta$.

Let H represent one of these remaining graphs. Suppose that for some line l in H, there is a propagator connecting x to a reflection of y in ∂T_η. Then we see that the graph H with l omitted has positive degree [(3.16) ensured this], and the degree of H is greater than or equal to -1. We can analyse H in the manner described in Sect. 3.4 of [I]. The internal propagators are decomposed with the relation $G_k^\eta = \sum_{j=0}^{k-1} G_{(j)}^\eta$, and we can restrict attention to one particular ordering of the internal lines. After inserting the appropriate bounds, we get the following factor for the line l:

$$\exp\bigl[-\delta_0(L^j\eta)^{-1}\{|x-y| + \operatorname{dist}(\{x,y\},\partial\square')\}\bigr]\,. \tag{3.17}$$

Following the methods of [I], we shrink the lines in H to x and the final sum of integers is

$$\begin{aligned}
&\sum_{j=0}^{j_0}(L^j\eta)^{D(H)}\exp\bigl[-\delta(L^j\eta)^{-1}\operatorname{dist}(\{x\},\partial\square')\bigr] \\
&\qquad \leqq C\exp\bigl[-\delta(L^{j_0}\eta)^{-1}\operatorname{dist}(\{x\},\partial\square')\bigr] \\
&\qquad\quad \cdot\begin{cases} (L^{j_0}\eta)^{D(H)}\,, & D(H) > 0\,, \\ 1 + |\ln[\operatorname{dist}(\{x\},\partial\square')]|\,, & D(H) = 0\,, \\ |\operatorname{dist}(\{x\},\partial\square')|^{D(H)}\,, & D(H) < 0\,, \end{cases}
\end{aligned} \tag{3.18}$$

where j_0 is the smallest integer on an external line. Suppose now that x is connected to another vertex z by a scalar field line with integer j_1; then for all $0 \leqq \alpha < 1$,

$$\begin{aligned} &|\operatorname{dist}(\{x\}, \partial\square')|^{-\alpha} |G^{\eta}_{(j_1)}(\square'; z, x)| \\ &\qquad \leqq C(L^{j_1}\eta)^{2-d-\alpha} \exp\left[-\delta_0 (L^{j_1}\eta)^{-1} |z-x|\right]. \end{aligned} \tag{3.19}$$

This bound follows from the representation (3.9) and Proposition 3.7 of [I]. Therefore, we have the additional factor $|\operatorname{dist}(\{x\}, \partial\square')|^{\alpha}$ available when we sum over x, and so we get

$$\begin{aligned} &\sum_{x \in \square'} \eta^d \chi(x) |\operatorname{dist}(\{x\}, \partial\square')|^{D(H)+\alpha} \\ &\qquad \cdot \exp\left[-\delta (L^{j_1}\eta)^{-1} \{|z-x| + \operatorname{dist}(\{x\}, \partial\square')\}\right] \\ &\qquad \leqq C(L^{j_1}\eta)^{D(H)+\alpha} \exp\left[-\delta (L^{j_1}\eta)^{-1} \operatorname{dist}(\{z\}, \partial\square')\right]. \end{aligned} \tag{3.20}$$

Since $D(H) \geqq -1$, we can use any α with $0 < \alpha < 1$. This extra degree of convergence has been borrowed from a line external to H; as usual, we have transferred the divergence to a bigger graph. This is exactly how divergent diagrams are renormalized in Sect. 3.5 of [I], so we see that H produces no divergence. The same reasoning holds when one vertex of H is restricted to the region $\eta\mathbb{Z}^d \setminus T_\eta$. Furthermore, we may extract a convergence factor $L^{-\gamma k}$ and so prove convergence of the graph as $k \to \infty$.

The other graphs in δm^2 can be treated similarly; in fact, the cancellation of the logarithmically divergent graphs does not require D b.c. The Ward-Takahashi identities hold for our choice of boundary conditions, since this choice is gauge-invariant. We will prove in Appendix A that the operators $\Delta^{(k)}$ and $\Delta^{(k)}(0)$ converge as $k \to \infty$.

With these modifications the arguments of Sects. 3.5 and 3.6 of [I] apply in C_J. Two points are settled in Sect. 6.1: the bounds of [I] on the graphs with one vertex at a point of the block lattice, and (3.98) of [I] for the mass counterterm units of order four, are carried over to the averaging (3.2) by comparing the model with the averaging function f to the model of Theorem 3.4 of [I] through the image representation (3.9). The terms of $S^{(k),1}$ which depend neither on the fields nor on the sources are common to the numerator and the denominator of the ratio and need not be bounded. Theorem 2.1 for $h = 0$ then follows from these bounds and those of Sect. 6.1.

Notice that when $h \neq 0$ the sources also enter the part of the effective action which is independent of the fields. This part does not converge as it stands when $d = 3$: the graphs of order e^2 which are linear in h and contain no field have degree zero and are of order $\ln \varepsilon^{-1}$, and this is why (2.13) carries the constant a_ε, the term of order e^2 in $\langle |\phi(0)|^2 \rangle$ on $\varepsilon\mathbb{Z}^d$, which removes their divergence at order e^2. The convergence of the generating functional with this constant, for w in the set of Theorem 2.1, is established in Sect. 6.6, the bounds of Sect. 6.1 being carried out there for the graphs which contain the source h and at least one field and for the field-free graphs linear in h of order higher than e^2 and λ, while the two field-free families of degree zero, of orders e^2 and λ, are summed explicitly, the first with the constant a_ε.

3.2. Correlation Inequalities and Monotonicity in the Volume

By using an appropriate collection of correlation inequalities, we can prove that $\langle \mathcal{O}(g,h) \rangle^{C_J}_{N,D}$ is monotonic in J, for suitable g, h. The reader is referred to [BFS 1] for a derivation of the inequalities (3.24)–(3.26), which are quoted below.

For two operators $\mathcal{O}$ and $\mathcal{O}'$, and $0 \leqq s \leqq 2$, define the ratio of expectations

$$\mathscr{E}^J(\mathcal{O}; \mathcal{O}'; s) = \frac{\langle \exp[\mathcal{O} + (s-1)\mathcal{O}'] \rangle^{C_J}_{N,D}}{\langle \exp[(s-1)\mathcal{O}'] \rangle^{C_J}_{N,D}}. \tag{3.21}$$

We are concerned with operators of the following form:

$$A(g) = \sum_{b \in C_J^*} \varepsilon^d A(b) g(b)\,, \tag{3.22}$$

$$r^2(h) = \sum_{x \in C_J} \varepsilon^d r^2(x) h(x)\,. \tag{3.23}$$

The inequalities (3.24)–(3.26) of the following proposition are derived in [BFS 1] by the method of duplicate variables. We shall need them, and the inequalities (3.27)–(3.29), for the action (2.5) modified as follows. The coefficient of each term $\frac{1}{2}\varepsilon^{d-2}|A_\mu(b_+)-A_\mu(b_-)|^2$ is replaced by a number $\kappa_b \geqq 0$, and the coefficient ε^{d-2} of each term $r(x)r(y)\cos(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y))$ by a number $J_{xy} \geqq 0$, the diagonal parts $\frac{1}{2}\varepsilon^{d-2}(r(x)^2 + r(y)^2)$ being kept; the quadratic form of the vector field may be any positive definite form, which then replaces $\frac{1}{2}\langle A, (-\Delta^\varepsilon + \mu_0^2)A\rangle$ in the hypothesis on u in (3.24) and (3.27); a mass term $\sum_x \varepsilon^d m_x^2 r(x)^2$ with $m_x^2 \geqq 0$ may be added, and $r \equiv 0$ may be imposed at some of the sites. For this class all six inequalities are proved in Sect. 6.1.

In (3.26) and (3.29) the operator is then $J_{xy}r(x)r(y)\cos(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y))$. Because $\mu_0 > 0$ and $\lambda > 0$ all expectations are finite. We shall use this below and in Sect. 6.

The hypothesis on u in the following proposition holds when $|A(u)|^2$ is one of the terms $\frac{1}{2}\varepsilon^{d-2}|A_\mu(b_+) - A_\mu(b_-)|^2$ of the action, since then the difference is at least $\frac{1}{2}\mu_0^2 \sum_x \varepsilon^d |A(x)|^2$, and for every u after multiplication by a small constant; these are the cases used below.

Proposition 3.1. *For real functions g, h with $h \geqq 0$, for any complex function u on the bonds such that $\frac{1}{2}\langle A, (-\Delta^\varepsilon + \mu_0^2)A\rangle - |A(u)|^2$ is a positive definite quadratic form, and for $\varepsilon > 0$, $0 \leqq s \leqq 2$,*

$$\frac{d}{ds}\mathscr{E}^J(r^2(h) + iA(g);\ |A(u)|^2;\ s) \leqq 0\,, \tag{3.24}$$

$$\frac{d}{ds}\mathscr{E}^J(r^2(h) + iA(g);\ r^2(x);\ s) \geqq 0\,, \tag{3.25}$$

$$\frac{d}{ds}\mathscr{E}^J(r^2(h) + iA(g);\ \varepsilon^{d-2}r(x)r(y)\cos(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y));\ s) \geqq 0\,, \tag{3.26}$$

$$\frac{d}{ds}\mathscr{E}^J(-r^2(h) + A(g);\ |A(u)|^2;\ s) \geqq 0\,, \tag{3.27}$$

$$\frac{d}{ds}\mathscr{E}^J(-r^2(h) + A(g);\ r^2(x);\ s) \leqq 0\,, \tag{3.28}$$

$$\frac{d}{ds}\mathscr{E}^J(-r^2(h) + A(g);\ \varepsilon^{d-2}r(x)r(y)\cos(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y));\ s) \leqq 0\,. \tag{3.29}$$

In (3.26) and (3.29) the coefficient of $r(x)r(y)\cos(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y))$ in the exponent of the integrand is $s\varepsilon^{d-2} \geqq 0$ for $0 \leqq s \leqq 2$; at $s = 0$ the coupling of the bond is removed. Using Proposition 3.1, we can deduce the following theorem.

Theorem 3.2. *For g real and $h \geqq 0$, both with compact support in $\mathbb{R}^d$, $\langle \mathcal{O}(ig, h)\rangle_{N,D}^{C_J}$ and $\langle \mathcal{O}(g, -h)\rangle_{N,D}^{C_J}$ are, respectively, monotonic increasing and decreasing in J, for J so large that* $\operatorname{supp} g$ *and* $\operatorname{supp} h$ *lie in C_J at distance greater than* 1 *from ∂C_J.*

Proof of Theorem 3.2. We establish monotonicity for a non-zero lattice spacing ε; continuity gives the same result for the continuum limit. For given sources g and h, we take J large enough so that $\operatorname{supp} g$ and $\operatorname{supp} h$ are bounded away from ∂C_J. Recall that

$$:\phi^2:(h) = r^2(h) - \sum_{x \in C_J} \varepsilon^d [2C^\varepsilon(x,x) + a_\varepsilon] h(x)\,, \tag{3.30}$$

where $C^\varepsilon = (-\Delta^\varepsilon + m^2)^{-1}$ is computed with free boundary conditions, as in (2.13), and a_ε depends on ε only. When x is a distance $R > 1$ from ∂C_J, the difference between $C^\varepsilon(x,x)$ computed with free and D b.c. is less than $C\exp[-m'R]$, uniformly in $\varepsilon \leqq 1$, for some $m' > 0$ and C depending on m and d only. So taking $\mathrm{dist}(\mathrm{supp}\, h, \partial C_J) > 1$, the subtraction with free boundary conditions differs from the subtraction with D b.c. by a term which vanishes as $J \to \infty$, and the results of Sect. 3.1 hold with either. The advantage of the free subtraction is that $:\phi^2:(h)$ is then independent of J.

Consider the expression

$$\mathscr{E}^{J+1}(r^2(h) + iF(g);\ \tfrac{1}{2}\varepsilon^{d-2}|A_\mu(b_+) - A_\mu(b_-)|^2;\ s)\,, \tag{3.31}$$

where b is a bond in C^*_{J+1} (here $F(g) = A(\tilde{g})$ with $\tilde{g}$ real, so that Proposition 3.1 applies). By (3.24), the value of (3.31) at $s = 2$ is not greater than its value at $s = 1$. Examining the definition (3.21), this means that removing the term $\frac{1}{2}\varepsilon^{d-2}|A_\mu(b_+) - A_\mu(b_-)|^2$ from the action does not increase $\langle \mathcal{O}(ig,h)\rangle^{C_{J+1}}_{N,D}$. Similarly, (3.26), compared at $s = 0$ and $s = 1$, shows that removing a term $-\varepsilon^{d-2} r(x) r(y) \cos(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y))$ from the action also does not increase this expectation. If we remove all such terms coupling sites in C_J to sites in $C_{J+1}\backslash C_J$, then the resulting expectation factorizes, and the integral over fields in $C_{J+1}\backslash C_J$ cancels.

Recall from Sect. 2 that C_J contains its top layers, on which $r \equiv 0$, that a bond $\langle x,y\rangle$ with $r(y) = 0$ contributes $\frac{1}{2}\varepsilon^{d-2} r(x)^2$ to (2.5), and that the form of the vector field is the sum over the bonds contained in C_J, A being defined on all sites of C_J. Since the scalar field is set to zero on the top layers of C_J, we remove in addition, by (3.26), the cosine terms of all bonds with an endpoint in these layers (h vanishes there, since $\mathrm{dist}(\mathrm{supp}\, h, \partial C_J) > 1$); the scalar fields there then decouple and their integrals cancel, while the vector field on these sites keeps the terms of the action for C_J.

The remaining expression is exactly $\langle \mathcal{O}(ig,h)\rangle^{C_J}_{N,D}$: the constants E_0, E_1 cancel in the ratio (2.10), δm^2 was defined using free boundary conditions and so is the same number in both actions, and the subtraction in $:\phi^2:(h)$, with its constant a_ε, depends on ε only (Sect. 2). Therefore, we deduce the inequality

$$\langle \mathcal{O}(ig,h)\rangle^{C_J}_{N,D} \leqq \langle \mathcal{O}(ig,h)\rangle^{C_{J+1}}_{N,D}\,. \tag{3.32}$$

A similar proof shows that

$$\langle \mathcal{O}(g,-h)\rangle^{C_J}_{N,D} \geqq \langle \mathcal{O}(g,-h)\rangle^{C_{J+1}}_{N,D}\,. \tag{3.33}$$

3.3. Reflection Positivity

Reflection positivity is a crucial ingredient in the reconstruction theorem of Osterwalder and Schrader [OS 1]. It was proved in [BFS 1] for the lattice models considered there, whose vector field action is translation invariant; we prove it below for the measure (3.38). Given a rectangular parallelepiped $\Omega \subset \varepsilon\mathbb{Z}^d$, consider a hyperplane π which is perpendicular to the νth coordinate direction, and which lies midway between neighbouring sites of Ω. Define $\Omega_\pm$ to be the sites in Ω lying on opposite sides of π, and let ϱ denote reflection in π. Then we shall assume that

$$\varrho(\Omega_+) = \Omega_-\,. \tag{3.34}$$

We now introduce a map θ taking functionals with support in Ω_+ into functionals supported in Ω_- by

$$\theta(\mathscr{F})(A(b), \phi(x)) = \overline{\mathscr{F}}(A(\varrho b), \phi(\varrho x))\,. \tag{3.35}$$

The bar denotes complex conjugation and $\varrho b = \varrho\langle b_-, b_+\rangle = \langle \varrho b_-, \varrho b_+\rangle$. We will choose boundary conditions for the fields on Ω which are invariant under reflection in π. It is convenient to consider the vector field A as defined on oriented bonds, and to assume that

$$A(-b) = A(\langle b_+, b_-\rangle) = -A(b)\,. \tag{3.36}$$

We define a characteristic function on Ω by

$$\Omega(x) = \begin{cases} 1 & \text{if} \quad x \in \Omega\,, \\ 0 & \text{otherwise}\,. \end{cases} \tag{3.37}$$

The action that we consider on Ω is $\Big(\text{recall that } F(p) = \varepsilon^{-1} \sum_{b \in \partial p} A(b)\Big)$,

$$\begin{aligned} S^{\varepsilon}_{\Omega}(A,\phi) = {} & \tfrac{1}{2} \sum_{p \in \Omega^{**}} \varepsilon^d |F(p)|^2 + \tfrac{1}{2}\mu_0^2 \sum_{b \in \Omega^*} \varepsilon^d |A(b)|^2 \\ & + \tfrac{1}{2} \sum_{x \in \Omega} \varepsilon^d \Big| \sum_{y \in x^*} \varepsilon^{-1} \Omega(y) A(\langle x,y\rangle)\Big|^2 \\ & + \tfrac{1}{2} \sum_{\langle x,y\rangle} \varepsilon^{d-2} [(\Omega r)(x)^2 + (\Omega r)(y)^2 - 2(\Omega r)(x)(\Omega r)(y) \\ & \quad \cdot \cos(\theta(x) - e\varepsilon A(\langle x,y\rangle) - \theta(y))] \\ & + \sum_{x \in \Omega} \varepsilon^d [\tfrac{1}{2} m^2 r(x)^2 + \lambda r(x)^4 + \tfrac{1}{2}\delta m^2\, r(x)^2]\,. \end{aligned} \tag{3.38}$$

We have written x^* for the set of nearest neighbour sites of x on $\varepsilon\mathbb{Z}^d$. The sum over $\langle x,y\rangle$ runs over all bonds of $\varepsilon\mathbb{Z}^d$; a bond leaving Ω contributes $\frac{1}{2}\varepsilon^{d-2}(\Omega r)(x)^2$. We write $\langle \mathscr{F} \rangle^{\varepsilon}_{\Omega}$ for the expectation of a functional $\mathscr{F}$ in the measure defined by (3.38). The quadratic form for the vector field in the action (3.38) differs from the "unitary" form $\frac{1}{2}\langle A, (d^*d + \mu_0^2)A\rangle$ by a gauge-fixing term only (d is the lattice exterior derivative). Since the gauge-fixing term in (3.38) depends on the box, we do not appeal to [BFS 1] for reflection positivity but prove the following proposition directly.

Proposition 3.3. *For any gauge-invariant functional $\mathscr{F}$ supported in Ω_+, that is depending only on $A(b)$ for bonds b with both endpoints in Ω_+ and on $\phi(x)$, $x \in \Omega_+$, which is bounded or a finite linear combination of functionals $\mathcal{O}(zg, wh)$,*

$$\langle \theta(\mathscr{F})\mathscr{F} \rangle^{\varepsilon}_{\Omega} \geqq 0\,. \tag{3.39}$$

Proof. We first show that for gauge-invariant $\mathscr{F}$ the expectation in the measure (3.38) equals the expectation in the measure with the unitary form. Write

$$A = A^T + d\chi\,, \qquad d^*A^T = 0\,, \quad \sum_x \chi(x) = 0\,, \tag{3.40}$$

a linear change of variables with constant Jacobian. The form in (3.38) and the unitary form are

$$Q(A^T) + q(\chi) \qquad \text{and} \qquad Q(A^T) + \tfrac{1}{2}\mu_0^2 \sum_b \varepsilon^d |(d\chi)(b)|^2\,, \tag{3.41}$$

where Q is the unitary form restricted to A^T and q is a positive definite form in χ, since $\mu_0 > 0$. The gauge transformation $\theta(x) \to \theta(x) + e\chi(x)$ removes $d\chi$ from the hopping terms of (3.38) and leaves $\mathscr{F}$ and the measure $\prod_x d\phi(x)$ unchanged, so that the integral over ϕ of $\mathscr{F}e^{-S}$ depends on A^T only; the integrals over χ are then finite constants which cancel in the normalized expectations. This is the argument of [BFS 1].

It remains to prove (3.39) for the unitary form. We divide the variables into those of the sites and bonds of Ω_+, those of Ω_-, and the variables $A(b)$ of the bonds $b = \langle x^-, x^+\rangle$ crossing π, where $x^+ = \varrho x^- \in \Omega_+$, the crossing bonds being oriented from Ω_- to Ω_+, so that $A(b) = \alpha_b - \alpha'_b$ below. A crossing bond appears by itself only in its mass term $\frac{1}{2}\mu_0^2\varepsilon^d A(b)^2$, and we write

$$\int g(A(b))\, e^{-\frac{1}{2}\mu_0^2\varepsilon^d A(b)^2}\, dA(b) = C\, E\, g(\alpha_b - \alpha'_b)\,, \tag{3.42}$$

where α_b, α'_b are independent Gaussian variables of variance $(2\mu_0^2\varepsilon^d)^{-1}$ and E denotes their expectation.

A plaquette p crossing π contains two crossing bonds b, b' and two bonds $c \subset \Omega_+$, $\varrho c \subset \Omega_-$; writing $X_p = \varepsilon^{-1}[A(c) - \alpha_{b'} + \alpha_b]$, and X'_p for the same expression in the variables of Ω_- and α', we have $F(p) = \pm(X'_p - X_p)$, so that

$$\tfrac{1}{2}\varepsilon^d F(p)^2 = \tfrac{1}{2}\varepsilon^d X_p^2 + \tfrac{1}{2}\varepsilon^d X_p'^2 - \varepsilon^d X_p X'_p \,. \tag{3.43}$$

A crossing hopping term is

$$-\varepsilon^{d-2} r(x^-) r(x^+) \cos(\theta(x^-) - e\varepsilon A(b) - \theta(x^+)) = -\varepsilon^{d-2} \sum_{i=1,2} \psi_b^i \psi_b'^i \,, \tag{3.44}$$

where $\psi_b = \phi(x^+) e^{ie\varepsilon\alpha_b}$ and $\psi'_b = \phi(x^-) e^{ie\varepsilon\alpha'_b}$, with real and imaginary parts ψ_b^i, $\psi_b'^i$. Let ξ denote the variables of Ω_+ together with α, and ξ' the variables of Ω_- transported to Ω_+ by ϱ, together with α'; by the symmetry of Ω the terms of the action involving one side only are the same function S_+ of ξ and of ξ', and the a priori measures coincide. Hence, if S' denotes the action without the mass terms of the crossing bonds,

$$\begin{aligned} e^{-S'} &= \mathscr{G}(\xi)\, \mathscr{G}(\xi') \exp\Big[\sum_\iota \gamma_\iota Y_\iota(\xi) Y_\iota(\xi') \Big] \,, \\ \mathscr{G}(\xi) &= \exp\Big[- S_+(\xi) - \tfrac{1}{2}\varepsilon^d \sum_p X_p(\xi)^2 \Big] \,, \end{aligned} \tag{3.45}$$

where $\{(\gamma_\iota, Y_\iota)\}$ runs over (ε^d, X_p) for the crossing plaquettes and $(\varepsilon^{d-2}, \psi_b^i)$ for the crossing bonds: all $\gamma_\iota > 0$ and all Y_ι are real. For $\mathscr{F}$ supported in Ω_+ we have $\mathscr{F} = \mathscr{F}(\xi)$ and $\theta(\mathscr{F}) = \overline{\mathscr{F}(\xi')}$, and we expand the exponential:

$$Z\, \langle \theta(\mathscr{F}) \mathscr{F} \rangle_\Omega^\varepsilon = C \sum_n \prod_\iota \frac{\gamma_\iota^{n_\iota}}{n_\iota!} \Big| \int d\xi\, \mathscr{F}(\xi) \mathscr{G}(\xi) \prod_\iota Y_\iota(\xi)^{n_\iota} \Big|^2 \geqq 0 \,, \tag{3.46}$$

the two integrals of each term being complex conjugates because the measures and the functions coincide on the two copies. The interchange of sum and integral is justified by

$$\sum_n \prod_\iota \frac{\gamma_\iota^{n_\iota} |Y_\iota Y'_\iota|^{n_\iota}}{n_\iota!} \leqq \exp\Big[\tfrac{1}{2} \sum_\iota \gamma_\iota (Y_\iota^2 + Y_\iota'^2) \Big] \,, \tag{3.47}$$

the plaquette parts of which cancel against $\mathscr{G}$ exactly, while the remaining factor

$$\exp\Big[\tfrac{1}{2}\varepsilon^{d-2} \sum_b r(x^+)^2 \Big] \tag{3.48}$$

is integrable against the quartic term of S_+. This is the positive cross-term criterion of [GJ 1], [FILS 1], and completes the proof of Proposition 3.3.

We shall use the correlation inequalities of the last section to compare the measure (3.38) with the measure (2.5) on cubes of different sizes below. The inequalities (3.24)–(3.29) hold for the action (3.38), and for the actions obtained from it and from (2.5) by removing or adding terms of the kinds considered below, since these actions are all in the class described before Proposition 3.1.

We return now to the construction of the infinite volume limit. For any sources g, h and for $z, w \in \mathbb{C}$,

$$|\langle \mathcal{O}(zg, wh) \rangle_{N,D}^{\varepsilon, C_J}| \leqq \langle \mathcal{O}(\operatorname{Re} zg, \operatorname{Re} wh) \rangle_{N,D}^{\varepsilon, C_J} \,. \tag{3.49}$$

We can write $\operatorname{Re} wh = h_+ - h_-$ with $h_+, h_- \geqq 0$ smooth of compact support: for w in a ball B of radius r_0 about a point w_0 with $2\sup(\operatorname{Re} w_0 h)_+ < \sigma_+$ we take one h_- for all

$w \in B$, namely $h_- = \rho_\delta * \sup_{w' \in B}(\mathrm{Re}\, w'h)^- + \delta\eta$, with ρ_δ a mollifier at a scale not exceeding $\delta/[(1+|w_0|+r_0)(1+\|\nabla h\|_\infty)]$ and $\eta \in C_0^\infty$ equal to 1 on $\mathrm{supp}\, h$, and $h_+ = \mathrm{Re}\, wh + h_-$; then $h_+ \geqq 0$ and $\sup h_+ \leqq \sup(\mathrm{Re}\, w_0 h)_+ + 2r_0 \sup|h| + 2\delta$ for $w \in B$, so that $2\sup h_+ < \sigma_+$ on B when r_0 and δ are small (for $wh \geqq 0$ we take $h_- = 0$). Then using Schwarz's inequality and Theorem 3.2,

$$\begin{aligned} (3.49) &\leqq [\langle \mathcal{O}(2\,\mathrm{Re}\, zg, -2h_-)\rangle_{N,D}^{\varepsilon,C_J}]^{1/2} \cdot [\langle \mathcal{O}(0,2h_+)\rangle_{N,D}^{\varepsilon,C_J}]^{1/2} \\ &\leqq [\langle \mathcal{O}(2\,\mathrm{Re}\, zg, -2h_-)\rangle_{N,D}^{\varepsilon,C_{J_0}}]^{1/2} \cdot [\langle \mathcal{O}(0,2h_+)\rangle_{N,D}^{\varepsilon,C_J}]^{1/2} \qquad (3.50) \\ &\leqq e^{CJ_0^d} [\langle \mathcal{O}(0,2h_+)\rangle_{N,D}^{\varepsilon,C_J}]^{1/2}, \end{aligned}$$

where $J_0 \leqq J$ is such that a neighbourhood of $\mathrm{supp}\, g \cup \mathrm{supp}\, h$ containing $\mathrm{supp}\, h_-$ lies in C_{J_0-1}, and the factor "C" in (3.50) depends on g, h, z, w (this is the ultra-violet stability bound); the first factor in (3.50) has the fixed source $-2h_-$ on B and is bounded on B uniformly in ε, by Theorem 2.1 and the convexity of its logarithm in $\mathrm{Re}\, z$, so that the bounds (3.50), (3.53) are locally uniform in (z,w) and in $J \geqq J_0$, as Vitali's theorem requires. We now fix ν and a plane $\pi = \{x_\nu = c\}$, where c is a half-odd multiple of some ε_{K_0}; since $L = 5$ is odd, π lies midway between neighbouring sites of $\varepsilon_K \mathbb{Z}^d$ for every $K \geqq K_0$. For each J, we consider $\langle \mathcal{O}(0,2h_+)\rangle_{\Omega_J}^{\varepsilon}$ defined with the action $S_{\Omega_J}^{\varepsilon}(A,\phi)$ given by (3.38), with the free subtraction of (2.13), the Wick constant and δm^2 being those of the infinite lattice and the same for every region, on the box

$$\Omega_J = C_{J+n} + (c + \varepsilon/2)e_\nu\,, \qquad Mn \geqq |c| + R\,, \tag{3.51}$$

where C_{J+n} is the set of sites $\{-M(J+n) \leqq x_\mu < M(J+n)\}$ of (2.7), with an even number of layers in each direction, so that Ω_J satisfies (3.34) with respect to π; here n is the smallest integer satisfying (3.51), where $R = 2 + \pi\sqrt{d}/(2\mu_0)$ is the constant (6.50) fixed in Sect. 6.4, chosen so that C_J with one layer of sites lies in Ω_J and C_{J+2n} contains every site within distance $\pi\sqrt{d}/(2\mu_0)$ of Ω_J, for every $\varepsilon \leqq 1$. Then we have

$$\langle \mathcal{O}(0,2h_+)\rangle_{N,D}^{\varepsilon,C_J} \leqq \langle \mathcal{O}(0,2h_+)\rangle_{\Omega_J}^{\varepsilon} \leqq \langle \mathcal{O}(0,2h_+)\rangle_{N,D}^{\varepsilon,C_{J+2n}}\,. \tag{3.52}$$

The same two inequalities hold for every $\mathcal{O}(ig,h)$ with $h \geqq 0$, with the same proof, and the reversed inequalities for $\mathcal{O}(g,-h)$; they are applied below to the pairs $\theta(\mathscr{F}_i)\mathscr{F}_j$, whose scalar sources satisfy $2\sup_x(\mathrm{Re}(\bar{w}_i \theta h_i + w_j h_j))_+ < \sigma_+$, which is the amplitude restriction of Theorem 2.3(ii). The first inequality follows as in the proof of Theorem 3.2. The quadratic form for the vector field in (3.38) is the form of (2.5) on the bonds internal to Ω_J together with a positive form in the variables of the bonds next to $\partial\Omega_J$, a sum of squares of linear forms in A to each of which (3.24) applies; removing this form, the terms coupling C_J to $\Omega_J \setminus C_J$ and, for the cubes of Sect. 2, the cosine terms of the bonds with an endpoint in the top layers of C_J, on which the scalar field vanishes in (2.5), does not increase the expectation, by (3.24), (3.26); the integrals over the fields at the sites of $\Omega_J \setminus C_J$ and of the top layers then factorize and cancel.

The second inequality is not available with C_{J+1} in place of C_{J+2n}: on one and the same box, once the bonds leaving it are integrated out, the quadratic form of (3.38) exceeds the form of (2.5) by a positive form on the bonds next to the boundary, so that (3.24) gives there the reverse inequality for every $\mathcal{O}(ig,h)$ with $h \geqq 0$, and at $e = 0$ the inequality fails strictly for the functionals $e^{itF(g)}$ (Lemma 6.13). A margin of order μ_0^{-1} is needed.

On the larger cube we proceed as follows. We remove from the action of C_{J+2n} the hopping terms of all bonds not internal to Ω_J, which does not increase the expectation by (3.26); the scalar field outside Ω_J then factorizes and cancels, and the matter of Ω_J couples to the vector field only through gauge-invariant functions of A on Ω_J, which on a box are functions of the field strength F on the plaquettes of Ω_J (on a box every A with $dA = 0$ is a gradient $d\chi$).

We prove in Sect. 6.4 that, when $Mn \geqq |c| + R$, the covariance of F on these plaquettes in the Gaussian measure of C_{J+2n} is dominated by its covariance in the measure (3.38), as quadratic

forms on the exact 2-forms of Ω_J, on which both are nondegenerate; the two quadratic forms on the bonds of Ω_J are not comparable, only the laws of F are. Adding to the form of (3.38) the positive form in F which gives F the covariance of C_{J+2n} does not decrease the expectation, by (3.24), and since the matter of Ω_J and $\mathcal{O}(ig, h)$ depend on A only through F, the two expectations then coincide, which gives the second inequality.

For the second factor in (3.50) we need a bound uniform in J. We prove in Sect. 6.6, from the comparison of the cube with a torus (Sect. 6.1), the reflection positivity of the periodic model, the chessboard estimate and ultra-violet stability, that

$$\langle \mathcal{O}(0, 2h_+)\rangle_{N,D}^{\varepsilon, C_J} \leqq \exp\left[\kappa \sum_x \varepsilon^d \, 2h_+(x)\right] \qquad \text{for} \quad 2 \sup h_+ \leqq \sigma_+ \,, \tag{3.53}$$

for all $J \geqq J_0$ and all ε, where $r^2(h)$ in $\mathcal{O}$ is Wick ordered as in (2.13), with the free subtraction and the constant a_ε of Sect. 2, and κ, σ_+ depend on the coupling constants, on M, on the choice of the constants a_ε and on the auxiliary choices made in Sect. 6.6, uniformly in ε, and not on g, h, J. The bound (3.53) is established in Sect. 6.6 under the condition on the coupling constants of Sect. 6.8, which is assumed throughout for the results with $h \neq 0$, as stated in Sect. 2. Combining monotonicity (Theorem 3.2) and (3.50), (3.53), we deduce

$$\exists \lim_{J \to \infty} \langle \mathcal{O}(zg, wh)\rangle_{N,D}^{C_J} = \langle \mathcal{O}(zg, wh)\rangle \tag{3.54}$$

for $\operatorname{Re} z = 0$, w real and $wh \geqq 0$ with $2 \sup wh \leqq \sigma_+$. When h changes sign we write $h = h^+ - h^-$ with $h^\pm \geqq 0$ smooth of compact support, and apply the argument to $\mathcal{O}(zg, w_1 h^+ + w_2 h^-)$, $w_1, w_2 \geqq 0$. The scalar source is then non-negative, so that the monotone limits exist for $\operatorname{Re} z = 0$, $w_1, w_2 > 0$, $2\sup(w_1 h^+ + w_2 h^-) < \sigma_+$, an open subset of $i\mathbb{R} \times \mathbb{R}^2$ and hence a set of uniqueness in $\mathbb{C}^3$; by (3.50), (3.53) the family is bounded on $\{2\sup(\operatorname{Re} w_1 h^+ + \operatorname{Re} w_2 h^-)_+ < \sigma_+\}$, Vitali's theorem gives the limit there, and we set $w_1 = w$, $w_2 = -w$. (Notice that we took the continuum limit before taking the infinite volume limit; all our bounds and monotonicity results are uniform in ε.)

Vitali's theorem then extends (3.54) to a function entire in z and holomorphic in w on the open set where $2\sup_x(\operatorname{Re} w\, h(x))_+ < \sigma_+$, which contains the tube $\mathcal{D}$: the bounds (3.50)–(3.53) are uniform in J there, and the set on which the monotone limits exist is a set of uniqueness.

Furthermore, for each J, $\langle \mathcal{O}(zg, wh)\rangle_{\Omega_J}^{\varepsilon_K}$ converges on some subsequence as $K \to \infty$, by Montel's theorem, the subsequence being chosen for the finitely many pairs $\theta(\mathscr{F}_i)\mathscr{F}_j$ of a given family, each pair being a functional $\mathcal{O}(\sum_k z_k g_k, \sum_k w_k h_k)$ with two complex coefficients to which the argument of (3.54) applies: each h_k is written as $h_k^+ - h_k^-$ and the scalar source as $\sum_k (w_k^+ h_k^+ + w_k^- h_k^-)$, the cube expectations are monotone where $\operatorname{Re} z_k = 0$ and $w_k^\pm \geqq 0$ and bounded where $2\sup_x(\sum_k \operatorname{Re} w_k^+ h_k^+ + \operatorname{Re} w_k^- h_k^-)_+ < \sigma_+$, Vitali's theorem in the several variables gives the limit there, and we set $w_k^+ = w_k$, $w_k^- = -w_k$; the bounds (3.50)–(3.53) hold for the box as well, the comparisons (3.52) holding for both factors in (3.50) in the directions of Theorem 3.2.

This limit is reflection positive, positivity being preserved under limits; also (3.52) implies that as $J \to \infty$ it converges to (3.54), for $\operatorname{Re} z = 0$, w real and $wh \geqq 0$ by the two-sided comparison, and then, by Vitali's theorem, on the whole set $\{2\sup(\operatorname{Re} wh)_+ < \sigma_+\}$, since the bounds (3.50)–(3.53) hold for the box uniformly in J, for sign-changing h through the family (h^+, h^-) as above. Therefore, (3.54) is reflection positive for the reflection in π. The planes π of this kind are dense, and the reflections in the remaining planes $\{x_\nu = c\}$ follow from the translation invariance proved below.

Translation invariance does not follow from the uniqueness of the limit (3.54) alone, since a translation moves the cubes. Let T be the translation by $b \in \varepsilon_{K_0}\mathbb{Z}^d$ with $|b|_\infty \leqq M$, and let $K \geqq K_0$. The lattice expectations are covariant under T, and $C_{J-1} \subset T^{-1}C_J \subset C_{J+1}$, so that the proof of Theorem 3.2, with the top layers of $T^{-1}C_J$ and of $C_{J\pm1}$ decoupled first by

removing the cosine terms at their sites, which does not increase $\langle \mathcal{O}(ig, h)\rangle$ by (3.26), gives, for $J \geqq J_0 + 1$, $\operatorname{Re} z = 0$, w real and $wh \geqq 0$,

$$\langle \mathcal{O}(zg, wh)\rangle_{N,D}^{\varepsilon_K, C_{J-1}} \leqq \langle \mathcal{O}(zTg, wTh)\rangle_{N,D}^{\varepsilon_K, C_J} \leqq \langle \mathcal{O}(zg, wh)\rangle_{N,D}^{\varepsilon_K, C_{J+1}}\,, \tag{3.55}$$

and the reversed inequalities in (3.55) for z real, w real and $wh \leqq 0$. Letting $K \to \infty$ and then $J \to \infty$, the outer terms of (3.55) converge to (3.54), so (2.19) holds at these points for every such b, by composition for every b in the dense set $\bigcup_K \varepsilon_K \mathbb{Z}^d$, and for all z and all w in the set of Theorem 2.1 by the identity theorem, for sign-changing h through the family $\mathcal{O}(zg, w_1 h^+ + w_2 h^-)$ as above. For arbitrary b the invariance follows from the continuity of (3.54) in the sources, which is proved in Sect. 6.3 for $h = 0$ and in Sect. 6.6 on the tube. Vitali's theorem allows us to deduce (2.21) on the tube.

4. Rotation Invariance

As explained in Sect. 2, in this section we will consider the model on a torus D_J defined by

$$D_J = \{x \in \mathbb{R}^d :\ -25MJ \leqq x_\mu < 25MJ,\ \mu = 1, \dots, d\}\,. \tag{4.1}$$

We denote by $Z^\varepsilon(D_J, g, h)$ the generating functional on the lattice T_ε with spacing ε and the "usual" orientation. We will denote by $Z^{R(\varepsilon)}(D_J, g, h)$ the generating functional on the lattice $T_{R(\varepsilon)}$ with the same spacing ε, but rotated by the angle θ_0 as shown in Fig. 1; each lattice samples the 2-form g at the centers of its own plaquettes, the mass term ω of Theorem 2.4 is included by taking $h = -\omega$, and the counterterm δm^2, defined with free boundary conditions, is the same number on both lattices.

The class of mass terms and tori for which the comparison is proved is the following: J is a large multiple of a fixed power of 5; $\omega \geqq 0$ is a smooth function on the torus which vanishes on a neighbourhood of the support of g and satisfies

$$\sum_x \varepsilon^d \omega(x) \geqq c_0 |D_J| \tag{4.2}$$

for a fixed $c_0 > 0$ depending on the coupling constants; and the support of g and the torus are unions of blocks of the final scale of the renormalization transformation (D_J is a union of blocks of $R(5^j \varepsilon \mathbb{Z}^d)$ for every $j \leqq K$, since $5R$ is an integer matrix and the sides of D_J are multiples of 25).

For $\omega = 0$ and $g = 0$ the results of [Ba 1–4] apply to both these models (for the rotated lattice see below), and we have

$$e^{-C_1 |D_J|} \leqq Z^\varepsilon(D_J, 0, 0),\ Z^{R(\varepsilon)}(D_J, 0, 0) \leqq e^{C_2 |D_J|}\,, \tag{4.3}$$

where C_1, C_2 do not depend on ε or J; with sources the constants depend also on $|D_J|$.[1]

For the rotated lattice the torus D_J is not a product of intervals along the lattice axes. The arguments of [Ba 1–4] go through for it, the torus entering only through the embedding of the block lattices and the images of the propagators, and the bounds which run the renormalization transformation to its final scale hold under a lower bound on J at that scale which depends on the mass term and is met for the mass terms used in Sect. 6.7. The transformation on these tori is carried out, for both lattices and for the modified transformation introduced below, in [M]; the estimates of that analysis which we use are stated below where they are used. At a non-zero mass term the lower bound in (4.3) is not what the argument below requires; in its place we use the lower bound (4.9), which is established in Sect. 6.7.

[1] As in (2.24) of [I].

In order to compare these models we consider a modified block spin transformation. Each lattice site y in $T_{R(5\varepsilon)}$ defines a block of 5^d sites on T_ε, the sites of T_ε in the cube of side 5ε centered at y with sides parallel to the rotated axes (the lattice $R(5\varepsilon\mathbb{Z}^d)$ has index 5^d in $\varepsilon\mathbb{Z}^d$), which we denote by $\hat{B}(y)$; these blocks are symmetric about their sites and partition T_ε. We introduce corresponding averaging operators

$$(\hat{Q}A)_\mu(y) = 5^{-d} \sum_{x\in\hat{B}(y)} f_\mu \cdot A(x)\,, \tag{4.4}$$

$$(\hat{Q}(A)\phi)(y) = 5^{-d} \sum_{x\in\hat{B}(y)} U(A(\hat{\Gamma}_{y,x}))\phi(x)\,, \tag{4.5}$$

where $f_\mu = Re_\mu$ are the unit vectors along the axes of the rotated lattice and $A(x)$ is the vector (2.2) of the components of A at x, so that $\hat{Q}A$ averages over the block the components of A along the rotated axes, a constant vector field passes through the average unchanged, $\hat{Q}\hat{Q}^* = 1$ and $\hat{Q}^*\hat{Q}$ is a projection, as for the averaging of [I]; and where the contour $\hat{\Gamma}_{y,x}$ is chosen to lie entirely inside $\hat{B}(y)$, which is connected, as a fixed function of $x - y$.

The averaging (4.4) is covariant under the gauge transformations (2.4) with linear χ, the average over a block of $f_\mu \cdot \partial\chi$ being then the lattice derivative of a block average of χ, and not under general χ; the averaging (4.5), with the transporters along the contours $\hat{\Gamma}_{y,x}$, is covariant under every gauge transformation. The comparison of the two transformations uses the gauge transformations with linear χ only, in the constant-background step of Sect. 3.4 of [I], and no gauge fixing, the vector field having the mass $\mu_0 > 0$.

Using the operators (4.4), (4.5) we can generate from $Z^\varepsilon(D_J, g, h)$ an effective action on the lattice $T^{(1)}_{R(5\varepsilon)}$, which we write $\hat{S}^{(1),R(5\varepsilon)}(T^{(1)}, A_1, \phi_1, g, h)$.

By applying succeeding steps of the renormalization transformation in the usual way, that is without rotating the blocks, we generate effective actions $\hat{S}^{(k),R(5^k\varepsilon)}(T^{(k)}, A_k, \phi_k, g, h)$ on the lattices $T^{(k)}_{R(5^k\varepsilon)}$; in these steps the contours of [I] are made of bonds of T_ε which follow the rotated axes as closely as possible, and all blocks, here and for the rotated lattice, are centered at the sites of the next lattice rather than rooted at a corner as in [Ba 1]. The estimates of [Ba 1–4] for the fluctuation integrals hold for centered blocks and for contours of bounded length inside them, and for the averaging (4.4), the constants depending on the contours only through their length; in [M] they are carried to centered blocks by a translation of the fine lattice. In Sect. 5 we will prove the following proposition.

Proposition 4.1. *For some $\sigma > 0$ and ϖ,*

$$\begin{aligned} Z^\varepsilon(D_J, g, h) \geqq & \int (dA_k)(d\phi_k)\chi_k(A_k)\chi_k(\phi_k) \\ & \cdot \exp[-\hat{S}^{(k),R(5^k\varepsilon)}(T^{(k)}, A_k, \phi_k, g, h) - C(1+K)^\varpi (L^k\varepsilon)^\sigma |D_J|] \end{aligned} \tag{4.6}$$

for $0 \leqq k \leqq K - J_1$, where J_1 is the integer called J in Theorem 3.1 of [I] *and χ_k is defined in Sect. 3 of* [I].

The proof, in Sect. 5.3, follows the lower bound of [Ba 1] step by step for the modified transformation; the operations of Sect. 3 of [Ba 1] on the covariances and the renormalization bound of Proposition 1 of [Ba 3] apply to the modified transformation as they do to the usual one, as shown in [M].

We can also perform the usual renormalization transformations on $Z^{R(\varepsilon)}(D_J, g, h)$, in the frame of the rotated lattice. This produces effective actions $S^{(k),R(5^k\varepsilon)}(T^{(k)}, A_k, \phi_k, g, h)$ on the same lattices $T^{(k)}_{R(5^k\varepsilon)}$, which we can compare with the previous actions.

Theorem 4.2. *Let $h = 0$ or $h = -\omega$ with ω as above, and let $k = K - [\vartheta K]$ with $\vartheta > 0$ sufficiently small (depending on γ, β below). Then for K sufficiently large*

$$\begin{aligned}\chi_k(A_k)\chi_k(\phi_k)\Big|\hat{S}^{(k),1}(T^{(k)}, A_k, \phi_k, g, h) - S^{(k),1}(T^{(k)}, A_k, \phi_k, g, h) + c_K \sum_x \varepsilon^d h(x)\Big| \\ \leqq C((1+K)^{\varpi} L^{-\gamma k}(L^k\varepsilon)^{-\beta} + (L^k\varepsilon)^{\sigma})|D_J|\,,\end{aligned} \tag{4.7}$$

where $0 < \gamma < 1$, $0 < \sigma, \beta$, ϖ and C depend on g, h, the terms of the two actions which do not depend on the fields are included, and c_K is a constant which depends neither on the fields, nor on g, h, nor on the torus, and which tends to zero as $K \to \infty$. The sum $\sum_x \varepsilon^d h(x)$ runs over the sites of T_ε in D_J; over those of $T_{R(\varepsilon)}$ it differs by $O(\varepsilon^N)$ for every N, which is absorbed in the right side.

The bound (4.7) for the terms of the two actions which do not depend on the fields holds, for the vacuum graphs of non-positive degree of order higher than λ^2, by the comparison of the infrared parts of their counterterms, and for the vacuum graphs of positive degree by the comparison of the values of the two transformations at the k-th step, with lines of index below k only, the divergent subgraphs being renormalized by their mass counterterm units and the counterterm E_1, one and the same number for both transformations, cancelling in the difference; and $c_K \to 0$, c_K being the difference of the averages over a block of the final scale of the two coefficients of $\varepsilon^d h(x)$ in the parts of the two actions which depend neither on the fields nor on g, computed with the propagators at $A = 0$ on the infinite lattices, while the difference between c_K and the corresponding term on the torus is bounded by the right side of (4.7) by the comparison of Appendix B; the comparison of the field-independent terms and the convergence $c_K \to 0$ are taken from [M].

Proof. The proof is almost identical to the proof of Theorem 3.4 in [I], assuming that we have corresponding results on convergence of propagators. These results will be established in Appendix B. The only significant difference in the rest of the proof is that regions are decoupled on the infinite lattice of each transformation, the periodic propagators on D_J being written as image sums of the propagators on the infinite lattice with the periodic lifts of the fields (the image terms are $O(e^{-\delta_0 s/4})$, s the shortest period in the unit scale of the k-th lattice, which grows like $5^{[\vartheta K]}$); on the infinite lattice the propagators at $A = 0$ have an explicit Fourier representation, and a constant background B is removed by the gauge transformation with $\chi(x) = B \cdot x$, under which (4.4) and (4.5) are covariant, since $\partial\chi = B$ and $B(\hat{\Gamma}_{y,x}) = B \cdot (x - y)$ for every contour inside a block.

The two actions are compared term by term: the quadratic forms and the normalization factors through the Fourier symbols of Appendix B; the interaction terms by the expansion of Sect. 3.4 of [I], made for each transformation separately, the lines of the two expansions being compared through the propagators of Appendix B and the renormalized classes of Sect. 3.5 of [I] being formed for each transformation with its own counterterms; and the constant c_K is the difference of the averages over a block of the final scale of the two coefficients of $\varepsilon^d h(x)$ in the parts of the two actions which depend neither on the fields nor on g, computed with the propagators at $A = 0$ on the infinite lattices.

The terms linear in h which depend on g are compared with the other interaction terms, their degree being positive, and the torus-dependent remainder of c_K is bounded by the right side of (4.7) by the comparison of Appendix B. The vertices of the two expansions are summed over two different fine lattices, of which $T^{(1)}_{R(5\varepsilon)}$ is a common sublattice; the lines are compared after the sums over the sites of a block of $T^{(1)}$ have been carried out. The block sums of the two propagators differ by the bounds (B.12)–(B.14), uniformly in the regular backgrounds of Sect. 3.4 of [I], and the block sums of $h(x)G^{\varepsilon}_k(x,x)$ differ from $h(y)$ times the block sum of the diagonal by a term of second order in h: the diagonal is the sum over the scales $j < k$ of functions which are symmetric about the centers of the blocks of $R(5^{j+1}\varepsilon\mathbb{Z}^d)$ and of size

$(L^j\varepsilon)^{2-d}$, so that the linear terms of the Taylor expansion of h cancel on those blocks and the remainders sum to $C\|\partial^2 h\|_\infty (L^k\varepsilon)^{4-d}|D_J|$.

We can now establish Theorem 2.4. First, using the results of [Ba 2] (as stated in Theorem 3.1 of [I]) for the rotated lattice, with k as in Theorem 4.2,

$$Z^{R(\varepsilon)}(D_J,g,h) \leqq e^{C(L^k\varepsilon)^\sigma |D_J|} \int (dA_k)(d\phi_k)\chi_k(A_k)\chi_k(\phi_k)\exp[-S^{(k),1}(T^{(k)},A_k,\phi_k,g,h)] + e^{-cp(L^k\varepsilon)^2 + C|D_J|}\varepsilon^{-q}\,, \tag{4.8}$$

where $c>0$ and q depend on g, h and $|D_J|$, and $p(L^k\varepsilon)$ is the large field parameter of Theorem 3.1 of [I], for which $p(L^k\varepsilon)^2$ grows faster than any multiple of K when $k = K - [\vartheta K]$; the second term is the contribution of the large fields, in which the mass term is kept as a source. The large field term keeps this form for every ε_K, in particular where the counterterm of this paper lies below that of the torus convention of [I], and for the tori and the stopping scales of Sect. 6.7; this is shown in [M]. We prove in Sect. 6.7 (Theorem 6.23) that, in the torus convention of [I],

$$Z^{R(\varepsilon)}(D_J,g,h),\ Z^\varepsilon(D_J,g,h) \geqq e^{-C|D_J|}\varepsilon^{q'} \tag{4.9}$$

for some q' depending on g, h and $|D_J|$, and likewise on the torus RD_J below; the functional of this paper and the large field term of (4.8) are the torus-convention ones at the mass term shifted by a constant u_ε, bounded uniformly in ε, times one and the same factor independent of the fields and of g ([M]); (4.9) is used at this shifted mass term, which is real and bounded uniformly in ε and so admitted by Theorem 6.23, so that, since $p(L^k\varepsilon)^2$ grows faster than any multiple of K, the second term in (4.8) is at most $\delta_K Z^{R(\varepsilon)}(D_J,g,h)$ with $\delta_K \to 0$. By Theorem 4.2 and Proposition 4.1,

$$\begin{aligned} \chi_k(A_k)\chi_k(\phi_k)\exp[-S^{(k),1}] &\leqq \chi_k(A_k)\chi_k(\phi_k)\exp\Big[-\hat S^{(k),1} - c_K\sum_x \varepsilon^d h(x) + \theta_K|D_J|\Big]\,, \\ \int (dA_k)(d\phi_k)\chi_k(A_k)\chi_k(\phi_k)\exp[-\hat S^{(k),1}] &\leqq e^{C(1+K)^\varpi (L^k\varepsilon)^\sigma |D_J|} Z^\varepsilon(D_J,g,h)\,, \end{aligned} \tag{4.10}$$

where θ_K is the right-hand side of (4.7) divided by $|D_J|$, and the condition $k \leqq K - J_1$ of Proposition 4.1 holds for K large; here χ_k is the same function of A_k, ϕ_k for both transformations, since the restrictions of Sect. 3 of [I] are imposed on the block fields and their derivatives on $T^{(k)}$ only,[2] and Proposition 4.1 is proved in Sect. 5.3 with this χ_k. Collecting these bounds,

$$Z^{R(\varepsilon)}(D_J,g,h) \leqq (1+\delta'_K)\, e^{-c_K\sum_x \varepsilon^d h(x)}\, Z^\varepsilon(D_J,g,h)\,, \qquad \delta'_K \to 0\,, \tag{4.11}$$

since $(1+K)^\varpi (L^k\varepsilon)^\sigma \to 0$ and $\theta_K \to 0$ for $k = K - [\vartheta K]$ with $\vartheta < \gamma/(\gamma+\beta)$.[3]

To complete the proof, we must prove the reverse inequality. We use a symmetry which exchanges the two lattices. Let ρ be the reflection $x_\nu \to -x_\nu$ in one of the two coordinates of the plane of rotation, and let $S = R\rho$. Then $\rho R\rho = R^{-1}$, so that $S^2 = R\rho R\rho = 1$; S maps $\varepsilon\mathbb{Z}^d$ onto $R(\varepsilon\mathbb{Z}^d)$ and $R(\varepsilon\mathbb{Z}^d)$ onto $\varepsilon\mathbb{Z}^d$; and S maps D_J onto the torus RD_J, which is fitted by both lattices, since $5R$ is an integer matrix and the sides of D_J are multiples of 25 (Sect. 2). Under S the two lattice models on D_J are carried into the two lattice models on RD_J, with the sources transformed (Sg is the transform of the 2-form g under S, as in Lemma 6.6, and $Sh = h\circ S^{-1}$) and each lattice sampling the transformed 2-form at its own plaquettes:

$$Z^{R(\varepsilon)}(D_J,g,h) = Z^\varepsilon(RD_J,Sg,Sh)\,, \qquad Z^\varepsilon(D_J,g,h) = Z^{R(\varepsilon)}(RD_J,Sg,Sh)\,. \tag{4.12}$$

The bound (4.11) holds on the torus RD_J as it does on D_J, with the same constant c_K, both models being periodic models on a torus fitted by both lattices (it is now the usual lattice

[2] As are those of (2.55) of [Ba 2].

[3] (3.22) of [I].

which sees a torus whose sides are not parallel to its axes, the case treated above for the rotated lattice on D_J). The mass term Sh lies in the class of Sect. 4, since S preserves $\int \omega$ and the neighbourhood of $\operatorname{supp} g$.

The support of Sg is a union of cubes with sides parallel to the usual axes, while the final blocks of the two transformations on RD_J are rotated cubes, and an axis cube is never a union of rotated cubes; so the bound (4.11) on RD_J for the sources (Sg, Sh) is not the transport of (4.11) by S, which returns (4.11) on D_J, but the forward bound on the second torus, which is proved in [M] for the transported sources, whose supports are unions of blocks of the final scale of the transported transformations. Applied on RD_J to the sources Sg, Sh and read through the identities (4.12), (4.11) gives

$$Z^{\varepsilon}(D_J, g, h) \leqq (1 + \delta'_K)\, e^{-c_K \sum_x \varepsilon^d (Sh)(x)}\, Z^{R(\varepsilon)}(D_J, g, h)\,, \tag{4.13}$$

where the sum is over the sites of the usual lattice in RD_J; it differs from the sum in (4.11) by $O(\varepsilon^N)$ for every N, both being Riemann sums of $\int h$ over a torus, and since c_K is bounded the two multipliers agree up to a factor which tends to one. Therefore

$$(1 + \delta'_K)^{-1} e^{c_K \sum_x \varepsilon^d (Sh)(x)} \leqq \frac{Z^{R(\varepsilon)}(D_J, g, h)}{Z^{\varepsilon}(D_J, g, h)} \leqq (1 + \delta'_K)\, e^{-c_K \sum_x \varepsilon^d h(x)}\,. \tag{4.14}$$

In the normalized expectations

$$\langle \mathcal{O}(zg, 0) \rangle_p^{\varepsilon_K, D_J, \omega} = Z^{\varepsilon_K}(D_J, zg, -\omega) / Z^{\varepsilon_K}(D_J, 0, -\omega) \tag{4.15}$$

the bounds (4.14), for the sources (zg, h) and $(0, h)$ with z real, give

$$(1 + \delta'_K)^{-2} e^{-2c_K \int \omega} (1 + o(1)) \leqq \frac{\langle \mathcal{O}(zg, 0) \rangle_p^{R(\varepsilon_K), D_J, \omega}}{\langle \mathcal{O}(zg, 0) \rangle_p^{\varepsilon_K, D_J, \omega}} \leqq (1 + \delta'_K)^2 e^{2c_K \int \omega} (1 + o(1))\,, \tag{4.16}$$

the factors $e^{\pm c_K \sum \varepsilon^d h}$ of numerator and denominator compounding rather than cancelling. Both ends tend to one, since $c_K \to 0$ (Theorem 4.2) and $\int \omega$ is fixed; hence

$$\langle \mathcal{O}(zg, 0) \rangle_p^{R(\varepsilon_K), D_J, \omega} \big/ \langle \mathcal{O}(zg, 0) \rangle_p^{\varepsilon_K, D_J, \omega} \to 1 \qquad (K \to \infty)\,. \tag{4.17}$$

Both expectations lie between 1 and e^{Cz^2} (Jensen's inequality, $\langle F(g) \rangle = 0$ by the symmetry $(A, \theta) \to (-A, -\theta)$, and the Gaussian domination bound of Sect. 6.2), so their difference tends to zero; the limit of the expectation on the usual lattice exists by Proposition 4.3 below, and so the limit on the rotated lattice exists and is equal to it. For complex z the same follows from Vitali's theorem, the expectations being bounded on strips $|\operatorname{Re} z| \leqq r$ uniformly in K. This proves (2.24).

Proposition 4.3. *For ω and J as above and g a real 2-form supported in D_J away from the support of ω, the limit $\lim_{K\to\infty} \langle \mathcal{O}(zg, 0) \rangle_p^{\varepsilon_K, D_J, \omega}$ exists for every real z.*

Proof. This is the argument of Sect. 3.2 of [I], made on the ratio of the generating functionals $Z^{\varepsilon_K}(D_J, zg, -\omega)$ and $Z^{\varepsilon_K}(D_J, 0, -\omega)$. For each of the two, Z say, the lower bound of Theorem 3.1 of [I], (4.8) for the usual lattice and (4.9) show that

$$\begin{aligned} e^{-C(L^k \varepsilon)^{\sigma} |D_J|}\, I \leqq Z \leqq (1 - \delta_K)^{-1} e^{C(L^k \varepsilon)^{\sigma} |D_J|}\, I\,, \\ I = \int (dA_k)(d\phi_k) \chi_k(A_k) \chi_k(\phi_k) \exp[-S^{(k),1}]\,, \end{aligned} \tag{4.18}$$

with $k = K - [\vartheta K]$; in the ratio of the two integrals the terms of the effective action which depend neither on the fields nor on g, and in particular the term linear in ω, which does not

converge as $\varepsilon \to 0$, cancel, and the remaining terms are compared, for K and $K+1$, on the common lattice of the two transformations, as follows.

The counterterm δm^2 of this paper is defined with free boundary conditions (Sect. 3.1), while the results of [I] are stated for the counterterm defined on the torus; the two differ by a constant u_ε, bounded uniformly in ε, up to a field-independent constant which cancels in the ratio, so that the model of this paper is the model of [I] with the scalar source shifted by u_ε, and Theorem 3.4 of [I], which is stated for cubic tori such as D_J, is taken at the shifted source for the terms of the effective action which depend on the fields or on g; the term linear in h which depends on neither, the only term of non-positive degree without a counterterm, which is not covered by Theorem 3.4 of [I] (Sect. 6.6), has cancelled in the ratio above; the convergence of u_ε as $\varepsilon \to 0$ is part of the uniformity statements below.

Theorem 3.4 of [I] holds with the scalar source $-\omega + u$, as in Sect. 6.6, with constants uniform over the family of sources $-\omega + u$ with $|u|$ bounded, (4.9) holds at the source $-\omega$ on the usual lattice, the lattice expectations are equicontinuous in u uniformly in K, and the shift u_ε converges as $\varepsilon \to 0$. The models at ε_K and ε_{K+1} are compared on a common lattice: when $k(K+1) = k(K)+1$ the two k-th lattices coincide, and when $k(K+1) = k(K)$ one further step of the transformation is applied to I, its large field remainder being absorbed by (4.9), and the one-step bound of Theorem 3.4 of [I] gives a convergent telescoping sum, as in [M]. The ratio of the two integrals in (4.18) therefore converges, which proves the proposition.

It remains to prove that $\theta_0/2\pi$ is irrational. We know that $\cos\theta_0 = \frac{4}{5}$. So suppose that $\theta_0 = \dfrac{p}{q} 2\pi$, where p, q are integers. Then

$$1 = \cos q\theta_0 = \begin{cases} (\cos\theta_0)^q \sum_{n=0}^{1/2q} \binom{q}{2n} + P_{q-2}(\cos\theta_0) & \text{for } q \text{ even}, \\ (\cos\theta_0)^q \sum_{n=0}^{1/2(q-1)} \binom{q}{2n+1} + P'_{q-2}(\cos\theta_0) & \text{for } q \text{ odd}, \end{cases} \tag{4.19}$$

where P_{q-2}, P'_{q-2} are polynomials of degree $q-2$. It is easy to show that

$$\sum_{n=0}^{1/2q} \binom{q}{2n} = \sum_{n=0}^{1/2(q-1)} \binom{q}{2n+1} = 2^{q-1}. \tag{4.20}$$

Therefore, for some integers $a_0, \dots, a_{q-1}$, $\cos\theta_0$ is a solution of the equation

$$2^{q-1}x^q + \sum_{n=0}^{q-1} a_n x^n = 0. \tag{4.21}$$

We will now prove that $4/5$ is not a solution of (4.21). Let $x = M/N$, where M, N are mutually prime. Then

$$2^{q-1}M^q + \sum_{n=0}^{q-1} a_n M^n N^{q-n} = 0. \tag{4.22}$$

Therefore, N divides $2^{q-1}M^q$, and so by assumption $N = 2^s$ for some s. Therefore, $x = 4/5$ is not a root of (4.21).

5. Ultra-violet Stability

5.1. Regularity and Decay of Operators with D b.c.

In this section we will extend Theorem 3.3 of [I] to include operators defined with D b.c. Recall that we modified the averaging operator in (3.2) by including the averaging function f defined

by (3.8). Theorem 3.3 of [I] was proved in the paper [Ba 4], and we will modify suitably that proof below.

For any C_J defined by (2.7), we denote by T_ε the lattice in C_J with spacing ε. After rescaling to the η-lattice ($\eta = L^{-k}$), it is written T_η. We denote by $T^{(k)}$ the sublattice of T_η with unit spacing. Let $\Omega \subset T_\eta$ be such that $\Omega^{(k)}$ is a union of large blocks on $T^{(k)}$. We will write $\partial\Omega$ to mean that part of the boundary of Ω which does not belong to ∂T_η. Recall that $G_k(A)$ is defined with D b.c. on ∂T_η, and $G_k(\Omega, A)$ with D b.c. on $\partial\Omega \cup \partial T_\eta$. Also a vector field configuration A is regular on Ω if

$$|\partial^\eta_\mu A_\nu(x)| \leqq C(e(L^k\varepsilon)^{2-d/2})^{\beta-1} \tag{5.1}$$

for $\mu, \nu = 1, \dots, d$, $\beta > 0$, and $x \in \Omega$.

Proposition 5.1. *Let A be regular on Ω (for (5.3), on T_η), and $f : \Omega \to \mathbb{R}$. Let $\varepsilon = L^{-K}$; then for all $k \leqq K$, for $x \in \Omega$ and some $\delta_0 > 0$,*

$$|G_k(\Omega, A)f(x)| \leqq C \exp[-\delta_0 \operatorname{dist}(x, \operatorname{supp} f)] \, \|f\| \,, \tag{5.2}$$

$$\begin{aligned} &|G_k(\Omega, A)f(x) - G_k(A)f(x)| \\ &\quad \leqq C \exp[-\delta_0 \operatorname{dist}(x, \operatorname{supp} f) - \delta_0 \operatorname{dist}(x, \partial\Omega) - \delta_0 \operatorname{dist}(\operatorname{supp} f, \partial\Omega)] \, \|f\| \,. \end{aligned} \tag{5.3}$$

Corresponding bounds hold for $D_{A,\mu}G_k(\Omega, A)$ and the α-Hölder derivatives of $G_k(\Omega, A)$, $D_{A,\mu}G_k(\Omega, A)$ for $0 < \alpha < 1$, when Ω is a rectangular parallelepiped, and for a general Ω at the points at distance at least $2M$ (in the maximum norm) from the points of $\partial\Omega$ at which Ω is not locally convex, that is from the re-entrant edges and from the edges and vertices at which two blocks of Ω meet only along an edge or at a point; these are the only points at which they are used in Sect. 5.2. At such a point the derivative of the Green's function with D b.c. need not be bounded uniformly in η.

Proof. We will mimic the random walk representation presented in [Ba 4] in order to prove Proposition 5.1. For each $j \in \mathbb{Z}^d$ define the set

$$\square_j = \Omega \cap \{\text{the union of large blocks with } Mj \text{ at one corner}\} \,. \tag{5.4}$$

We introduce a partition of unity on $\mathbb{Z}^d$ as follows. For each $j \in \mathbb{Z}^d$, define

$$h_j(x) = \prod_{\mu=1}^{d} h\Big(\frac{x_\mu}{M} - j_\mu\Big) \,, \tag{5.5}$$

where h is C^∞, supported on $[-2/3, 2/3]$ and $h = 1$ on $[-1/3, 1/3]$. Also we choose h so that

$$\sum_{j \in \mathbb{Z}^d} h_j^2 = 1 \,. \tag{5.6}$$

We define the operator

$$G_0 = \sum_j h_j G_k(\square_j, A) h_j \,; \tag{5.7}$$

this will serve as a first approximation to $G_k(\Omega, A)$. A straightforward calculation (see [Ba 4]) shows that

$$\begin{aligned} &((-\Delta^{\eta,D}_{A,\Omega} + m^2(L^k\varepsilon)^2 + a_k P_k(A))G_0)(x, x') \\ &\quad = \sum_j [h_j(x)((-\Delta^{\eta,D}_{A,\square_j} + m^2(L^k\varepsilon)^2 + a_k P_k(A))G_k(\square_j, A))(x, x')h_j(x')] \\ &\qquad - \sum_{b \in st(x)} (\partial^\eta h_j)(b)(D_A G_k(\square_j, A))(b, x')h_j(x') \\ &\qquad + (-\Delta^\eta h_j)(x) G_k(\square_j, A; x, x')h_j(x') \\ &\qquad + (R_k(A, \partial^\eta h_j)G_k(\square_j, A))(x, x')h_j(x') \,, \end{aligned} \tag{5.8}$$

where $st(x)$ is the set of bonds connected to x, and we have included a superscript to denote D b.c. Also $R_k(A,\partial^\eta h) = [a_k P_k(A), h]$ is the commutator of the averaging term with the multiplication by h,

$$R_k(A,\partial^\eta h)\phi(x) = a_k \sum_{x'\in B^k(y^k(x))} \eta^d f(x) f(x') U(A(\Gamma^{(k)}_{x,y^k(x),x'})) \cdot (h(x') - h(x))\phi(x')\,, \tag{5.9}$$

so that $\|R_k(A,\partial^\eta h_j)\|_\infty \leqq CM^{-1}$, since $\sum_{x'\in B^k(y)} \eta^d f(x') = 1$ and h_j varies by $O(M^{-1})$ over a unit block.

When each $\square_j$ is a rectangular parallelepiped, which is the case when Ω is one, Lemma 5.3 below bounds the three terms of $K_j G_k(\square_j, A)h_j$ on L_∞ by CM^{-1}, CM^{-2} and CM^{-1}; since a point lies in the support of at most 2^d of the functions h_j, $\|R\|_\infty \leqq CM^{-1}$ and the series (5.12) converges on L_∞ for M sufficiently large.

For a general union of large blocks Ω the sets $\square_j$ at the re-entrant corners of Ω are not rectangular and Lemma 5.3 does not apply to them; in that case we sum (5.12) on L_2. By Lemma 5.2 and (5.9), $\|K_j G_k(\square_j, A)h_j\|_2 \leqq CM^{-1}$ for every j, so (5.12) converges on L_2 for M large, and a term of (5.13) is bounded at the point x through the bound of $G_k(\square, A)$ from L_2 to L_∞, uniform in η, which holds for every $\square_j$: with $C' = (-\Delta^{\eta,D}_{A,\square} + m^2(L^k\varepsilon)^2 + 1)^{-1}$ and $W = 1 - a_k P_k(A)$ we have $G_k(\square, A) = C' + C'WG_k(\square, A)$, and, for $d = 3$, $|C'(x,z)| \leqq C(\eta + |x-z|)^{-1} e^{-\delta|x-z|}$, since the expansion of C' in the hopping terms of $-\Delta^\eta_A$ is bounded term by term by that of the free operator with mass at least 1; hence $\sum_z \eta^d |C'(x,z)|^2 \leqq C$ for $d = 3$, and $\|G_k(\square, A)g\|_\infty \leqq C(1 + \|W\|_2 \|G_k(\square, A)\|_2)\|g\|_2 \leqq C\|g\|_2$.

A walk of n steps from the block of x to a block meeting $\operatorname{supp} f$ therefore contributes at most $C(CM^{-1})^n \|h_{\omega_n} f\|_2 \leqq C(M)(CM^{-1})^n \|f\|_\infty$, and (5.2), (5.3) follow as before, with C depending on M, the walks of $G_k(A)$ in (5.3) visiting blocks of $T_\eta \setminus \Omega$, on which A is regular by hypothesis.

The bounds on $D_{A,\mu} G_k(\Omega, A)$ and on the Hölder derivatives at a point x of a general Ω at distance at least $2M$ from the points of $\partial\Omega$ at which Ω is not locally convex follow from those on a rectangular $\square \subset \Omega$, a sum of a few large blocks containing $\Omega \cap \{\operatorname{dist}(\cdot, x) \leqq M\}$, with $\partial\square \setminus \partial\Omega$ at distance at least M from x. Let h be of the form (5.5), equal to 1 within distance $M/2$ of x and supported in $\square$ away from $\partial\square \setminus \partial\Omega$, so that $|\partial^\eta h| \leqq CM^{-1}$ and $|\Delta^\eta h| \leqq CM^{-2}$, and let $u = G_k(\Omega, A)f$. Then $hu = G_k(\square, A)(hf) + G_k(\square, A)\{[-\Delta^\eta_A, h]u + R_k(A, \partial^\eta h)u\}$, the two operators agreeing on functions supported in $\square$ away from $\partial\square \setminus \partial\Omega$, where $R_k(A, \partial^\eta h) = [a_k P_k(A), h]$ is the operator (5.9), of norm CM^{-1} on L_∞, and $[-\Delta^\eta_A, h]u = \sum_\mu \{D^{\eta *}_{A,\mu}(w_\mu u_\mu) + v_\mu u'_\mu\}$ with u_μ, u'_μ translates of u by one lattice step, transported by $U(A)$, and $|w_\mu| \leqq CM^{-1}$, $|v_\mu| \leqq CM^{-2}$, supported where h varies, at distance between $M/2$ and M from x.

The first term is bounded, with its derivatives and their decay, by Proposition 5.1 for the rectangular $\square$; the terms $v_\mu u'_\mu$ and $R_k u$ by the same bounds and (5.2) for u; and the terms $D^{\eta *}_{A,\mu}(w_\mu u_\mu)$ by (5.2) for u and the bound $|(D^\eta_{A,\mu} G_k(\square, A) D^{\eta *}_{A,\nu} g)(x)| \leqq Ce^{-\delta_0 \operatorname{dist}(x, \operatorname{supp} g)} \|g\|_\infty$ for g supported at distance at least $M/2$ from x, with its Hölder analogue, which follows from the decay and regularity of the kernel of $G_k(\square, A)$ off the diagonal. The factor $e^{-\delta_0 \operatorname{dist}(x, \operatorname{supp} f)}$ is recovered from (5.2), with C depending on M, and the same localization applied to $G_k(\Omega, A)f - G_k(A)f$ with (5.3) gives the bounds for the difference.

If we define the operators

$$(K_j\phi)(x) = \square_j(x)\Big\{ \sum_{b\in st(x)} (\partial^\eta h_j)(b)(D^\eta_A \phi)(b) + (\Delta^\eta h_j)(x)\phi(x) - R_k(A, \partial^\eta h_j)\phi(x) \Big\}\,, \tag{5.10}$$

$$R = \sum_j K_j G_k(\square_j, A) h_j\,, \tag{5.11}$$

then (5.8) implies that

$$G_k(\Omega, A) = G_0(I - R)^{-1} = \sum_{n=0}^{\infty} G_0 R^n \,. \tag{5.12}$$

We can now deduce the random walk representation. Define a path $\omega = \{\omega_0, \omega_1, \dots, \omega_n\}$, $\omega_i \in \mathbb{Z}^d$, where ω_i, ω_{i+1} are vertices of the same unit cube. Then (5.12) can be written

$$G_k(\Omega, A) = \sum_{\omega} h_{\omega_0} G_k(\square_{\omega_0}, A) h_{\omega_0} K_{\omega_1} G_k(\square_{\omega_1}, A) h_{\omega_1} \dots K_{\omega_n} G_k(\square_{\omega_n}, A) h_{\omega_n} \,. \tag{5.13}$$

The expansion (5.13) also bounds the difference between $G_k(\Omega, A)$ and an operator without the D b.c. on ∂T_η. Let $\mathbb{T}$ be a torus whose side is a multiple of M, of which T_η is a union of blocks, let $\tilde{\Omega}$ be a union of large blocks of $\mathbb{T}$ with $\tilde{\Omega} \cap T_\eta = \Omega$, and suppose that A is regular on $\tilde{\Omega}$. We write $G_k^0(\tilde{\Omega}, A) = (-\Delta_{A,\tilde{\Omega}}^{\eta,D} + m^2 (L^k \varepsilon)^2 + a_k P_k(A))^{-1}$ for the operator with D b.c. on $\partial\tilde{\Omega}$ only, with the averaging function f of (3.8) in both operators.

Its expansion (5.13) uses the blocks of $\mathbb{T}$, to which Lemma 5.2 applies, and Lemma 5.3 when $\tilde{\Omega}$ is rectangular, the summation on L_2 above giving the same bounds otherwise, and it coincides with the expansion of $G_k(\Omega, A)$ except for the walks which visit a block meeting ∂T_η. Such a walk of n steps, starting in a block containing x and ending in a block meeting $\operatorname{supp} g$, satisfies $nM \geqq c^{-1} \max\{\operatorname{dist}_{\mathbb{T}}(x, \operatorname{supp} g), \operatorname{dist}(x, \partial T_\eta) + \operatorname{dist}(\operatorname{supp} g, \partial T_\eta)\} - cM$, c depending on d only, where $\operatorname{dist}_{\mathbb{T}}$ is the distance in $\mathbb{T}$, since a step moves the block by at most $M\sqrt{d}$. Therefore, for $g : \Omega \to \mathbb{C}$ and $x \in \Omega$ as in Proposition 5.1,

$$\begin{aligned} &|G_k(\Omega, A) g(x) - G_k^0(\tilde{\Omega}, A) g(x)| \\ &\quad \leqq C \exp[-\delta_0(\operatorname{dist}_{\mathbb{T}}(x, \operatorname{supp} g) + \operatorname{dist}(x, \partial T_\eta) + \operatorname{dist}(\operatorname{supp} g, \partial T_\eta))] \, \|g\|_\infty \,, \end{aligned} \tag{5.14}$$

where δ_0 is a smaller constant than in (5.3) and C depends on M. The distance in $\mathbb{T}$ cannot be replaced by the distance in T_η when $\mathbb{T}$ is T_η with its opposite faces identified, since G_k^0 then connects Ω across the identified faces; when the side of $\mathbb{T}$ is at least twice that of T_η the two distances agree on T_η. This is a bound on the operators on L_∞; a bound of this form on the kernels fails at coincident points adjacent to ∂T_η, where the two kernels differ by the image term of the free propagator, which is of order η^{2-d}.

In [Ba 4], it is shown that Proposition 5.1 follows from the following two lemmas.

Lemma 5.2. *Let $\square$ be a sum of a few large blocks. Then*

$$\begin{aligned} \|G_k(\square, A) f\|_2, \ \|D_{A,\mu}^{\eta} G_k(\square, A) f\|_2, \\ \|G_k(\square, A) D_{A,\mu}^{\eta *} f\|_2, \ \|D_{A,\mu}^{\eta} G_k(\square, A) D_{A,\nu}^{\eta *} f\|_2 \leqq C \|f\|_2 \,. \end{aligned} \tag{5.15}$$

Lemma 5.3. *Let $\square$ be a rectangular parallelepiped which is a sum of a few large blocks. Then*

$$\|G_k(\square, A) f\|_\infty \leqq C \|f\|_\infty \,. \tag{5.16}$$

Corresponding bounds hold for $D_{A,\mu}^{\eta} G_k(\square, A)$ and the α-Hölder derivatives of $G_k(\square, A)$, $D_{A,\mu}^{\eta} G_k(\square, A)$ for $0 < \alpha < 1$.

Proof of Lemma 5.2. Since A is regular on $\square$, we can write

$$A = B + A' \,, \tag{5.17}$$

where B is a constant vector field and

$$|A'|, \ |\partial_\mu^\eta A'| \leqq C (L^k \varepsilon)^{(2-d/2)(\beta-1)} \,. \tag{5.18}$$

In the paper [I], we deduced the following expansion for $G_k(\square, A)$:

$$G_k(\square, A) = G_k^{1/2}(\square, B)(I - G_k^{1/2}(\square, B)V_k(A', B)G_k^{1/2}(\square, B))^{-1}G_k^{1/2}(\square, B)\,. \tag{5.19}$$

If we assume that Lemma 5.2 holds with A replaced by B, then (5.19) implies (5.15). Furthermore, we can gauge transform from B to the zero vector field configuration. The lemma then follows by getting lower bounds on quadratic forms. We see that

$$\langle \phi, (-\Delta_\square^{\eta,D} + m^2(L^k\varepsilon)^2 + a_k P_k)\phi\rangle \geqq \langle \phi, (-\Delta_\square^{\eta,N} + m^2(L^k\varepsilon)^2 + a_k P_k)\phi\rangle\,, \tag{5.20}$$

where $-\Delta_\square^{\eta,N}$ has N b.c. on $\partial\square$. The proof then proceeds exactly as in [Ba 4]; we separate $-\Delta_\square^{\eta,N}$ across unit blocks using N b.c. On each block $P_k = Q_k^* Q_k$ is the operator $f\langle f, \cdot\rangle$, with $\sum_\square \eta^d f = 1$; it does not vanish on the constants, and the other eigenvalues of $-\Delta_\square^{\eta,N}$ are at least 4. Writing $\phi = c + \psi$ with c constant on the block and $\sum_\square \eta^d \psi = 0$, we obtain as in the proof of (5.45) below

$$-\Delta_\square^{\eta,N} + a_k P_k \geqq \tfrac{1}{2}(-\Delta_\square^{\eta,N}) + m_0^2\,, \tag{5.21}$$

where m_0 is $O(1)$ uniformly in η, depending on a_k and f.

Proof of Lemma 5.3. By expanding (5.19), we get

$$G_k(\square, A) = \sum_{n=0}^{\infty} G_k(\square, B)(V_k(A', B)G_k(\square, B))^n\,. \tag{5.22}$$

Once again, we assume that Lemma 5.3 holds with A replaced by B. By combining (5.22) with the bound

$$\|V_k(A', B)G_k(\square, B)\|_\infty \leqq Ce(L^k\varepsilon)^\beta\,, \tag{5.23}$$

we then deduce the required result. Recall that (see [I])

$$\begin{aligned} -V_k(A', B) = {} & D_B^{\eta *} F_{1,k}(A')U(B) + U^*(B)F_{1,k}(A')^* D_B^\eta \\ & + |F_{1,k}(A')|^2 + a_k F_{2,k}(A', B)^* Q_k(B) \\ & + a_k Q_k^*(B)F_{2,k}(A', B) + a_k F_{2,k}(A', B)^* F_{2,k}(A', B) \end{aligned} \tag{5.24}$$

(the operators $F_{1,k}$ and $F_{2,k}$ are defined in [I]). As mentioned in [Ba 4], $V_k(A', B)$ is a first order differential operator acting on the function to its right. The only obstacle to this interpretation is the term $D_B^{\eta *} F_{1,k}(A')$. Explicitly computing it, we get

$$\begin{aligned} & (D_B^{\eta *} F_{1,k}(A')\phi)(x) \\ & \quad = \sum_{\mu=1}^{d} \eta^{-1}\{U(B)\eta^{-1}(U(A'(\langle x - \eta e_\mu, x\rangle)) - 1)\phi(x)\chi(x)\chi(x - \eta e_\mu) \\ & \qquad - \eta^{-1}(U(A'(\langle x, x + \eta e_\mu\rangle)) - 1)\phi(x + \eta e_\mu)\chi(x)\chi(x + \eta e_\mu)\}\,, \end{aligned} \tag{5.25}$$

where

$$\chi(x) = \begin{cases} 1 & \text{if} \quad x \in \square \backslash \partial\square\,, \\ 0 & \text{otherwise}\,. \end{cases} \tag{5.26}$$

In Appendix A, we will show that

$$|\eta^{-1}(G_k(\square, B)\phi)(x)| \leqq C\,, \tag{5.27}$$

when $x + \eta e_\mu$ or $x - \eta e_\mu$ lies on $\partial\square$ [this is a consequence of the fact that $G_k(\square, B)$ has D b.c. on $\partial\square$]. Therefore, (5.25) is a regular operator on $G_k(\square, B)$, and so (5.23) holds. It remains to prove (5.16) for a constant vector field. By gauge covariance, this is equivalent to having zero

vector field. In Appendix A, we will show how these bounds follow from an explicit Fourier representation.

The next proposition gives a lower bound on the effective Laplacian which is crucial in the proof of ultra-violet stability. Again, we assume that $\Omega^{(k)}$ is a union of large blocks on $T^{(k)}$, and we denote by $\overline{\Omega}^{(k)}$ that part of $\partial\Omega^{(k)}$ which intersects $\partial T^{(k)}$.

Proposition 5.4. *When A is regular on Ω,*

$$\langle\phi, \Delta^{(k)}(\Omega, A)\phi\rangle \geqq \gamma_0\Big\{ \sum_{x\in\overline{\Omega}^{(k)}} |\phi(x)|^2 + \sum_{\langle x,x'\rangle\in\Omega^{(k)*}} |U(A(\langle x,x'\rangle))\phi(x') - \phi(x)|^2 + m^2(L^k\varepsilon)^2 \sum_{x\in\Omega^{(k)}} |\phi(x)|^2\Big\} - C(e(L^k\varepsilon)^{2-d/2})^{2-\alpha} \sum_{x\in\Omega^{(k)}} |\phi(x)|^2 \,. \tag{5.28}$$

Proof. The bound (5.28) is established in [Ba 4] for N b.c., but without the term $\gamma_0 \sum_{x\in\overline{\Omega}^{(k)}} |\phi(x)|^2$. This term is present only because we are using D b.c. Since

$$-\Delta_A^{\eta,D} \geqq -\Delta_A^{\eta,N} \,, \tag{5.29}$$

the remaining terms on the right-hand side of (5.28) follow from the bound on the operator with N b.c. (It is important to notice that the proof presented in [Ba 4] requires only L_2-bounds on operators.) So we will prove the lower bound for the first term on the right-hand side of (5.28).

We may separate $\Delta^{(k)}(\Omega, A)$ by Neumann boundary conditions across the boundaries of each unit block in $\Omega^{(k)}$. On each block away from $\overline{\Omega}^{(k)}$, we estimate the operator by zero. So let $\square$ be the block on Ω defined by some $x \in \overline{\Omega}^{(k)}$. Then

$$\langle\phi, \Delta^{(k)}(\square, A)\phi\rangle = a_k|\phi(x)|^2 - a_k^2\langle\phi, Q_k(A)G_k(\square, A)Q_k^*(A)\phi\rangle \,. \tag{5.30}$$

In fact (5.30) is bounded below by $\gamma_1|\phi(x)|^2$ for every vector field A, where $\gamma_1 = a_k(1 + a_kC_f^2)^{-1}$, $C_f^2 = \sum_{B^k(x)} \eta^d f(x')^2$, f the averaging function of (3.8). To see this, recall that (5.30) is

$$\min_{\Phi} \big\{\langle\Phi, (-\Delta_{A,\square}^{\eta,D} + m^2(L^k\varepsilon)^2)\Phi\rangle + a_k|\phi(x) - (Q_k(A)\Phi)(x)|^2\big\} \,, \tag{5.31}$$

the minimum over Φ on $\square$, where $-\Delta_{A,\square}^{\eta,D}$ has D b.c. on $\partial\square \cap \partial T_\eta$ and N b.c. on the other sides of $\square$. Since Φ vanishes on ∂T_η and $|U(A(b))| = 1$, we have $|\Phi(b_+)| - |\Phi(b_-)| \leqq |U(A(b))\Phi(b_+) - \Phi(b_-)|$ on each bond normal to ∂T_η, and summing along each line of η^{-1} sites normal to ∂T_η gives the first of the bounds

$$\sum_{\square} \eta^d|\Phi|^2 \leqq \langle\Phi, -\Delta_{A,\square}^{\eta,D}\Phi\rangle \,, \qquad |(Q_k(A)\Phi)(x)|^2 \leqq C_f^2 \sum_{\square} \eta^d|\Phi|^2 \,, \tag{5.32}$$

the second being Schwarz's inequality. Minimizing over Φ in (5.31) with (5.32) gives the bound with $\gamma_1 = a_k(1 + a_kC_f^2)^{-1}$. When A is regular, (5.30) can also be expanded, as follows.

Since A is regular on Ω, we can write $A = B + A'$, with B constant and A' small. We can then gauge transform (5.30) to remove B, and expand about $A' = 0$ in the usual way (see Sect. 3.4 of [I]). Using the L_2-bounds of Lemma 5.2, the terms in the expansion of order e^2 and higher can be bounded by $C(e(L^k\varepsilon)^{2-d/2}p(L^k\varepsilon))^2|\phi(x)|^2$. The term of order e is proportional to $\mathrm{Re}(\phi^*(x)i\phi(x)) = 0$. So we are left with the piece independent of A, which is

$$a_k|\phi(x)|^2 - a_k^2\langle\phi, Q_kG_k(\square)Q_k^*\phi\rangle \,. \tag{5.33}$$

For simplicity, we change $G_k(\square)$ so that $-\Delta_\square^\eta$ has D b.c. on one side of $\square$ only, and N b.c. on the other sides, and we drop the mass term $m^2(L^k\varepsilon)^2$; both changes decrease $-\Delta_\square^\eta$, hence

increase $G_k(\square)$ and decrease (5.33), so it suffices to bound the modified (5.33) from below. We denote this side by B; it contains η^{1-d} sites. We will use the integral representation of (5.33):

$$\begin{aligned} & C \exp\left[-\tfrac{1}{2}(5.33)\right] \\ & \quad = \int d\phi'|_{\square} \exp\left[-\tfrac{1}{2} a_k |\phi(x) - (Q_k \phi')(x)|^2 - \tfrac{1}{2}\langle \phi', -\Delta_{\square}^{\eta} \phi' \rangle\right], \end{aligned} \tag{5.34}$$

where C is some constant. Clearly, for some A_0,

$$(5.34) = C \exp\left[-\tfrac{1}{2} A_0 |\phi(x)|^2\right]. \tag{5.35}$$

In order to evaluate A_0, we will Fourier transform (5.34). Since (5.34) does not mix the components of $\phi(x)$, it is sufficient to consider each component separately. Then

$$\int d\phi \exp - \left[i\xi\phi + \tfrac{1}{2} A_0 \phi^2\right] = \sqrt{2\pi A_0^{-1}} \exp\left[-\tfrac{1}{2} A_0^{-1} \xi^2\right]. \tag{5.36}$$

If we change ϕ' to $\phi + \phi'$ in (5.34), we get

$$\begin{aligned} \int d\phi'|_{\square} \exp \Bigg[& -\tfrac{1}{2} a_k |Q_k \phi'(x)|^2 - \tfrac{1}{2}\langle \phi', -\Delta_{\square}^{\eta,N} \phi' \rangle \\ & - \tfrac{1}{2} \sum_{y \in B} \eta^{d-2} |\phi'(y) + \phi(x)|^2 \Bigg], \end{aligned} \tag{5.37}$$

where $-\Delta_{\square}^{\eta,N}$ has N b.c. on every side of $\square$. We can write

$$\begin{aligned} \sum_{y \in B} \eta^{d-2} |\phi'(y) + \phi(x)|^2 = & \sum_{y \in B} \eta^d \Big| \eta^{-1} \phi'(y) - \sum_{y' \in B} \eta^{d-2} \phi'(y') \Big|^2 \\ & + \eta^{-1} \Big| \phi(x) + \sum_{y \in B} \eta^{d-1} \phi'(y) \Big|^2. \end{aligned} \tag{5.38}$$

If we define

$$\begin{aligned} \langle \phi', A\phi' \rangle = & \, a_k |Q_k \phi'(x)|^2 + \langle \phi', -\Delta_{\square}^{\eta,N} \phi' \rangle \\ & + \sum_{y \in B} \eta^d \Big| \eta^{-1} \phi'(y) - \sum_{y' \in B} \eta^{d-2} \phi'(y') \Big|^2, \end{aligned} \tag{5.39}$$

then the Fourier transform (5.36) of (5.34) becomes

$$\begin{aligned} & \int d\phi \exp(-i\xi\phi) \int d\phi' \exp\left[-\tfrac{1}{2} \eta^{-1} \Big| \phi(x) + \sum_{y \in B} \eta^{d-1} \phi'(y) \Big|^2 - \tfrac{1}{2} \langle \phi', A\phi' \rangle \right] \\ & \quad = \int d\phi \exp(-i\xi\phi - \tfrac{1}{2} \eta^{-1} \phi^2) \int d\phi' \exp\left[i\xi \sum_{y \in B} \eta^{d-1} \phi'(y) - \tfrac{1}{2} \langle \phi', A\phi' \rangle \right] \\ & \quad = C \exp\left[-\tfrac{1}{2} \eta \xi^2 - \tfrac{1}{2} \xi^2 \langle f, A^{-1} f \rangle\right], \end{aligned} \tag{5.40}$$

where

$$f(y) = \sum_{y' \in B} \eta^{d-1} \delta^{\eta}(y - y'). \tag{5.41}$$

Therefore, we deduce

$$A_0^{-1} = \eta + \langle f, A^{-1} f \rangle. \tag{5.42}$$

From (5.39), dropping the last term, which is positive, we have

$$\langle \phi, A\phi \rangle \geqq \langle \phi, -\Delta_{\square}^{\eta,N} \phi \rangle + a_k |Q_k \phi(x)|^2 \,. \tag{5.43}$$

Write $\phi = c + \psi$ with c constant on $\square$ and $\sum_{\square} \eta^d \psi = 0$, so that $Q_k \phi(x) = c + Q_k \psi(x)$, since the averaging function of (3.8) has $\sum_{\square} \eta^d f = 1$. Then

$$|Q_k \psi(x)|^2 \leqq C_f^2 \|\psi\|^2 \,, \qquad \langle \psi, -\Delta_{\square}^{\eta,N} \psi \rangle \geqq 4 \|\psi\|^2 \,, \qquad |c + Q_k \psi(x)|^2 \geqq \tfrac{1}{2}|c|^2 - |Q_k \psi(x)|^2 \,, \tag{5.44}$$

with C_f^2 as after (5.30), since the non-zero eigenvalues of $-\Delta_{\square}^{\eta,N}$ are $4\eta^{-2} \sum_{\mu} \sin^2(\pi k_\mu \eta/2) \geqq 4$. Keeping only the fraction $t = \min\{1, (a_k C_f^2)^{-1}\}$ of the term $a_k |Q_k \phi(x)|^2$ in (5.43) and using (5.44), we see that

$$\langle \phi, A\phi \rangle \geqq \langle \phi, (\tfrac{1}{2}(-\Delta_{\square}^{\eta,N}) + m_0^2)\phi \rangle \,, \tag{5.45}$$

with $m_0^2 = \frac{1}{2} \min\{2, a_k, C_f^{-2}\}$, uniformly in η.

Since $f = \eta^{-1} \mathbf{1}_B$ is constant along B and $-\Delta_{\square}^{\eta,N}$ is the sum of the one-dimensional Neumann Laplacians in the coordinate directions, $(\frac{1}{2}(-\Delta_{\square}^{\eta,N}) + m_0^2)^{-1} f$ depends only on the coordinate normal to B, and

$$\langle f, A^{-1} f \rangle \leqq \langle f, (\tfrac{1}{2}(-\Delta_{\square}^{\eta,N}) + m_0^2)^{-1} f \rangle = G_1(0,0) \,, \tag{5.46}$$

where G_1 is the kernel of $(\frac{1}{2}(-\Delta^{\eta,N}) + m_0^2)^{-1}$ on the one-dimensional lattice $\{0, \eta, \dots, 1-\eta\}$. For any g on this lattice, $|g(0)| \leqq |g(y)| + \sum_{y'<y} \eta |\partial^\eta g(y')|$, so that, squaring and averaging over y,

$$|g(0)|^2 \leqq 2\|g\|^2 + 2\langle g, -\Delta^{\eta,N} g \rangle \leqq 4 \max\{m_0^{-2}, 1\} \langle g, (\tfrac{1}{2}(-\Delta^{\eta,N}) + m_0^2) g \rangle \,. \tag{5.47}$$

For $g = G_1(\cdot, 0)$ the right side is $4 \max\{m_0^{-2}, 1\} g(0)$, hence $G_1(0,0) \leqq 4 \max\{m_0^{-2}, 1\}$. Therefore

$$A_0^{-1} \leqq \eta + 4 \max\{m_0^{-2}, 1\} = O(1) \,, \tag{5.48}$$

and (5.28) follows, with $\gamma_0 = \frac{1}{2} \min\{\gamma_1, \gamma_N\}$, γ_N the constant of the bound of [Ba 4], by adding the lower bound $\gamma_1 |\phi(x)|^2$ and the lower bound with N b.c. with weights $\frac{1}{2}$.

5.2. Upper and Lower Bounds on the Generating Functional

We will now establish Theorem 3.1 of [I] for the generating functional $Z_{N,D}^{\varepsilon}(C_J, g, h)$ defined with N b.c. for the vector field and D b.c. for the scalar field. Of course, the proof follows by modifying suitably the material in [Ba 1] and [Ba 2].

Because the covariant Laplacian for the scalar field has D b.c. on ∂C_J, there is an additional positive factor in each effective action $S^{(k)}$ which keeps the scalar field ϕ_k small near $\partial T^{(k)}$. Specifically, we introduce the following bounds on the scalar field at each step of the renormalization transformation:

$$\prod_{x \in \Lambda_0^{(k)} \cap \partial T^{(k)}} \chi(|\phi_k(x)| \leqq C p(L^k \varepsilon)) \,, \tag{5.49}$$

where $\Lambda_0^{(k)}$ is the subset of $T^{(k)}$ on which all fields are small. If the bound in (5.49) is not satisfied at some point, then Proposition 5.4 shows that we get a small factor $\exp(-p(L^k\varepsilon)^2)$ from the integral.

We recall the small field conditions of the kth step in the form in which we use them.[4] The regions $\Lambda_7^{(k)} \subset \cdots \subset \Lambda_0^{(k)} \subset \Lambda_{-1}^{(k)}$ are unions of large blocks, and consecutive regions are separated by corridors of width

$$r(L^k \varepsilon) = R \left(1 + \ln(L^k \varepsilon)^{-1}\right)^{\varrho} \,, \qquad \varrho > 1 \,, \tag{5.50}$$

[4] They are the conditions of Sect. 2 of [Ba 2]: (2.2) there for the fields of the kth step, (2.7)–(2.8) for the regions and their corridors, (2.16) for neighbouring block fields; the exponent of the logarithm in p is fixed by the condition stated after (3.54) there, and the sum over the large field blocks is (2.78).

where R is taken sufficiently large; thus every point of $T^{(k)}$ within distance $3r(L^k\varepsilon)$ of $\Lambda_4^{(k)}$ lies in $\Lambda_0^{(k)}$. On $\Lambda_0^{(k)}$ the fields satisfy

$$\begin{aligned} |\phi_k(y)| &\leqq C\lambda(L^k\varepsilon)^{-1/4}p(L^k\varepsilon)\,, \qquad |(D_A\phi_k)(b)| \leqq p(L^k\varepsilon)\,, \\ |\phi_{k+1}(y) - (Q(A)\phi_k)(y)| &\leqq p(L^k\varepsilon)\,, \qquad |\partial A^{(k)}| \leqq Cp(L^k\varepsilon)\,, \end{aligned} \tag{5.51}$$

where $\lambda(L^k\varepsilon) = \lambda L^k\varepsilon$ is the coupling constant at the kth step, $\lambda(L^k\varepsilon)^{4-d}$ in the units of this section; consequently $|\phi_{k+1}(y') - U\phi_{k+1}(y)| \leqq Cp(L^k\varepsilon)$ on neighbouring blocks, and $|\phi_{k+1}| \leqq C\lambda(L^k\varepsilon)^{-1/4}p(L^k\varepsilon)$. A block y at which $|\phi_k(y)| \geqq (2\lambda(L^k\varepsilon))^{-1/4}p(L^k\varepsilon)$ is a large field block. The exponent of the logarithm in $p(\cdot)$ exceeds ϱ, so that

$$r(L^k\varepsilon) < p(L^k\varepsilon) \tag{5.52}$$

for $L^k\varepsilon$ sufficiently small, depending on R.

The additional bounds (5.49) are enough to ensure that the proof of the upper bound goes through; the lower bound is treated at the end of this section. First, (5.49) ensures that the fluctuation fields ϕ' are always small. Recall that the field ϕ' at the kth step is defined by[5]

$$\phi_k = \phi' + aL^{-2}C^{(k)}_{\Lambda_4^{(k)}}(B^k(\Lambda_2^{(k)}), B^{(k+1),\eta})Q^*(B^{(k+1),\eta})\phi_{k+1}\,. \tag{5.53}$$

As shown in [Ba 2] for the operator defined without the D b.c., and as we show below for $C^{(k)}_{\Lambda_4^{(k)}}$, when $\operatorname{dist}(x, \partial\Lambda_4^{(k)} \setminus \partial T^{(k)}) > p(L^k\varepsilon)$ and $\operatorname{dist}(x, \partial T^{(k)}) \geqq r(L^k\varepsilon)$, $x \in \Lambda_4^{(k)}$, we have

$$\begin{aligned} &aL^{-2}C^{(k)}_{\Lambda_4^{(k)}}(B^k(\Lambda_2^{(k)}), B^{(k+1),\eta})Q^*(B^{(k+1),\eta})\phi_{k+1}(x) \\ &\quad = U(B^{(k+1),\eta}(\Gamma_{y,x}))\phi_{k+1}(y) + Cp(L^k\varepsilon)\,, \qquad x \in B(y)\,. \end{aligned} \tag{5.54}$$

The right side carries the transported block field $\phi_{k+1}(y)$ rather than $(Q^*(B^{(k+1),\eta})\phi_{k+1})(x)$, which vanishes on the top layers of the blocks, where $f = 0$. The identity rests on $Q1 = 1$, which holds for the averaging (3.2) by (3.1): then $P1 = Q^*1$ and $aL^{-2}CQ^*1 = 1 - C\Delta^{(k)}1$, and for the operator without the D b.c. we have $(-\Delta + m^2(L^k\varepsilon)^2 + a_kP)1 = m^2(L^k\varepsilon)^2 + a_kQ^*1$, so that $a_kQG_kQ^*1 = 1 - m^2(L^k\varepsilon)^2QG_k1$ and

$$\Delta^{(k)}1 = a_km^2(L^k\varepsilon)^2QG_k1 = O((L^k\varepsilon)^2) \tag{5.55}$$

by the L_∞ bound on G_k; the rest of the proof is that of [Ba 2].[6] Within $r(L^k\varepsilon)$ of $\partial T^{(k)}$ we do not use (5.54).

Therefore, suppose that $x \in \Lambda_4^{(k)}$, and $\operatorname{dist}(x, \partial T^{(k)}) < r(L^k\varepsilon)$. (We cannot take $\operatorname{dist}(x, \partial T^{(k)}) < p(L^k\varepsilon)$ here, since the contour $\Gamma_{x,y}$ in (5.56) must stay in $\Lambda_0^{(k)}$, where (5.49) is imposed.) Then using (5.49) and the bound on the covariant derivative of ϕ_k,

$$\begin{aligned} |\phi_k(x)| &= |U(B^{(k+1),\eta}(\Gamma_{x,y}))\phi_k(y) + \phi_k(x) - U(B^{(k+1),\eta}(\Gamma_{x,y}))\phi_k(y)| \\ &\leqq Cp(L^k\varepsilon)^2\,, \end{aligned} \tag{5.56}$$

where $y \in \partial T^{(k)}$ and $\Gamma_{x,y}$ is a contour connecting x and y. The block field ϕ_{k+1} satisfies the same bound on the blocks y within $2r(L^k\varepsilon)$ of $\partial T^{(k)}$ which lie within $r(L^k\varepsilon)$ of $\Lambda_4^{(k)}$. For a block y meeting $\partial T^{(k)}$, (5.49) at a point of $B(y) \cap \partial T^{(k)}$ and the bound on $D_A\phi_k$ on the bonds of $B(y)$ give $|\phi_k| \leqq Cp(L^k\varepsilon)$ on $B(y)$, hence $|(Q(A)\phi_k)(y)| \leqq Cp(L^k\varepsilon)$ and, by (5.51), $|\phi_{k+1}(y)| \leqq Cp(L^k\varepsilon)$. For the other blocks the bound on neighbouring block fields, applied

[5] Eq. (2.110) of [Ba 2].

[6] Lemma 2.7 of [Ba 2], whose proof follows that of Lemma 2.5 there; the L_∞ bound is (2.25) of [Ba 1].

along the chain of blocks normal to $\partial T^{(k)}$, which contains at most $Cr(L^k\varepsilon)$ blocks and lies in $\Lambda_0^{(k)}$, gives $|\phi_{k+1}(y)| \leqq Cp(L^k\varepsilon)^2$, by (5.52). The remaining blocks are at distance at least $r(L^k\varepsilon)$ from x, and by the exponential decay of $C^{(k)}_{\Lambda_4^{(k)}}$ and the bound on ϕ_{k+1} they contribute at most $Ce^{-\delta r(L^k\varepsilon)}\lambda(L^k\varepsilon)^{-1/4}p(L^k\varepsilon) \leqq p(L^k\varepsilon)$ to the second term of (5.53), for R sufficiently large. It follows that for $x \in \Lambda_4^{(k)}$, with $\mathrm{dist}(x, \partial T^{(k)}) < r(L^k\varepsilon)$,

$$|\phi'(x)| \leqq Cp(L^k\varepsilon)^2\,, \tag{5.57}$$

as required.

For $x \in \Lambda_4^{(k)}$ with $\mathrm{dist}(x, \partial T^{(k)}) \geqq r(L^k\varepsilon)$ we compare $C^{(k)}_{\Lambda_4^{(k)}}(B^k(\Lambda_2^{(k)}), B^{(k+1),\eta})$ with the corresponding operator of [Ba 2] on a torus, defined with the same averaging operator (3.2) but without the D b.c. on $\partial T^{(k)}$. Let $\mathbb{T}$ be the torus of side twice that of $T^{(k)}$ obtained by reflecting $T^{(k)}$ in the planes $x_\mu = MJ - \varepsilon/2$ and $x_\mu = -MJ - \varepsilon/2$ of each direction, in the coordinates of (2.7); these planes carry the blocks of every scale onto blocks, since $2MJ$ is a multiple of $L^k\varepsilon$, and the bonds crossing the first join the Dirichlet layer $x_\mu = MJ - \varepsilon$ to its image, those crossing the second the layer $x_\mu = -MJ$ to its image. We continue the regions $\Lambda_j^{(k)}$, the vector field, the block fields $B^{(k+1)}$ and the scalar field ϕ_k to $\mathbb{T}$ by these reflections, each component of a vector field being reflected evenly and a bond crossing a mirror plane, which is its own image, being given the value of the adjacent bond of the same direction in the layer $x_\mu = MJ - \varepsilon$, respectively $x_\mu = -MJ$; on the blocks outside $T^{(k)}$ we put $\phi_{k+1} = Q(B^{(k+1)})\phi_k$ with the averaging (3.2), whose weight (3.8) is the same on every block of $\mathbb{T}$.

Every difference quotient of a continued vector field is then, up to sign, a difference quotient of the original field or zero, and the curvature of a plaquette straddling a mirror plane is minus a difference quotient of the original field; so the regularity (5.1), which (5.14) requires, is preserved, the continued regions are unions of large blocks of $\mathbb{T}$, and the conditions (5.51) on the vector fields and the block fields hold on the continued regions with constants depending only on the original ones, a block average over a mirror block differing from the reflected one only through the bonds crossing the mirror plane. The conditions on ϕ_k and its covariant differences hold by the reflection on all bonds of the continued region except the mirror bonds at the face points of $\Lambda_{-1}^{(k)} \setminus \Lambda_0^{(k)}$, where (5.49) is not imposed and the covariant difference is bounded by $2(2\lambda(L^k\varepsilon))^{-1/4}p(L^k\varepsilon)$ only. A covariant difference on a mirror bond is zero where the bond joins two Dirichlet layers, and at most $2|\phi_k(x)| \leqq Cp(L^k\varepsilon)$ where it joins a point $x \in \Lambda_0^{(k)} \cap \partial T^{(k)}$ to its image, by (5.49); on a mirror block ϕ_{k+1} is the average of the continued ϕ_k, and the condition on $\phi_{k+1} - Q\phi_k$ holds there with a larger constant, by the bound on neighbouring block fields. The exceptional bonds are at distance at least $r(L^k\varepsilon)$ from every point at which (5.54) is used, and in the proof of (5.54) the conditions on the scalar fields enter only through the kernel of $C^{(k)}_{\Lambda_4^{(k)}}$ summed against the differences $\phi_{k+1}(y') - U\phi_{k+1}(y)$, so that their contribution is bounded by $Ce^{-\delta r(L^k\varepsilon)}\lambda(L^k\varepsilon)^{-1/4}p(L^k\varepsilon) \leqq p(L^k\varepsilon)$, as in (5.58) below. On $T^{(k)}$ the distance in $\mathbb{T}$ is the ordinary distance, and (5.14) applies with $\tilde{\Omega}$ the continued region.

By (5.14), a bound on the operators on L_∞ which carries the distance from x to $\partial T^{(k)}$, applied to the functions $Q_k^*(A)\delta_{y'}$, which are bounded by $\sup f$ and supported in one block, the kernels of the operators $\Delta^{(k)}(\Omega, A) = a_k - a_k^2 Q_k(A)G_k(\Omega, A)Q_k^*(A)$ of (5.30) with and without the D b.c. differ by at most $C\exp[-\delta_0(|y-y'| + \mathrm{dist}(y, \partial T^{(k)}) + \mathrm{dist}(y', \partial T^{(k)}))]$. Both covariances decay exponentially, by the lower bound of Proposition 5.4, whose proof with N b.c. applies to the operator without the D b.c. as well, and by the random walk expansion of [Ba 4]. By that expansion the covariance of the operator of [Ba 2] on the continued region, restricted to $\Lambda_4^{(k)}$, differs from the covariance of its restriction to $\Lambda_4^{(k)}$ by the walks which leave $\Lambda_4^{(k)}$, which are bounded by $C\exp[-\delta(\mathrm{dist}(y, \partial T^{(k)}) + \mathrm{dist}(y', \partial T^{(k)}))]$; and the two operators on $\Lambda_4^{(k)}$ are compared by the resolvent identity, which gives the same bound for the difference of

the covariances, with a smaller constant $\delta_1 > 0$ in place of δ_0, since $|y-u|+|u-u'|+|u'-y'| \geqq |y-y'|$ and $|y-u| + \mathrm{dist}(u, \partial T^{(k)}) \geqq \mathrm{dist}(y, \partial T^{(k)})$ for the intermediate points u, u'. Since $|\phi_{k+1}| \leqq C\lambda(L^k\varepsilon)^{-1/4} p(L^k\varepsilon)$ on the small field region, the left-hand side of (5.54) changes by at most

$$Ce^{-\delta_1 r(L^k\varepsilon)}\lambda(L^k\varepsilon)^{-1/4} p(L^k\varepsilon) \leqq p(L^k\varepsilon) \tag{5.58}$$

for R sufficiently large, R depending on λ through $\ln \lambda^{-1}$ only. Since (5.54) holds for the operator of [Ba 2] at such x, it holds for $C^{(k)}_{\Lambda_4^{(k)}}$, and hence $|\phi'(x)| \leqq |\phi_k(x) - U(B^{(k+1),\eta}(\Gamma_{y,x}))\phi_{k+1}(y)| + Cp(L^k\varepsilon) \leqq Cp(L^k\varepsilon)^2$ for such x, by (5.51).

Second, (5.49) also ensures that a large scalar field near $\partial T^{(k)}$ produces a small damping factor in the integral. Recall that the dominant term in the effective action is

$$-\lambda(L^k\varepsilon)^{4-d} \sum_{x \in B^k(\Lambda_7^{(k-1)\prime})} \eta^d |\phi^{(k)}(x)|^4 , \tag{5.59}$$

where the field $\phi^{(k)}(x)$ is given by

$$\phi^{(k)}(x) = (a_k G_k(\Lambda_7^{(k-1)\prime}, A^{(k)}) Q_k^*(A^{(k)})\phi_k)(x) . \tag{5.60}$$

Suppose that the distance of $x \in B^k(\Lambda_7^{(k-1)\prime})$ from $\partial T^{(k)}$ is $r(L^k\varepsilon)$ or greater, and let $\tilde{G}_k(\Lambda_7^{(k-1)\prime}, A^{(k)})$ be the operator of [Ba 2] on the doubled torus $\mathbb{T}$ with the continued fields, as above, with the same boundary conditions on $\partial\Lambda_7^{(k-1)\prime}$ but without the D b.c. on $\partial T^{(k)}$. Then by (5.14), applied to the function $Q_k^*(A^{(k)})\phi_k$ with $\tilde{\Omega}$ the continued region, since $|\phi_k| \leqq C\lambda(L^k\varepsilon)^{-1/4} p(L^k\varepsilon)$ on $\Lambda_7^{(k-1)\prime}$ by the small field conditions of the previous step,

$$\begin{aligned} |a_k(G_k(\Lambda_7^{(k-1)\prime}, A^{(k)}) - \tilde{G}_k(\Lambda_7^{(k-1)\prime}, A^{(k)}))Q_k^*(A^{(k)})\phi_k(x)| &\leqq Ce^{-\delta_0 r(L^k\varepsilon)}\lambda(L^k\varepsilon)^{-1/4} p(L^k\varepsilon) \\ &\leqq C(L^k\varepsilon)^\alpha \end{aligned} \tag{5.61}$$

for R large, and for x at the distance from $\partial\Lambda_7^{(k-1)\prime}$ required in [Ba 2], which is the case for the large field blocks of the kth step, the comparison of $\tilde{G}_k(\Lambda_7^{(k-1)\prime}, A^{(k)})$ with $G_k(A^{(k)})$ is that of [Ba 2]. Hence we have

$$\begin{aligned} \phi^{(k)}(x) &= (a_k G_k(A^{(k)}) Q_k^*(A^{(k)})\phi_k)(x) + O(L^k\varepsilon)^\alpha \\ &= \phi_k(y) + Cp(L^k\varepsilon)|x-y| + O(L^k\varepsilon)^\alpha , \end{aligned} \tag{5.62}$$

where $G_k(A^{(k)})$ is the operator of [Ba 2] on the doubled torus, with periodic boundary conditions there, y is any point in $B^k(\Lambda_7^{(k-1)\prime})$, and the second line is a bound on the difference. Therefore, for any union Ω of large blocks of $\Lambda_7^{(k-1)\prime}$ at distance $r(L^k\varepsilon)$ or greater from $\partial T^{(k)}$ and at the distance from $\partial\Lambda_7^{(k-1)\prime}$ required in [Ba 2], we deduce

$$-\lambda(L^k\varepsilon) \sum_{x \in B^k(\Omega)} \eta^d |\phi^{(k)}(x)|^4 \leqq -\tfrac{1}{2}\lambda(L^k\varepsilon) \sum_{y \in \Omega} |\phi_k(y)|^4 + C(L^k\varepsilon)^\alpha |\Omega| , \tag{5.63}$$

where $\alpha > 0$. This bound allows us to extract a small factor for each large field $\phi_k(y)$ at distance $r(L^k\varepsilon)$ or greater from $\partial T^{(k)}$.

For a large field block y within $r(L^k\varepsilon)$ of $\partial T^{(k)}$ the bound (5.63) is not available. By (5.14), applied as in (5.61), $|\phi^{(k)}(x) - \tilde{\phi}^{(k)}(x)| \leqq Ce^{-\delta_0 s}\lambda(L^k\varepsilon)^{-1/4} p(L^k\varepsilon)$ for x at distance s from $\partial T^{(k)}$, where $\tilde{\phi}^{(k)}$ is the field of the operator of [Ba 2] on the doubled torus, for which $\tilde{\phi}^{(k)}(x) = \phi_k(y) + Cp(L^k\varepsilon)|x-y| + O(L^k\varepsilon)^\alpha$ on $B^k(y)$, as in (5.62). Let s_1 be such that $Ce^{-\delta_0(s_1-1)} \leqq \frac{1}{8}$. For a large field block y at distance s_1 or more from $\partial T^{(k)}$ we have $|\phi^{(k)}(x)| \geqq \frac{1}{2}|\phi_k(y)|$ on

$B^k(y)$ for $L^k\varepsilon$ small, so that the quartic term (5.59) gives the factor $\exp[-\lambda(L^k\varepsilon)|\phi_k(y)|^4/16] \leqq \exp[-p(L^k\varepsilon)^2]$ for $p(L^k\varepsilon)^2 \geqq 32$, which holds for $L^k\varepsilon$ small, as in [Ba 2].

For a large field block y within s_1 of $\partial T^{(k)}$ we use instead the quadratic form of $\Delta^{(k)}(\Lambda_7^{(k-1)\prime}, A^{(k)})$, a positive term of $S^{(k)}$ to which Proposition 5.4 applies with $\Omega = \Lambda_7^{(k-1)\prime}$. We use (5.28) with the constant $\frac{1}{3}\gamma_0$ in place of γ_0, leaving the remaining third of the lower bound to the arguments of [Ba 2] on the large field region, where the quadratic form of $\Delta^{(k)}$ is only estimated from below. This leaves at our disposal twice the terms

$$\tfrac{1}{3}\gamma_0\Big\{ \sum_{z\in\overline{\Omega}^{(k)}} |\phi_k(z)|^2 + \sum_{b\in\ell} |U(A^{(k)}(b))\phi_k(b_+) - \phi_k(b_-)|^2 \Big\}, \tag{5.64}$$

the first copy paying for the large field blocks on the lines ℓ and the second for the negative term of (5.28). Here ℓ is the line of bonds of $\Omega^{(k)*}$ normal to $\partial T^{(k)}$ from the large field block y on that normal which is farthest from $\partial T^{(k)}$ within s_1 of it, ending at a point z of a face block, and $\overline{\Omega}^{(k)}$ is the part of $\partial\Omega^{(k)}$ on $\partial T^{(k)}$, as in Proposition 5.4; ℓ has fewer than s_1 bonds, and since at most d such lines end at a given z, each is given the share $\frac{1}{d}$ of the first term of (5.64). The line ℓ lies in $\Omega^{(k)}$: $\Omega^{(k)} = \Lambda_7^{(k-1)\prime}$ and $T^{(k)}$ are unions of the large blocks of the previous step, cubes of side M in the units of the kth lattice placed on one grid,[7] and $M > s_1$, s_1 being fixed before M; so the cube containing y, which lies within s_1 of $\partial T^{(k)}$, has a face on $\partial T^{(k)}$ and contains ℓ. For $k = 0$, where $\Omega^{(0)} = T^{(0)}$, this is trivial. The same holds for the lines used below.

If $|\phi_k(z)| \geqq \frac{1}{2}|\phi_k(y)|$, this share is at least $\frac{1}{12d}\gamma_0(2\lambda(L^k\varepsilon))^{-1/2}p(L^k\varepsilon)^2$. Otherwise, since $|U(A^{(k)}(b))\phi_k(b_+) - \phi_k(b_-)| \geqq \big||\phi_k(b_+)| - |\phi_k(b_-)|\big|$, Schwarz's inequality gives

$$\sum_{b\in\ell} |U(A^{(k)}(b))\phi_k(b_+) - \phi_k(b_-)|^2 \geqq \frac{(|\phi_k(y)| - |\phi_k(z)|)^2}{s_1} \geqq \frac{(2\lambda(L^k\varepsilon))^{-1/2}p(L^k\varepsilon)^2}{4s_1}\,. \tag{5.65}$$

In either case the terms of (5.64) available to ℓ amount to at least $2s_1p(L^k\varepsilon)^2$ for $L^k\varepsilon \leqq \varepsilon_0$, ε_0 depending on λ, since $\lambda(L^k\varepsilon) = \lambda L^k\varepsilon$ and s_1 does not depend on the scale; this holds at every step of the transformation, whose scales lie below the scale L^{-J} of Theorem 3.1 of [I], J large depending on the coupling constants. Since $S^{(k)}$ contains $\frac{1}{2}\langle\phi_k, \Delta^{(k)}\phi_k\rangle$, this quadratic form gives at least the factor $\exp[-s_1p(L^k\varepsilon)^2]$, at least one factor $\exp[-p(L^k\varepsilon)^2]$ for each of the large field blocks on ℓ, whose number is at most s_1. The lines through distinct normals are disjoint, so these factors multiply.

The negative term of (5.28) is absorbed, at distance s_1 or more from $\partial T^{(k)}$, by the quartic term, as in [Ba 2], the comparison with the field of [Ba 2] holding there with an error at most $\frac{1}{8}\lambda(L^k\varepsilon)^{-1/4}p(L^k\varepsilon)$, and within s_1 of $\partial T^{(k)}$ by the second third of the bracket of (5.28): along a line of bonds normal to $\partial T^{(k)}$ of length s_1 ending at a point z of a face block,

$$\sum_w |\phi_k(w)|^2 \leqq 2s_1^2\Big\{|\phi_k(z)|^2 + \sum_b |U(A^{(k)}(b))\phi_k(b_+) - \phi_k(b_-)|^2\Big\}, \tag{5.66}$$

the sums running over the sites and the bonds of the line; every point within s_1 of $\partial T^{(k)}$ lies on at most d such lines, and the coefficient $C(e(L^k\varepsilon)^{2-d/2})^{2-\alpha}$ of the negative term times $2d^2s_1^2$ is at most $\frac{1}{3}\gamma_0$ for $L^k\varepsilon$ small. The proof of the upper bound goes through.

In the proof of the lower bound a bound of the form (5.49) is inserted only at the last step, where the fields are restricted to the small field region; at the intermediate steps none is inserted. There the small field conditions at the $(k+1)^{\text{th}}$ step are those of [I],[8] with the bound on the Laplacian of the background field replaced by a bound $C_1'p(L^{k+1}\varepsilon)$ on its covariant derivative

[7] (2.7)–(2.8) of [Ba 2]; the large blocks of [Ba 1] have the side ML in the units of their own lattice.

[8] (A.1) of [I]; the cut-off $p_1(L^k\varepsilon)$ on the fluctuation field is that of [Ba 1].

on every bond of C_J, the bonds to ∂C_J included, and a bound $C_1'' p(L^{k+1}\varepsilon)$ on $\phi_{k+1} - Q\phi^{(k+1)}$; together with the cut-off $|\phi'| \leqq p_1(L^k\varepsilon)$ on the fluctuation field, and for L sufficiently large, a suitable choice of the constants C_1', C_1'' and $L^k\varepsilon$ sufficiently small, these conditions, together with the bound $|\partial A^{(k)}| \leqq Cp(L^k\varepsilon)$ on the vector field, are preserved by the transformation. Since the background field vanishes on ∂C_J, it is bounded by $CL^{-3/2}p(L^k\varepsilon)$ on the blocks adjacent to $\partial T^{(k)}$, hence so is $Q\phi^{(k+1)}$ there, and the field $aL^{-2}C^{(k)}_{\Lambda_4^{(k)}}Q^*\phi_{k+1}$ of (5.53) differs from it by at most $CL^{-5/2}p(L^k\varepsilon)$; so $|\phi_k(x)| \leqq Cp(L^k\varepsilon) + p_1(L^k\varepsilon)$ for $x \in \partial T^{(k)}$, and the restriction (5.49) at the last step is compatible with the characteristic functions of [Ba 1], which, together with that restriction, is all that the D b.c. add to the argument of [Ba 1].

5.3. Lower Bound for the Rotated Model

The proof of Proposition 4.1 follows by verifying the properties used in [Ba 1] to prove a lower bound. We will write $\hat{G}_k(A)$, $\hat{C}^{(k)}(A)$ etc. to denote the propagators of the modified transformation of Sect. 4, that is of the averaging operators (4.4), (4.5) followed by the remaining steps of the transformation. The following proposition gives the required properties of $\hat{G}_k(A)$.

Proposition 5.5. *Let T_η be a torus fitted by both the usual and the rotated lattice, let A be regular on T_η, and $f : T_\eta \to \mathbb{R}^N$. Then*

$$|(\hat{G}_k(A)f)(x)| \leqq C \exp(-\delta_0 \operatorname{dist}(x, \operatorname{supp} f))\|f\|_\infty \,, \tag{5.67}$$

$$|(D_{A,\mu}\hat{G}_k(A)f)(x)| \leqq C \exp[-\delta_0 \operatorname{dist}(x, \operatorname{supp} f)]\|f\|_\infty \,, \tag{5.68}$$

$$\begin{aligned} &\frac{1}{|x-y|^\alpha}|U(A(\Gamma_{x,y}))(D^\eta_{A,\mu}\hat{G}_k(A)f)(y) - (D^\eta_{A,\mu}\hat{G}_k(A)f)(x)| \\ &\quad \leqq C_\alpha \exp[-\delta_0 \operatorname{dist}(\{x,y\}, \operatorname{supp} f)]\|f\|_\infty \,, \end{aligned} \tag{5.69}$$

where $0 < \alpha < 1$, for all x, $y \in T_\eta$; the constants C, δ_0 are independent of k, and C_α depends also on α.

Proposition 5.5 guarantees that the effective action $\hat{S}^{(k),R(5^k\varepsilon)}(T^{(k)}, A_k, \phi_k, g, h)$ has the necessary properties to act as a small, local perturbation of the free field measure at each step of the renormalization transformation. In order to apply the results of [Ben 1] to perform a cluster expansion at each step of the renormalization transformation, we need the following results concerning $\hat{C}^{(k)}(A)$. For $\Omega \subset T_1^{(k)}$, define

$$\hat{C}^{(k)}_\Omega(A) = ((\hat{\Delta}^{(k)}(A) + aL^{-2}P(A))|_\Omega)^{-1} \,. \tag{5.70}$$

Proposition 5.6. *Let $\Omega \subset T_1^{(k)}$ be a sum of big blocks; then for $x, y \in \Omega$,*

$$|\hat{C}^{(k)}_\Omega(A; x, y)| \leqq C \exp[-\delta_0|x-y|] \,. \tag{5.71}$$

Proof of Proposition 5.5. Once again we use a random walk representation. This time, we use blocks $\{\square_j\}$ which are small scale versions of D_J, that is they can be fitted by both the usual lattice and the rotated lattice; we take their sides to be multiples of $5^{k+1}\varepsilon$, multiples of 5 on the rescaled lattice, so that, $5R$ being an integer matrix, they are fitted by the rotated block lattices $R(5^j\varepsilon\mathbb{Z}^d)$ for every $j \leqq k$, as D_J is. Our first approximation is the operator

$$\hat{G}_0 = \sum_j h_j \hat{G}_k(\square_j, \tilde{A}) h_j \,, \tag{5.72}$$

where $\hat{G}_k(\square, \tilde{A})$ has periodic boundary conditions on the block $\square$. Also, we take $\tilde{A}$ equal to A in the interior of $\square$, and changing regularly to a constant configuration near $\partial\square$; here B is

the value of A at the center of $\square$, and $\tilde{A}$ is taken regular on $\square$ in the sense of (5.1), so that $A' = \tilde{A} - B$ satisfies (5.18) on $\square$, whose side is bounded. We then deduce the representation

$$\hat{G}_k(A) = \sum_{\omega} h_{\omega_0} \hat{G}_k(\square_{\omega_0}, \tilde{A}_{\omega_0}) h_{\omega_0} K_{\omega_1} \hat{G}_k(\square_{\omega_1}, \tilde{A}_{\omega_1}) \dots K_{\omega_n} \hat{G}_k(\square_{\omega_n}, \tilde{A}_{\omega_n}) h_{\omega_n} \,. \tag{5.73}$$

To deduce Proposition 5.5, we need the following lemmas.

Lemma 5.7.

$$\begin{gathered} \|\hat{G}_k(\square, \tilde{A}) f\|_2, \ \|D^{\eta}_{A,\mu} \hat{G}_k(\square, \tilde{A}) f\|_2, \\ \|\hat{G}_k(\square, \tilde{A}) D^{\eta *}_{A,\mu} f\|_2, \ \|D^{\eta}_{A,\mu} \hat{G}_k(\square, \tilde{A}) D^{\eta *}_{A,\nu} f\|_2 \leqq C \|f\|_2 \,. \end{gathered} \tag{5.74}$$

Lemma 5.8.

$$\|\hat{G}_k(\square, \tilde{A}) f\|_\infty \leqq C \|f\|_\infty \,. \tag{5.75}$$

Corresponding bounds hold for $D^{\eta}_{A,\mu} \hat{G}_k(\square, \tilde{A})$ and the α-Hölder derivatives of these operators, $0 < \alpha < 1$.

Proof. As with Lemma 5.2, Lemma 5.7 follows from the lower bound

$$\langle \phi, (-\Delta^{\eta}_{\square,B} + m^2 (L^k \varepsilon)^2 + a_k \hat{P}_k(B)) \phi \rangle \geqq C \langle \phi, \phi \rangle \,. \tag{5.76}$$

On $\square$ the field $\tilde{A}$ differs from the constant B by A', which is small where it is not zero, by the regularity of A and the construction of $\square$. On a bond $b = \langle x, x + \eta e_\mu \rangle$ of $\square$ we have $|D^{\eta}_{\tilde{A},\mu} \phi(x)|^2 \geqq \frac{1}{2} |D^{\eta}_{B,\mu} \phi(x)|^2 - \eta^{-2} |U(\tilde{A}(b)) - U(B(b))|^2 |\phi(x + \eta e_\mu)|^2$ and $|U(\tilde{A}(b)) - U(B(b))| = |\exp[ie(L^k\varepsilon)^{2-d/2} \eta A'(b)] - 1| \leqq e(L^k\varepsilon)^{2-d/2} \eta |A'(b)|$, so that the last term is at most $(e(L^k\varepsilon)^{2-d/2} \sup_\square |A'|)^2 |\phi(x + \eta e_\mu)|^2$; and $|\hat{Q}_k(\tilde{A}) \phi(y) - \hat{Q}_k(B) \phi(y)|$ is bounded by the supremum over the contours Γ of the block $\hat{B}_k(y)$, of length at most C, of $|U(\tilde{A}(\Gamma)) - U(B(\Gamma))| \leqq Ce(L^k\varepsilon)^{2-d/2} \sup_\square |A'|$, times $(\sum_{\hat{B}_k(y)} \eta^d |\phi|^2)^{1/2}$. Since $e(L^k\varepsilon)^{2-d/2} \sup_\square |A'| \leqq Ce(L^k\varepsilon)^{(2-d/2)\beta}$ by (5.18) and the construction of $\square$, (5.76) for $\tilde{A}$ follows from (5.76) for B with a quarter of the constant, for $L^k\varepsilon$ small, and Lemma 5.7 follows as Lemma 5.2 did.

To prove (5.76), we partition $\square$ into cubes $\{\Delta_i\}$ on the η-lattice, with sides parallel to the coordinate planes on that lattice. The size of Δ is chosen so that each cube contains fully at least one block of 5^{kd} sites defined by $\hat{P}_k$. We then estimate (5.76) from below by omitting $\hat{P}_k$ except on one block fully contained in each Δ_i, and omitting the (derivative)^2 terms on bonds not contained in a single Δ_i. We then get

$$\begin{aligned} (5.76) &\geqq \sum_i \langle \phi, (-\Delta^{\eta,N}_{\Delta_i} + m^2 (L^k\varepsilon)^2 + a_k \hat{P}_k(\Delta_i)) \phi \rangle \\ &\geqq \sum_i C \langle \phi|_{\Delta_i}, \phi|_{\Delta_i} \rangle \,, \end{aligned} \tag{5.77}$$

Here we have used N b.c. on each Δ_i and kept one block of $\hat{P}_k(B)$ in each cube; both terms dropped are non-negative. On a cube B is gauged to zero by $\phi \mapsto e^{ieB\cdot x} \phi$ (here and below e stands for the charge $e(L^k\varepsilon)^{2-d/2}$ of the kth step), since the contours defining $\hat{Q}_k(B)$ lie inside the block, so that $|D^{\eta}_{B,\mu} \phi(b)| = |\partial^{\eta}_{\mu}(e^{ieB\cdot x}\phi)(b)|$ for bonds $b \subset \Delta_i$ and $|\hat{Q}_k(B)\phi(y)| = |\hat{Q}_k(0)(e^{ieB\cdot x}\phi)(y)|$; on the torus $\square$ itself this is not possible, since a constant field has a holonomy there, which is why we separate first. The second inequality of (5.77) holds since $-\Delta^{\eta,N}_{\Delta_i}$ annihilates exactly the constant functions on Δ_i and its non-zero eigenvalues are not less than $4/R^2$, where R is the side of Δ_i, while $\hat{P}_k(\Delta_i)$, which averages over a block $\hat{B}_k(y) \subset \Delta_i$, vanishes on no constant: writing the gauged field $\psi = e^{ieB\cdot x}\phi = c + \psi'$ on Δ_i, with c the mean of ψ and $\sum_{\Delta_i} \eta^d \psi' = 0$, so that $\langle \phi, -\Delta^{\eta,N}_{\Delta_i,B} \phi \rangle = \langle \psi, -\Delta^{\eta,N}_{\Delta_i} \psi \rangle \geqq (4/R^2) \|\psi'\|^2$,

$$\langle \phi, \hat{P}_k(\Delta_i) \phi \rangle = |c + (\hat{Q}_k(0)\psi')(y)|^2 \geqq \tfrac{1}{2}|c|^2 - C_f^2 \|\psi'\|^2 \,, \qquad C_f^2 = \sum_{\hat{B}_k(y)} \eta^d f^2 \,, \tag{5.78}$$

so that keeping only the fraction $\min\{1, 2/(a_k C_f^2 R^2)\}$ of the last term in (5.77) gives (5.77) with

$$C = \min\{a_k, 2C_f^{-2}R^{-2}\}/(2R^d)\,, \tag{5.79}$$

Here f is the weight of $\hat{Q}_k$; with periodic boundary conditions the averaging (4.4), (4.5) has uniform weights, $f = 1$ (Sect. 3.1), so that $C_f = 1$, and a_k is bounded below uniformly in k;[9] thus C does not depend on k. This gives (5.74) (the other bounds follow as before).

To prove Lemma 5.8 let $C^{(1)}(\square, \tilde{A}) = (-\Delta^\eta_{\square,\tilde{A}} + 1)^{-1}$ on the block and $V = a_k \hat{P}_k(\tilde{A}) + m^2(L^k\varepsilon)^2 - 1$, so that

$$\begin{aligned} \hat{G}_k(\square, \tilde{A}) &= C^{(1)}(\square, \tilde{A}) - C^{(1)}(\square, \tilde{A}) V \hat{G}_k(\square, \tilde{A})\,, \\ D^\eta_{\tilde{A},\mu} \hat{G}_k(\square, \tilde{A}) f &= D^\eta_{\tilde{A},\mu} C^{(1)}(\square, \tilde{A}) g\,, \qquad g = f - V\hat{G}_k(\square, \tilde{A}) f\,. \end{aligned} \tag{5.80}$$

By the expansion of $C^{(1)}(\square, \tilde{A})$ in the hopping terms of $-\Delta^\eta_{\tilde{A}}$, every path contributing a phase, its kernel is bounded in modulus by that of $C^{(1)}(\square, 0)$, which is at most $C(\eta + |x-z|)^{2-d} e^{-\delta|x-z|}$ uniformly in η for $d = 3$, by its Fourier representation; hence $\sum_z \eta^d |C^{(1)}(\square, \tilde{A})(x, z)|^2 \leqq C$, and by Lemma 5.7

$$\|\hat{G}_k(\square, \tilde{A}) f\|_\infty \leqq \|C^{(1)}(\square, \tilde{A}) f\|_\infty + C\|V\hat{G}_k(\square, \tilde{A}) f\|_2 \leqq C\|f\|_\infty\,, \tag{5.81}$$

the constant depending on the size of $\square$, which is fixed; this is (5.75). The bound on $D^\eta_{\tilde{A},\mu} \hat{G}_k(\square, \tilde{A})$ follows from the Hölder bound proved below, $D^\eta_{\tilde{A},\mu} C^{(1)}(\square, \tilde{A})$ mapping L_∞ into the Hölder space.

For a constant field B the kernel of $\hat{G}_k(\square, B)$, which the Hölder bound uses, is the sum of the translates $\hat{G}_k(B)(x, y + \ell n)$, $n \in \mathbb{Z}^d$, of the kernel of the operator on $\eta\mathbb{Z}^d$, where ℓ is the side of $\square$. Since $\square$ is fitted by the rotated lattice and by its block lattices at every scale, the translations by ℓn map each of these lattices onto itself; so they preserve the blocks, their contours and the field B, the operator $-\Delta^\eta + m^2(L^k\varepsilon)^2 + a_k \hat{P}_k(B)$ on $\eta\mathbb{Z}^d$ commutes with them, and $\hat{G}_k(\square, B) f = \hat{G}_k(B) f_{\mathrm{per}}$, where f_{per} is the periodic extension of f and the sum of translates converges by the decay proved in Appendix B.

On $\eta\mathbb{Z}^d$ the field B is gauged to zero by $\phi \mapsto e^{ieB\cdot x}\phi$, so that

$$\hat{G}_k(B)(x, y) = e^{-ieB\cdot x} \hat{G}_k(0)(x, y) e^{ieB\cdot y}\,, \tag{5.82}$$

and the bounds of the lemma for $\tilde{A} = B$ follow also from the corresponding bounds for the operator on $\eta\mathbb{Z}^d$ with $\tilde{A} = 0$ acting on bounded functions, which we prove in Appendix B.

The Hölder clause of Lemma 5.8 follows from (5.75) and (5.80), in which $\|g\|_\infty \leqq C\|f\|_\infty$ by (5.75), since $\hat{P}_k(\tilde{A})$ is bounded on L_∞. For $\tilde{A} = 0$ the kernel of $C^{(1)}$ on $\eta\mathbb{Z}^d$ has first differences bounded by $C(\eta + |x-z|)^{1-d} e^{-\delta_1|x-z|}$ and second differences by $C(\eta + |x-z|)^{-d} e^{-\delta_1|x-z|}$, uniformly in η, by its Fourier representation; for $|x - y| = t \leqq 1$ we split the sum over z in $D^\eta_\mu C^{(1)} g(x) - D^\eta_\mu C^{(1)} g(y)$ at $|z - x| = 2\sqrt{d}\, t$, bound the near part by the first differences and the far part by the second differences along a lattice path from y to x, and obtain

$$|D^\eta_\mu C^{(1)} g(x) - D^\eta_\mu C^{(1)} g(y)| \leqq Ct(1 + \ln t^{-1})\|g\|_\infty \leqq C_\alpha t^\alpha \|g\|_\infty\,, \qquad 0 < \alpha < 1\,, \tag{5.83}$$

since $t(1 + \ln t^{-1}) \leqq C_\alpha t^\alpha$ on $(0, 1]$; for $t > 1$ the Hölder quotient is bounded by $2\|D^\eta_\mu C^{(1)} g\|_\infty$. The periodic block is handled by the periodization of Lemma 5.8, and a constant field B by the gauge transformation (5.82), the Hölder quotient being taken with the transporter $U(B(\Gamma_{x,y}))$. For $\tilde{A} = B + A'$ we expand $C^{(1)}(\square, \tilde{A})$ about $A' = 0$ as in (5.22); the expansion contains no terms $F_{2,k}$, since $C^{(1)}$ contains no averaging operator, and, the block having no boundary, (5.25)–(5.27)

[9] (2.15) of [Ba 1].

are not needed. Its terms contain the operators $D^{\eta}_{B,\mu} C^{(1)}(\Box, B) D^{\eta^*}_{B,\nu}$, which are not bounded on L_∞ uniformly in η, so the series is summed in the Hölder norm of (5.69). By (5.82) the twisted Hölder quotients of $D^{\eta}_{B,\mu} C^{(1)}(\Box, B) D^{\eta^*}_{B,\nu}$ are those of $\partial^{\eta}_{\mu} C^{(1)} \partial^{\eta^*}_{\nu}$ at $B = 0$, as in (5.83). The kernel K of the latter is a second difference of the kernel of $(-\Delta^{\eta} + 1)^{-1}$, so that $|K(x,z)| \leqq C(\eta + |x-z|)^{-d} e^{-\delta_1 |x-z|}$, its first differences in x are bounded by $C(\eta + |x-z|)^{-d-1} e^{-\delta_1 |x-z|}$, and $\sum_z \eta^d K(x,z) = 0$. For $|x-y| = t \leqq 1$ and $Q = \{z : \ |z-x|_\infty \leqq 2\sqrt{d}\, t\}$,

$$\begin{aligned}(K\psi)(x) - (K\psi)(y) = & \sum_{z \in Q} \eta^d K(x,z)(\psi(z) - \psi(x)) - \sum_{z \in Q} \eta^d K(y,z)(\psi(z) - \psi(y)) \\ & + \sum_{z \notin Q} \eta^d [K(x,z) - K(y,z)](\psi(z) - \psi(x)) \\ & + (\psi(x) - \psi(y)) \sum_{z \in Q} \eta^d K(y,z)\,. \end{aligned} \tag{5.84}$$

The first three sums are bounded by $C t^{\alpha} \|\psi\|_{\alpha}$ by the kernel bounds, the third along a lattice path from y to x as in (5.83); in the last, $\partial^{\eta^*}_{\nu}$ is summed by parts onto the faces of Q, which are at distance at least $(2\sqrt{d}-1)t$ from y and carry the first difference of the kernel of $(-\Delta^{\eta}+1)^{-1}$, bounded there by $C(\eta + t)^{1-d}$, and the faces having $C(t/\eta)^{d-1}$ sites, $|\sum_{z \in Q} \eta^d K(y,z)| \leqq C$ uniformly in η and t; for $t > 1$ the quotient is bounded by $2\|K\psi\|_\infty \leqq C\|\psi\|_\alpha$, and the periodic block is handled by periodization. Further, $D^{\eta}_{B,\mu} C^{(1)}(\Box, B)$ and $C^{(1)}(\Box, B) D^{\eta^*}_{B,\nu}$ map L_∞ into the Hölder space by (5.83) and the same estimate, and the multiplication operators $F_{1,k}(A')$ of (5.24), that is multiplication by $\eta^{-1}(U(\tilde{A}(b)) - U(B(b)))$, have Hölder norm at most $Ce(L^k\varepsilon)^{2-d/2} \sup_{\Box}(|A'| + |\partial^{\eta} A'|) \leqq Ce(L^k\varepsilon)^{(2-d/2)\beta}$ by (5.18), since $|U(\tilde{A}(b)) - U(B(b))| \leqq e(L^k\varepsilon)^{2-d/2}\eta |A'(b)|$ and, for parallel bonds b, b', $|U(\tilde{A}(b)) - U(\tilde{A}(b')) - U(B(b)) + U(B(b'))| \leqq e(L^k\varepsilon)^{2-d/2}\eta |A'(b) - A'(b')|$. The series therefore converges in the norm of the operators from L_∞ to the Hölder space for $L^k\varepsilon$ small, and the transporter $U(\tilde{A}(\Gamma_{x,y}))$ differs from $U(B(\Gamma_{x,y}))$ by at most $Ce(L^k\varepsilon)^{(2-d/2)\beta}|x-y|$. This gives the Hölder clause of Lemma 5.8, and the clause (5.69) of Proposition 5.5 then follows from Lemmas 5.7, 5.8 and the representation (5.73) in the same way as Proposition 5.1 follows from Lemmas 5.2, 5.3, by the method of [Ba 4].

Therefore, Proposition 5.5 holds for $\hat{G}_k(A)$ as required. To prove Proposition 5.6, we use the method presented in Chap. 3 of [Ba 4]. This method requires only L_2-bounds on the operators $\hat{G}_k(\Omega, A)$, and these may be established in the same way as Lemma 5.7. Hence (5.71) holds for the set Ω.

6. The Osterwalder-Schrader Axioms

In this section we complete the proofs of the results stated in Sect. 2: the axioms of [OS 1–2] for the Schwinger functions of the field strength, the construction of the limit measure, the results with scalar sources, and rotation invariance. Theorems 2.1 and 2.2, the continuum limit and the infinite volume limit with the boundary conditions of Sect. 2, are proved in Sect. 3, using the ultra-violet stability bounds of Sect. 5.

In Sect. 6.1 we relate the expectations with Dirichlet boundary conditions to those on a box. The regularity bounds are established in Sect. 6.2, the axioms of symmetry, translation invariance and reflection positivity in Sects. 6.3 and 6.4, the limit measure and its decomposition into ergodic components in Sect. 6.5, the results with scalar sources in Sect. 6.6, and in Sect. 6.7 the rotation invariance of the infinite volume limit, Theorem 2.5, from Theorems 2.4 and 6.24; the estimates taken from [M] are stated in Sects. 4 and 6.7 where they are used.

Throughout, $d = 3$ and $L = 5$, as in Sect. 2, and the vector field carries the fixed mass $\mu_0 > 0$ of (2.5). We write ε for ε_K when no confusion can arise; g always denotes a real 2-form, evaluated at the centers of the plaquettes as in (2.12), and h a real function.

The expectations are those of Sects. 2–4. In the expectation $\langle\cdot\rangle_{N,D}^{\varepsilon,C_J}$ of (2.10) the scalar field vanishes on the layers $x_\mu = -MJ - \varepsilon$ and $x_\mu = MJ - \varepsilon$, $\mu = 1,2,3$, the first just outside the cube $\{-MJ \leqq x_\mu < MJ\}$ of (2.7) and the second its last layer, so that the sites carrying a scalar variable are those with $-MJ \leqq x_\mu \leqq MJ - 2\varepsilon$.

These layers are $2MJ$ apart, a multiple of $L^k\varepsilon$ for $k \leqq K$, as (3.4) requires. Since MJ is a multiple of $L^k\varepsilon$, each layer is the last layer of a block $B_k(y) = \{x : \ y_\mu \leqq x_\mu < y_\mu + L^k\varepsilon\}$, $y \in L^k\varepsilon\mathbb{Z}^d$, of every scale $L^k\varepsilon$, $k \leqq K$. The averaging function (3.8) vanishes on the last layer of every block in each direction and is constant on the remaining sites, so it vanishes on the two layers. Hence the reflections in these layers carry the support of (3.8) in each block onto its support in the mirror block, $P_k(0)$ is invariant, and the representation (3.9) holds; this is shown in Sect. 6.1. If the layers were placed at the corners of the blocks, the reflections in them would not map the support of (3.8) onto itself, $P_k(0)$ would not be invariant, and the argument of Sect. 6.1 would not apply.

The expectation $\langle\cdot\rangle_\Omega^\varepsilon$ of (3.38) is the one in which every site of the box Ω carries a scalar variable; and the periodic expectations of Sects. 2 and 4 carry, when this is indicated, a mass term $\sum_x \varepsilon^d \omega(x) r(x)^2$ with ω smooth and non-negative as in (4.2), which is the term entering Theorem 2.4; up to a constant independent of the fields it is the source (2.14) with $w = -1$ and $h = \omega$, in the Wick ordering (2.13). The relation between the first two expectations is made explicit in Sect. 6.1.

6.1. The Correlation Inequalities and the Two Cubes

All the expectations listed above are of the form

$$\langle \mathscr{F} \rangle = Z^{-1} \int \mathscr{F} \exp\Big[\sum_{\langle x,y\rangle} c_{xy}\, r(x) r(y) \cos(\theta(x) - \theta(y) - e\varepsilon A_{\langle x,y\rangle}) \Big] \cdot \prod_x d\rho_x(r(x))\, d\theta(x)\; d\gamma(A)\,, \tag{6.1}$$

where the sum runs over a set of bonds, $c_{xy} \geqq 0$, each ρ_x is a finite measure on $[0,\infty)$ with $\int e^{\alpha r^2} d\rho_x < \infty$ for all $\alpha \geqq 0$ (a point mass at $r = 0$ is allowed, and this is how a site at which the scalar field vanishes is included), and γ is a centered non-degenerate Gaussian measure for the vector field. We call such a measure a BFS measure.

For the action (2.5) with the boundary conditions of Sect. 2 we have $c_{xy} = \varepsilon^{d-2}$ on the bonds inside C_J between sites at which the scalar field is not set to zero; $d\rho_x$ contains the factor $\exp[-\varepsilon^d(\frac{1}{2}(m^2 + \delta m^2)r^2 + \lambda r^4)]$ together with the diagonal parts $\frac{1}{2}\varepsilon^{d-2} r(x)^2$ of all the bonds at x, including the bonds leading out of C_J or to a site where the scalar field vanishes; and γ is the Gaussian measure of the vector field with N b.c., which is non-degenerate because $\mu_0 > 0$. A mass term $\sum_x \varepsilon^d \omega(x) r(x)^2$ with $\omega \geqq 0$ multiplies $d\rho_x$ by $e^{-\varepsilon^d \omega(x) r^2}$, and the action (3.38) and the periodic actions are of the same form. All these measures are invariant under $(A,\theta) \to (-A,-\theta)$, which changes the sign of $F(g)$ and fixes r; hence $\langle \mathcal{O}(ig,h)\rangle$ is real.

The Wick constant in (2.13) is that of the infinite lattice $\varepsilon\mathbb{Z}^d$, and δm^2, a_ε depend on ε only; so the c-number factor $\exp[-w\sum_x \varepsilon^d h(x)(2(-\Delta^\varepsilon + m^2)^{-1}(x,x) + a_\varepsilon)]$ of $\mathcal{O}(zg, wh)$ is the same for every expectation of this section on the same lattice, and the inequalities proved below for the expectations of $\exp[\pm r^2(h) + \cdots]$ carry over to $\mathcal{O}$. If the Wick constant depended on the region, the c-number factor would not be the same on the two sides of the inequalities; this is why the free constant is used throughout.

We now extend Proposition 3.1 to this class and prove the inequalities (3.27)–(3.29).

Lemma 6.1. *Let $\langle\cdot\rangle$ be a BFS measure, g a real function on the bonds, $h \geqq 0$ a function on the sites, u a complex function on the bonds such that the quadratic form of γ minus $|A(u)|^2$ is*

positive definite, and $\Theta = \exp[r^2(h) + iA(g)]$, $\Theta^- = \exp[-r^2(h) + A(g)]$, *with* $A(g)$, $r^2(h)$ *as in* (3.22), (3.23). *Then, with* $\langle X; Y\rangle = \langle XY\rangle - \langle X\rangle\langle Y\rangle$,

$$\langle\Theta; |A(u)|^2\rangle \leqq 0\,, \qquad \langle\Theta; r(x)^2\rangle \geqq 0\,, \qquad \langle\Theta; r(x)r(y)\cos(\theta(x) - \theta(y) - e\varepsilon A_{\langle x,y\rangle})\rangle \geqq 0\,, \tag{6.2}$$

and the three inequalities hold with the opposite signs for Θ^-. *Hence removing* $s|A(u)|^2$, $0 \leqq s \leqq 1$, *from the action does not increase* $\langle\Theta\rangle$ *and does not decrease* $\langle\Theta^-\rangle$, *and decreasing a coefficient* c_{xy}, *or multiplying some* ρ_x *by* e^{-sr^2}, *does not increase* $\langle\Theta\rangle$ *and does not decrease* $\langle\Theta^-\rangle$; *every intermediate measure is a BFS measure.*

Proof. Let μ be the unnormalized measure and $\omega' = (r', \theta', A')$ an independent copy of $\omega = (r, \theta, A)$. Then for any two functions F, G,

$$(\textstyle\int d\mu)^2\langle F; G\rangle = \frac{1}{2}\iint [F(\omega) - F(\omega')][G(\omega) - G(\omega')]\, d\mu(\omega)\, d\mu(\omega')\,. \tag{6.3}$$

We write $r = u - v$, $r' = u + v$, $\theta = \theta_+ + \theta_-$, $\theta' = \theta_+ - \theta_-$, $A = a_+ + a_-$, $A' = a_+ - a_-$. Then $a_\pm$ are independent centered Gaussian variables with the covariance of γ halved; the radial measure $\prod_x d\rho_x(u_x - v_x)\, d\rho_x(u_x + v_x)$ is supported on $\{u \geqq |v|\}$ and is invariant under $v_x \to -v_x$ at each site; and the integral over $(\theta, \theta') \in [0, 2\pi)^2$ at a site is the integral over (θ_+, θ_-) with the same measure, since the map $(\theta_+, \theta_-) \to (\theta, \theta')$ has Jacobian 2 and, taken modulo 2π, covers the torus exactly twice, the integrand being 2π-periodic in θ and θ'. Put $\chi_{xy} = \theta_{+,x} - \theta_{+,y} - e\varepsilon a_{+,\langle x,y\rangle}$ and $\psi_{xy} = \theta_{-,x} - \theta_{-,y} - e\varepsilon a_{-,\langle x,y\rangle}$. For the generators of the action we have

$$\begin{aligned} r_x r_y \cos(\chi + \psi) + r'_x r'_y \cos(\chi - \psi) &= 2(u_x u_y + v_x v_y)\cos\chi\cos\psi + 2(u_x v_y + v_x u_y)\sin\chi\sin\psi\,,\\ r_x r_y \cos(\chi + \psi) - r'_x r'_y \cos(\chi - \psi) &= -2(u_x v_y + v_x u_y)\cos\chi\cos\psi - 2(u_x u_y + v_x v_y)\sin\chi\sin\psi\,, \end{aligned} \tag{6.4}$$

and, for real g and u,

$$\begin{aligned} r_x^2 + r_x'^2 = 2(u_x^2 + v_x^2)\,, \qquad r_x^2 - r_x'^2 &= -4u_x v_x\,,\\ \cos A(g) + \cos A'(g) &= 2\cos a_+(g)\cos a_-(g)\,,\\ \cos A(g) - \cos A'(g) = -2\sin a_+(g)\sin a_-(g)\,, \qquad A(u)^2 - A'(u)^2 &= 4a_+(u)a_-(u)\,. \end{aligned} \tag{6.5}$$

We first prove the inequalities for Θ. Let $\mathscr{K}$ be the cone of non-negative combinations of the functions $(-1)^{w+s}\, m\, V(\theta_+, a_+)V(\theta_-, a_-)$, where m is a monomial in (u, v) of total degree w in the variables v_x and V is a product of factors $\cos\chi_b$, $\sin\chi_b$, $\cos a(g_i)$, $\sin a(g_i)$, exactly s of them sines. $\mathscr{K}$ is closed under multiplication, and by (6.4) and (6.5) $f(\omega) + f(\omega')$ and $f(\omega) - f(\omega')$ lie in $\mathscr{K}$ for each of the generators $f = r_x^2$, $f = r_x r_y \cos(\theta_x - \theta_y - e\varepsilon A_{\langle x,y\rangle})$, $f = \cos A(g)$. Hence $F \pm F' \in \mathscr{K}$ for every polynomial F in the generators with non-negative coefficients, by

$$F_1F_2 \mp F_1'F_2' = \tfrac{1}{2}[(F_1 \mp F_1')(F_2 + F_2') + (F_1 \pm F_1')(F_2 - F_2')]\,, \tag{6.6}$$

and the exponential of the action, $\exp\sum_b c_b[f_b + f_b']$, lies in the closure of $\mathscr{K}$, the series converging absolutely.

The integral of an element of $\mathscr{K}$ is non-negative: it vanishes unless w is even, by $v_x \to -v_x$, and unless s is even, since $(\theta, a) \to (-\theta, -a)$ gives $\int V = (-1)^s \int V$; when both are even the sign is $+$, the radial integral of m is non-negative since $u_x \geqq |v_x|$, and the integral over the angles and the Gaussian variables factorises into $(\int V)^2 \geqq 0$, the measures of the $+$ and $-$ variables being the same.

By (6.3), $\langle F; G\rangle \geqq 0$ for F, G in the closure of the cone of these polynomials. Taking $F = e^{r^2(h)}\cos A(g)$, $h \geqq 0$, which is such a limit, and $G = r_x^2$, the hopping term, or $\cos tA(u_j)$, gives the second and third inequalities of (6.2) and, with

$$|A(u)|^2 = \lim_{t\to 0}\; 2t^{-2}\sum_{j=1,2}[1 - \cos tA(u_j)]\,, \qquad u = u_1 + iu_2\,, \tag{6.7}$$

the first; $e^{iA(g)}$ is replaced by $\cos A(g)$ by the symmetry $(A,\theta)\to(-A,-\theta)$.

These are the Ginibre inequalities of [BFS 1] for the measures (6.1).

For Θ^- we use, in place of (6.3), the identity

$$(\int d\mu)^2\langle F;G\rangle = \iint F(\omega)\,[G(\omega)-G(\omega')]\,d\mu(\omega)\,d\mu(\omega')\,, \tag{6.8}$$

which holds for any two functions F, G, with $F=\Theta^-$ and $G=Y$, where

$$\Theta^-(\omega) = e^{a_+(g)}e^{a_-(g)}e^{-\sum_x h_x(u_x^2+v_x^2)}e^{2\sum_x h_x u_x v_x}\,. \tag{6.9}$$

We expand the exponential of the first line of (6.4), summed over the bonds, and $e^{2\sum_x h_x u_x v_x}$ in power series, which converge absolutely since $h\geqq 0$. Every term of the integrand is then a sign σ times a monomial in (u,v) with a non-negative coefficient, times the even radial factor $e^{-\sum h_x(u_x^2+v_x^2)}$, times $V(\theta_+,a_+)V(\theta_-,a_-)$ with one real function V, a product of $e^{a(g)}$, of factors $\cos\chi_{xy}$ or $\sin\chi_{xy}$, and of $a(u)$ when $Y=A(u)^2$.

The integral over a_+,a_- factorises, since they are independent with the same distribution, and so does the integral over the angles; both give the square $(\int V)^2\geqq 0$. The radial integral of a monomial vanishes if its degree in some v_x is odd, and is non-negative otherwise, since $u_x\geqq|v_x|$. So every term has the sign σ, which is -1 for $Y=r_x^2$ and for the hopping term (the second line of (6.4)), and $+1$ for $Y=A(u)^2$ with u real; for complex $u=u_1+iu_2$, $|A(u)|^2=A(u_1)^2+A(u_2)^2$ is the sum of two such terms. This gives the three inequalities for Θ^-, that is (3.27)–(3.29).

The monotonicity statements follow by differentiating in the parameter: removing $s|A(u)|^2$ leaves the Gaussian non-degenerate by the hypothesis on u, and every intermediate measure is a BFS measure.

Lemma 6.2. *Let $\langle\cdot\rangle$ be a BFS measure, U a set of sites with $h=0$ on U and Θ, Θ^- as above. Then $\langle\Theta\rangle$ does not increase, and $\langle\Theta^-\rangle$ does not decrease, when the scalar field is set to zero on U.*

Proof. Setting $c_{xy}=0$ on every bond at a site of U, which by Lemma 6.1, bond by bond, does not increase $\langle\Theta\rangle$ and does not decrease $\langle\Theta^-\rangle$, makes the measure a product of the measure at the sites of U, which cancels in the expectations, and of the measure in which the scalar field vanishes at the sites of U, the diagonal parts of the cut bonds being kept at their other ends. It is the step which, prepended to the proof of Theorem 3.2, gives that theorem for the cubes of Sect. 2 (Lemma 6.4 below). We shall also use the following form of that proof.

Lemma 6.3. *Let $\langle\cdot\rangle_Y$ be the expectation* (2.10) *on a finite set Y of sites of $\varepsilon\mathbb{Z}^d$, or the periodic expectation on a torus, in either case possibly with a mass term and with sites at which the scalar field vanishes, and let $X\subset Y$ be a set of sites such that, on a torus, X and its nearest neighbours lie in one fundamental cell. Let $\langle\cdot\rangle_X$ be the expectation* (2.10) *on X whose single-site measures are those of $\langle\cdot\rangle_Y$, the diagonal parts of the bonds leaving X included. If every plaquette on which g does not vanish has its corners in X and every site on which h does not vanish lies in X, then for $h\geqq 0$*

$$\langle\mathcal{O}(ig,h)\rangle_X \leqq \langle\mathcal{O}(ig,h)\rangle_Y\,, \qquad \langle\mathcal{O}(g,-h)\rangle_X \geqq \langle\mathcal{O}(g,-h)\rangle_Y\,. \tag{6.10}$$

Proof. As in the proof of Theorem 3.2: remove from $\langle\cdot\rangle_Y$ the gradient terms of the vector field coupling X to $Y\setminus X$ and the cosine parts of the hopping terms between X and $Y\setminus X$, keeping the diagonal parts $\frac{1}{2}\varepsilon^{d-2}(r(x)^2+r(y)^2)$, the vector field remaining non-degenerate because $\mu_0>0$. By Lemma 6.1 the first expectation does not increase and the second does not decrease, and the result is a product whose X-factor is $\langle\cdot\rangle_X$, the diagonal parts of the cut bonds being exactly the Dirichlet terms of X and the remaining gradient terms the N b.c. on X. The c-number factors of $\mathcal{O}$ are the same on both sides, by the convention on the Wick constant stated above.

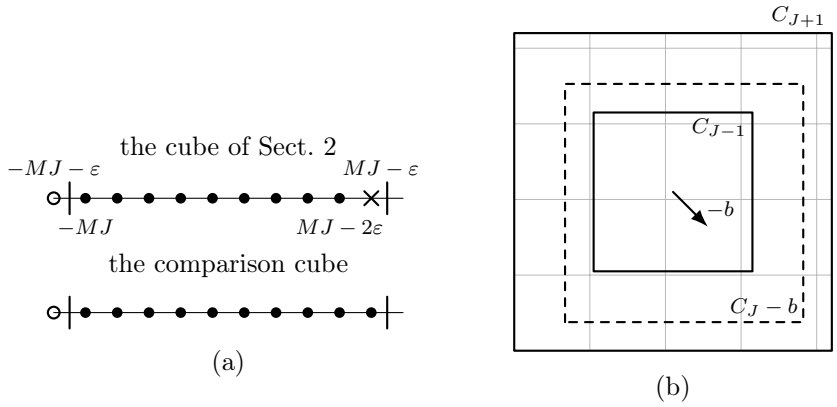


Fig. 2: (a) One direction of C_J in the two conventions of Sect. 6.1: in Sect. 2 the scalar field vanishes at $MJ-\varepsilon$ (cross) and $-MJ-\varepsilon$ (open circle), in the comparison cube at no site of C_J. (b) The cubes C_{J-1}, C_J-b and C_{J+1}, $|b|_\infty \leqq M$, on the block grid (Lemma 6.10)

Two conventions for the cube occur in this paper. Here and throughout, $C_J \cap \varepsilon\mathbb{Z}^d$ denotes the half-open cube $\{x \in \varepsilon\mathbb{Z}^d : -MJ \leqq x_\mu < MJ\}$ of (2.7), whose top layers are $x_\mu = MJ-\varepsilon$. In the expectation $\langle\cdot\rangle_{N,D}^{\varepsilon,C_J}$ of Sect. 2 the scalar field vanishes on the layers $x_\mu = -MJ-\varepsilon$ and $x_\mu = MJ-\varepsilon$, so that the two Dirichlet planes in each direction are $2MJ$ apart; this is the cube on which the renormalization transformation of Sect. 3.1 is carried out, and on which Theorem 2.1 is proved. In the proof of Theorem 3.2 and in the action (3.38) every site of $C_J \cap \varepsilon\mathbb{Z}^d$ carries a scalar variable; we denote this expectation by $\langle\cdot\rangle_{N,D,*}^{\varepsilon,C_J}$ and call it the comparison cube. The vector field is the same in both, with N b.c. on C_J; the action (3.38) has the scalar part of the comparison cube but differs in the quadratic form of the vector field at $\partial\Omega$, and the two are compared in Lemma 6.13.

With $Y = C_{J+1}$ and $X = C_J$, both in the convention of the comparison cube, Lemma 6.3 is the second step of the proof of Theorem 3.2, which compares the comparison cubes C_J and C_{J+1}. For the cubes of Sect. 2 we use Lemma 6.2 for the top layers of C_J, and then Lemma 6.3 with $X = C_J$ in the convention of the comparison cube and $Y = C_{J+1}$ in the convention of Sect. 2, whose top layers do not meet C_J; this is the chain (6.11) of Lemma 6.4, and it gives Theorem 3.2 for the cubes of Sect. 2. With $X = C_J$ and Y a torus containing C_J and its neighbours in one fundamental cell, and the mass term vanishing on C_J, it is the inequality between the comparison cube and the torus used in Sect. 2 and in Sect. 6.6, δm^2 and the Wick constant being defined with free boundary conditions for the torus as well as for the cubes.

Lemma 6.4. *For g real and $h \geqq 0$, both supported at distance greater than 1 from the complement of C_J,*

$$\langle\mathcal{O}(ig,h)\rangle_{N,D}^{\varepsilon,C_J} \leqq \langle\mathcal{O}(ig,h)\rangle_{N,D,*}^{\varepsilon,C_J} \leqq \langle\mathcal{O}(ig,h)\rangle_{N,D}^{\varepsilon,C_{J+1}}\,, \tag{6.11}$$

and the reversed inequalities hold for $\mathcal{O}(g,-h)$. Consequently both families are monotone in J, and for $h=0$ and z purely imaginary the limits $\lim_{K\to\infty}$ of the comparison cubes, along any subsequence on which they exist, lie between $\langle\mathcal{O}(zg,0)\rangle_{N,D}^{C_J}$ and $\langle\mathcal{O}(zg,0)\rangle_{N,D}^{C_{J+1}}$, the limits of Theorem 2.1, so that their limit as $J\to\infty$ is (2.18).

Proof. The first inequality is Lemma 6.2 with U the top layers of C_J, where h vanishes; the second is Lemma 6.3 with $Y = C_{J+1}$ in the convention of Sect. 2 and $X = C_J$, the top layers of C_{J+1} not meeting C_J, so that the X-factor is the comparison cube. The rest follows from Theorem 2.1, Theorem 3.2 and Lemma 6.5(i) below.

The limits of this paper do not depend on the choice of M among the multiples of 50 admitted in Sect. 2: two such values have a common multiple, which is again admitted; its cubes are cubes of both families, a subsequence of each, so the limits along the cubes of the

common multiple agree with those along the cubes of either.

Lemma 6.5. *For every expectation above, g_i real 2-forms, h_j real functions and $z \in \mathbb{C}^n$, $w \in \mathbb{C}^p$:*
(i)

$$\Big|\Big\langle \mathcal{O}\Big(\sum_i z_i g_i, \sum_j w_j h_j\Big)\Big\rangle\Big| \leqq \Big\langle \mathcal{O}\Big(\sum_i \mathrm{Re}\, z_i\, g_i, \sum_j \mathrm{Re}\, w_j\, h_j\Big)\Big\rangle\,, \tag{6.12}$$

and $\ln\langle \mathcal{O}(\sum_i x_i g_i, \sum_j y_j h_j)\rangle$ is a convex function of the real coefficients (x, y);
(ii) *for real sources,*

$$\langle \mathcal{O}(g,h)\rangle \langle \mathcal{O}(-g,-h)\rangle \geqq 1\,; \tag{6.13}$$

(iii) *for real g and $h_1, h_2 \geqq 0$,*

$$\langle \mathcal{O}(g, h_1 - h_2)\rangle \leqq \langle \mathcal{O}(2g, -2h_2)\rangle^{1/2} \langle \mathcal{O}(0, 2h_1)\rangle^{1/2}\,. \tag{6.14}$$

Proof. (i) is (3.49) and Hölder's inequality, the exponent of $\mathcal{O}$ being linear in the sources and the constant a_ε of (2.13) multiplying $\mathcal{O}$ by $\exp[-w\, a_\varepsilon \sum_x \varepsilon^d h(x)]$, a positive constant for real w; (ii) is Schwarz's inequality; (iii) is Schwarz's inequality with the constants distributed accordingly, as in (3.50).

Lemma 6.6. *Let P be a signed permutation matrix, so that $P\varepsilon\mathbb{Z}^d = \varepsilon\mathbb{Z}^d$, and for a 2-form g and a function h let Pg, Ph denote their transforms, $(Ph)(x) = h(P^{-1}x)$ and $(Pg)_x(u,v) = g_{P^{-1}x}(P^{-1}u, P^{-1}v)$. For a finite set X of sites let PX carry the transported boundary conditions, mass term and Dirichlet sites, the vector field of PX being attached to the image bonds Pb. Then*

$$\langle \mathcal{O}(zPg, wPh)\rangle_{PX} = \langle \mathcal{O}(zg, wh)\rangle_X\,, \tag{6.15}$$

and the same holds for the periodic expectations on a torus D and PD.

Proof. The map $\phi'(Px) = \phi(x)$, $A'(Pb) = \pm A(b)$ (the sign of the image bond) is a bijection of the configurations which preserves Lebesgue measure and carries the action, the plaquettes with their centers and the field strength into those of PX; a transposed pair of indices changes the sign of both $F(p)$ and $g(p)$.

For a permutation matrix and the cubes of Sect. 2 we have $PC_J = C_J$ and the Dirichlet layers are mapped onto the Dirichlet layers, so that the expectation of Sect. 2 is exactly invariant; for a reflection $x_\nu \to -x_\nu$ the site set $C_J \cap \varepsilon\mathbb{Z}^d$ is shifted by one layer, and the invariance of the limit is proved in Sect. 6.3 by a comparison with two neighbouring cubes.

We close this subsection with the two steps by which the comparison of the two models of Theorem 3.4 of [I], carried out on the cube C_J in Sect. 3.1, takes account of the averaging function (3.8), which is not constant but has average one on every block of side $L\eta$ of Sect. 3.1, η being the spacing of the lattice on which the comparison is made. The weight is removed first by the dipole identity

$$G_i(0)((f-1)\chi)(z) = \sum_c \sum_{\xi\in c} \eta^d (f(\xi)-1)\,[G_i(0)(z,\xi)\chi(\xi) - G_i(0)(z,\xi_c)\chi(\xi_c)]\,, \tag{6.16}$$

the sum over the blocks c of side $L\eta$ with corners ξ_c, which holds because $\sum_{\xi\in c}\eta^d(f(\xi)-1) = 0$ on every block.

The bracket is bounded by the first difference of $G_i(0)$ in the block times $|\chi|$, plus $|G_i(0)(z,\xi_c)|$ times the oscillation of χ in the block. Hence

$$|G_i(0)((f-1)\chi)(z)| \leqq C\eta \|\chi\|_*\,, \qquad |\partial_z G_i(0)((f-1)\chi)(z)| \leqq C\eta(1+\ln\eta^{-1})\|\chi\|_*\,, \tag{6.17}$$

where $\|\chi\|_* = \sup|\chi| + \sup_c \eta^{-1}\,\mathrm{osc}_c\,\chi$, the supremum over the blocks c; the logarithm arises only from the term containing $\sup|\chi|$.

These bounds rest on kernel bounds for the propagators $G_{(j)}$ of the scales $L^j\eta$ in the decomposition $G_i(0) = \sum_j G_{(j)}$ of [I],[10] with the Dirichlet planes on the top layers of Sect. 2 and the averaging (3.8): $|\nabla_\xi G_{(j)}(z,\xi)| \leqq C(L^j\eta)^{1-d}\exp[-\delta_0(L^j\eta)^{-1}|z-\xi|]$ and $|\nabla_z\nabla_\xi G_{(j)}(z,\xi)| \leqq C(L^j\eta)^{-d}\exp[-\delta_0(L^j\eta)^{-1}|z-\xi|]$. They bound the sum over ξ of $\eta^d|\nabla_\xi G_i(0)(z,\xi)|$, and the sums over the blocks c of $(L\eta)^d$ times the maximum over c of $|G_i(0)(z,\xi)|$ and of $|\nabla_\xi G_i(0)(z,\xi)|$, which give the first bound in (6.17) together with its oscillation term.

They bound, for each j, the sum over ξ of $\eta^d|\nabla_z\nabla_\xi G_{(j)}(z,\xi)|$ by C, whose sum over the $C\ln\eta^{-1}$ values of j gives the logarithm in the second. For $f = 1$ and free boundary conditions these are the bounds of [Ba 4] on the terms of the decomposition. The representation (3.9) expresses the operator with D b.c. on the top layers through the operator with free boundary conditions and the same weight f, so that the bounds needed here are those of [Ba 4] extended to the averaging operator (3.2) with the weight (3.8); for the single-scale operator this extension is made in Appendix A, and the terms of the decomposition are treated in the same way.

The same identity compares the weighted block averages of the two models block by block. The kernel bounds enter through the image representation (3.9), with the Dirichlet planes $x_\mu = -MJ - \varepsilon$ and $x_\mu = MJ - \varepsilon$ of Sect. 2. These are planes on which (3.8) vanishes, and $2MJ$ is a multiple of $L^k\varepsilon$.

Since Q_k is the average (3.2) over the block of side $L^k\varepsilon$ with the weight (3.8), the kernel of $Q_k^*Q_k$ vanishes unless both arguments lie in the support of f, and the reflection in either plane carries the support of f in each block onto the support of f in a block, leaving f unchanged: in each direction the support of f in a block of side $L^k\varepsilon$ consists of the sites at the offsets $0,\dots,L^k-2$ which are not congruent to $L-1$ modulo L, and the reflection in a top layer, which carries the offset j to the offset L^k-2-j of the mirror block, maps this set onto the corresponding set of the mirror block, f being constant on its support. So $P_k(0)$ commutes with the reflection, and (3.9) holds. If we had taken $f = 1$, or placed the corners of the blocks on the Dirichlet planes, the representation would not hold.

6.2. Regularity

The bounds of this subsection do not depend on the renormalization transformation; they rest on an integration over the scalar field at fixed vector field, which we now describe. Only the convergence in ε in Theorem 6.9(ii) uses Theorem 2.1. For a function v on the sites with values in $\mathbb{R}^d$ we write $A(v) = \sum_x \varepsilon^d v(x)\cdot A(x)$, with $A(x)$ as in (2.2); the field strength smeared with a 2-form g is of this form, $F(g) = A(v)$, with $v = \varepsilon^{-1}d^*g$ a lattice divergence of g. A current J on the bonds is regarded as such a function by (2.2), $\langle v, CJ\rangle$ being the corresponding pairing.

Lemma 6.7. *Let $\langle\cdot\rangle$ be a BFS measure* (6.1) *whose Gaussian measure γ has covariance C. Then for every real v and real s*

$$\langle e^{sA(v)}\rangle \leqq \exp[\tfrac{1}{2}s^2\langle v, Cv\rangle]\,, \qquad 0 \leqq \langle e^{iA(v)}\rangle \leqq 1\,. \tag{6.18}$$

Proof. We integrate over the scalar field at fixed A. Each factor of the hopping term is expanded in a Fourier series,

$$\exp[c_{xy}r(x)r(y)\cos(\theta(x)-\theta(y)-e\varepsilon A_{\langle x,y\rangle})] = \sum_n I_n(c_{xy}r(x)r(y))\, e^{in(\theta(x)-\theta(y)-e\varepsilon A_{\langle x,y\rangle})}\,, \tag{6.19}$$

where I_n is the modified Bessel function, $I_n(z) = I_{-n}(z) \geqq 0$ for $z \geqq 0$ and $\sum_n I_n(z) = e^z$, so that the series converges absolutely; since $e^{c_{xy}r(x)r(y)} \leqq e^{c_{xy}(r(x)^2+r(y)^2)/2}$ is cancelled by the diagonal parts of the hopping terms, $\lambda > 0$ controlling the rest, the series may be integrated term by term over the scalar field. The integral over the angles forces the integers to form a current J on the bonds with zero divergence (at a site where the scalar field vanishes the current

[10] (2.18) of [I].

vanishes on all its bonds), and the integral over the radial variables is positive. So the integral over the scalar field is $\sum_J w(J)\cos(e\varepsilon A(J))$ with $w(J) \geqq 0$, where $A(J) = \sum_b J_b A_b$.

The Gaussian integral of $e^{sA(v)}\cos(e\varepsilon A(J))$ equals

$$\exp[\tfrac{1}{2}s^2\langle v, Cv\rangle - \tfrac{1}{2}e^2\varepsilon^2\langle J, CJ\rangle]\cos(se\varepsilon\langle v, CJ\rangle)\,, \tag{6.20}$$

and dividing by the same sum at $s = 0$ gives the first bound, since $|\cos| \leqq 1$. For $e^{iA(v)}$ the same computation gives $e^{-\langle v,Cv\rangle/2}$ times a weighted average of $\cosh(e\varepsilon\langle v, CJ\rangle) \geqq 1$, which lies in $[0, 1]$ since the expectation is real and at most 1 in modulus.

Lemma 6.8. *Let C be the covariance of the vector field on C_J with N b.c., that is of the Gaussian measure of the expectations of Sect. 2 and of the comparison cube, and let g be a real 2-form all of whose plaquettes have their corners in $C_J \cap \varepsilon\mathbb{Z}^d$, $F(g) = A(v)$. Then*

$$\langle v, Cv\rangle \leqq \min\{2\|g\|^2,\ \mu_0^{-2}\|v\|^2\}\,, \qquad \|g\|^2 = \sum_p \varepsilon^d g(p)^2\,, \quad \|v\|^2 = \sum_x \varepsilon^d |v(x)|^2\,. \tag{6.21}$$

Proof. The Gaussian density is $e^{-Q(A)/2}$ with

$$Q(A) = \sum_{\mu,\nu}\sum \varepsilon^{d-2}(A_\mu(x + \varepsilon e_\nu) - A_\mu(x))^2 + \mu_0^2\|A\|^2\,, \tag{6.22}$$

the sum over the pairs $x, x + \varepsilon e_\nu$ in C_J, and $\langle v, Cv\rangle = \sup_A[2A(v) - Q(A)]$. Since $Q(A) \geqq \mu_0^2\|A\|^2$, the second bound follows.

For the first, let P_g be the set of plaquettes on which g does not vanish; for $p = (x; \mu < \nu)$ in P_g both gradient terms $(A_\nu(x + \varepsilon e_\mu) - A_\nu(x))^2$ and $(A_\mu(x + \varepsilon e_\nu) - A_\mu(x))^2$ occur in Q, and distinct plaquettes use distinct terms, so $\sum_{p\in P_g} \varepsilon^d F(p)^2 \leqq 2Q(A)$ by $(a - b)^2 \leqq 2a^2 + 2b^2$. Hence

$$\langle v, Cv\rangle \leqq \sup_A \Big[2\sum_{p\in P_g} \varepsilon^d g(p)F(p) - \tfrac{1}{2}\sum_{p\in P_g} \varepsilon^d F(p)^2\Big] \leqq 2\|g\|^2\,. \tag{6.23}$$

On a torus the constant is 1 in place of 2. Let $\partial_\mu f(x) = \varepsilon^{-1}(f(x + \varepsilon e_\mu) - f(x))$ and $\partial_\mu^* f(x) = \varepsilon^{-1}(f(x) - f(x - \varepsilon e_\mu))$ be the forward and backward lattice derivatives, so that $Q(A) = \sum_{\mu,\nu}\sum_x \varepsilon^d(\partial_\nu A_\mu)^2 + \mu_0^2\|A\|^2$. Summing by parts twice, $\sum_x \partial_\mu A_\nu\,\partial_\nu A_\mu = \sum_x \partial_\mu^* A_\mu\,\partial_\nu^* A_\nu$, and $\sum_x(\partial_\mu^* A_\mu)^2 = \sum_x(\partial_\mu A_\mu)^2$ by translation, so that

$$\sum_p \varepsilon^d F(p)^2 = \sum_{\mu\neq\nu}\sum_x \varepsilon^d(\partial_\mu A_\nu)^2 - \sum_x \varepsilon^d\Big(\sum_\mu \partial_\mu^* A_\mu\Big)^2 + \sum_\mu\sum_x \varepsilon^d(\partial_\mu A_\mu)^2 \leqq Q(A)\,. \tag{6.24}$$

For 2-forms in $C_0^\infty(\mathbb{R}^d)$ we write $\|g\|_2$ for the L^2 norm of the components and

$$\|g\|_4 = \sum_{\mu<\nu} \sup_{x,\,|\gamma|\leqq 4} (1 + |x|^2)^2 |D^\gamma g_{\mu\nu}(x)|\,, \tag{6.25}$$

a norm of the type $|\cdot|_m$ of [OS 2], applied to the components; since $\int(1 + |x|^2)^{-4}dx = \pi^2/8$ in $d = 3$,

$$\|g\|_2 \leqq (\pi/2\sqrt{2})\|g\|_4\,. \tag{6.26}$$

The lattice norms in (6.21) converge to $\|g\|_2$ and to $\|d^*g\|_2$ as $\varepsilon \to 0$, the components of $\varepsilon^{-1}d^*g$ being difference quotients of the sampled components of g, and are bounded by $C\|g\|_4$ uniformly in ε.

Theorem 6.9. *Let $g_1, \ldots, g_n$ be real 2-forms in $C_0^\infty(\mathbb{R}^d)$ supported at distance greater than 1 from the complement of C_{J_0}, $J \geqq J_0$, $z \in \mathbb{C}^n$, and $g(z) = \sum_i(\mathrm{Re}\, z_i)g_i$. Then:*

(i) *for all* ε *and* $J \geqq J_0$,

$$\Big|\Big\langle \mathcal{O}\Big(\sum_i z_i g_i, 0\Big)\Big\rangle_{N,D}^{\varepsilon,C_J}\Big| \leqq \exp\big[\min\{\|g(z)\|^2,\ \tfrac{1}{2}\mu_0^{-2}\|\varepsilon^{-1}d^*g(z)\|^2\}\big]\,, \tag{6.27}$$

and for a real 2*-form* u *all of whose plaquettes have their corners in* C_J

$$\langle F(u)\rangle_{N,D}^{\varepsilon,C_J} = 0\,, \qquad \langle F(u)^2\rangle_{N,D}^{\varepsilon,C_J} \leqq 2\|u\|^2\,; \tag{6.28}$$

(ii) *the limits* (2.17) *and* (2.18) *at* $h = 0$ *exist locally uniformly in* $z \in \mathbb{C}^n$ *and are entire functions of* z*; they satisfy* (6.27) *with the norms* $\|g(z)\|_2$, $\|d^*g(z)\|_2$*; for* z *purely imaginary* $\langle \mathcal{O}(zg,0)\rangle_{N,D}^{C_J}$ *is real, nondecreasing in* J *and lies in* $[0,1]$*; and* $\langle \mathcal{O}(g,0)\rangle \geqq 1$ *for real* g*;*

(iii) *the Schwinger functions* $S_n(g_1,\dots,g_n) = \partial_{z_1}\cdots\partial_{z_n}\langle \mathcal{O}(\sum_i z_i g_i, 0)\rangle|_{z=0}$ *satisfy, for real* g_i*,*

$$|S_n(g_1,\dots,g_n)| \leqq (\sqrt{2}\,e)^n (n!)^{1/2} \prod_{i=1}^n \|g_i\|_2 \leqq (e\pi/2)^n (n!)^{1/2} \prod_{i=1}^n \|g_i\|_4\,, \tag{6.29}$$

where e *here is the base of the natural logarithms and* $\|g\|_4$ *is the norm* (6.25)*;* S_n *extends to a continuous symmetric multilinear form on the Schwartz* 2*-forms, with the factor* 2^n *for complex test functions, and is therefore a tempered distribution on* $\mathbb{R}^{dn}$*. The bound* (6.27) *in the limit,* $|\langle \mathcal{O}(g,0)\rangle| \leqq \exp \|\operatorname{Re} g\|_2^2$ *for complex* g*, is the exponential bound of the regularity axiom of Sect. 6.1 of [GJ 1], with* $p = 2$*, and the linear growth condition (E0′) of [OS 2] holds;*

(iv) *for real* 2*-forms* u*,* u' *in* $C_0^\infty(\mathbb{R}^d)$*,*

$$|\langle \mathcal{O}(iu,0)\rangle - \langle \mathcal{O}(iu',0)\rangle| \leqq \sqrt{2}\,\|u - u'\|_2\,. \tag{6.30}$$

Proof. (i) Since the measure is positive, the modulus is at most $\langle e^{F(g(z))}\rangle$, and Lemma 6.7 with Lemma 6.8 gives (6.27); the expectation of Sect. 2 is a BFS measure whose Gaussian is that of Lemma 6.8, the sites where the scalar field vanishes carrying point masses. $\langle F(u)\rangle = 0$ by the symmetry $(A,\theta) \to (-A,-\theta)$, and the bound on $\langle F(u)^2\rangle$ follows from $e^x + e^{-x} \geqq 2 + x^2$ and (6.18) at $\pm s$, letting $s \to 0$.

(ii) At fixed J the functions $z \mapsto \langle \mathcal{O}(\sum z_i g_i, 0)\rangle_{N,D}^{\varepsilon,C_J}$ are entire and, by (i), bounded uniformly in ε on compact sets, since the lattice norm of $g(z)$ is bounded by $C\|g(z)\|_4$. By Theorem 2.1 they converge at every real point; $\mathbb{R}^n$ is a set of uniqueness for entire functions on $\mathbb{C}^n$, so by Montel's theorem and the identity theorem they converge locally uniformly on $\mathbb{C}^n$ to an entire function, and (6.27) passes to the limit.

For $z = it$, t real, the lattice expectations are real and of modulus at most 1 (Lemma 6.7), and nondecreasing in J by Theorem 3.2 and Lemma 6.4; so $J \mapsto \langle \mathcal{O}(it\cdot g, 0)\rangle_{N,D}^{C_J}$ converges, with values in $[0,1]$ by (6.18). The family of the limits at fixed J is bounded by (6.27), which does not depend on J, and converges on $(i\mathbb{R})^n$, a set of uniqueness; Montel's theorem gives the locally uniform convergence to an entire function as $J \to \infty$, which is (2.18) at $h = 0$. For real g, $\langle \mathcal{O}(g,0)\rangle \geqq 1$ by Jensen's inequality and $\langle F(g)\rangle = 0$, at every ε and J.

(iii) Normalize $\|g_i\|_2 = 1$ (if some $g_i = 0$ there is nothing to prove). On the polydisc $|z_i| \leqq r$ we have $\|g(z)\|_2 \leqq nr$, so the limit is bounded by $e^{n^2r^2}$ there, and Cauchy's estimate for the mixed derivative in the n variables gives

$$|S_n| \leqq r^{-n} e^{n^2r^2} = (2en)^{n/2} \leqq (\sqrt{2}\,e)^n (n!)^{1/2} \qquad \text{at } r^2 = 1/(2n)\,, \tag{6.31}$$

since $n^n \leqq e^n n!$. The second bound in (6.29) is $\|g\|_2 \leqq (\pi/2\sqrt{2})\|g\|_4$. The continuity in the L^2 norms gives the extension to the Schwartz 2-forms by density, and the multilinearity follows from that of the Taylor coefficients; for complex test functions each of the n arguments is split into real and imaginary parts. A continuous multilinear form on the Schwartz space is a tempered distribution on $\mathbb{R}^{dn}$ by the kernel theorem.

(iv) Let J_0 be such that u, u' are supported at distance greater than 1 from the complement of C_{J_0}. For $J \geqq J_0$, $|e^{ix} - e^{iy}| \leqq |x-y|$ and (i) give, at every ε,

$$|\langle e^{iF(u)}\rangle - \langle e^{iF(u')}\rangle| \leqq \langle F(u-u')^2\rangle^{1/2} \leqq \sqrt{2}\,\|u-u'\|\,, \tag{6.32}$$

and both sides converge.

The constants in (6.27) and (6.29) do not depend on e, λ, m or μ_0; the second bound in (6.27) is the only one which carries μ_0^{-2}.

The bound (6.29) gives also the linear growth condition of [OS 2]. Let $H_i = -\Delta_{x_i} + |x_i|^2$ act on the i-th variable of $F \in \mathcal{S}(\mathbb{R}^{dn})$, the components of the 2-forms being treated separately, and expand $F = \sum_\alpha c_\alpha h_{\alpha_1} \otimes \cdots \otimes h_{\alpha_n}$ in the Hermite functions h_α, $\alpha \in \mathbb{N}^d$, the eigenfunctions of $-\Delta + |x|^2$ on $\mathbb{R}^d$ with the eigenvalues $\lambda_\alpha = 2|\alpha| + d$ (Sect. 6.5), $H_i h_{\alpha_i} = \lambda_{\alpha_i} h_{\alpha_i}$; the expansion converges in $\mathcal{S}(\mathbb{R}^{dn})$, on which S_n is continuous. By (6.29) and Schwarz's inequality with the weights $\prod_i \lambda_{\alpha_i}^{\pm t}$, t an integer with $2t > d$,

$$|S_n(F)| \leqq (\sqrt{2}\,e)^n (n!)^{1/2} \sum_\alpha |c_\alpha| \leqq C^n (n!)^{1/2} \Big\| \prod_{i=1}^n H_i^t F \Big\|_{L^2(\mathbb{R}^{dn})}\,, \tag{6.33}$$

where C depends only on t and d, since $\sum_\alpha \lambda_\alpha^{-2t}$ converges for $2t > d$ and the $\binom{d}{2}$ components of the forms are summed; and the last norm is bounded by C^n times a Schwartz norm of F of order $2tn$ in the derivatives and $(2t+d)n$ in the polynomial weights, since $\|G\|_{L^2(\mathbb{R}^{dn})} \leqq c^n \sup_x \prod_i (1+|x_i|^2)^{d/2} |G(x)|$. This is the linear growth condition, with the index of the norm linear in n and a constant of factorial growth; we do not use it below.

6.3. Translations and the Cubic Group

No cube is invariant under a translation or a reflection, and the invariance of the limit (2.18) is obtained from a comparison of the translated cube with a smaller and a larger cube of the same family, in the direction of the correlation inequalities, followed by the limits $K \to \infty$ and $J \to \infty$. We denote by $T_b f$ the translate of f by b, $(T_b f)(x) = f(x-b)$.

We recall the convention of Sect. 2 (Fig. 2). The sites of C_J on the lattice $\varepsilon_K \mathbb{Z}^d$ are those with $-MJ \leqq x_\mu < MJ$; the scalar field is set to zero on the top layers $x_\mu = MJ - \varepsilon$ and, through the diagonal parts of the crossing bonds, on the layers $x_\mu = -MJ - \varepsilon$ outside, so that the sites where it is not set to zero are those with $-MJ \leqq x_\mu \leqq MJ - 2\varepsilon$; the vector field has N b.c. on all faces.

Lemma 6.10. *Let g be real and $h \geqq 0$, both supported at distance greater than 1 from the complement of C_{J_0}, and let $b \in \varepsilon_{K_0}\mathbb{Z}^d$ with $|b|_\infty \leqq M$. Then for $K \geqq K_0$ and $J \geqq J_0 + 1$*

$$\langle \mathcal{O}(ig,h)\rangle_{N,D}^{\varepsilon_K, C_{J-1}} \leqq \langle \mathcal{O}(iT_b g, T_b h)\rangle_{N,D}^{\varepsilon_K, C_J} \leqq \langle \mathcal{O}(ig,h)\rangle_{N,D}^{\varepsilon_K, C_{J+1}}\,, \tag{6.34}$$

and the reversed inequalities hold for $\mathcal{O}(g,-h)$.

Proof. By translation covariance the middle term is the expectation of $\mathcal{O}(ig,h)$ in the cube $C_J - b$, with the scalar field vanishing on its top layers. The Wick constant and the counterterm in $\mathcal{O}(ig,h)$ are those of Sect. 2, independent of the cube, so the three terms are expectations of one and the same function of the fields.

The sites of $C_J - b$ lie among those of C_{J+1}, and the sites where its scalar field is not set to zero among those where the scalar field of C_{J+1} is not set to zero; likewise the sites of C_{J-1} lie among those of $C_J - b$, and the sites where the scalar field of C_{J-1} is not set to zero among those where the scalar field of $C_J - b$ is not set to zero. All four inclusions hold since $|b_\mu| \leqq M$, with equality allowed: for $b_\mu = M$ the top layer of $C_J - b$ is the top layer of C_{J-1} and its outside layer that of C_{J+1}, and for $b_\mu = -M$ the top layer of $C_J - b$ is that of C_{J+1} and its outside layer that of C_{J-1}.

Starting from the expectation on C_{J+1} we set the scalar field to zero on the sites of C_{J+1} outside the sites of $C_J - b$ on which the scalar field is not set to zero (Lemma 6.2) and remove the couplings between $C_J - b$ and the rest (Lemma 6.3); the result is the middle term, and the first expectation does not increase. Starting from the middle term and doing the same for $C_{J-1} \subset C_J - b$ gives the left inequality. The sources lie inside C_{J-1}, at distance greater than 1 from its complement, so that they are not affected by any of these operations. For $\mathcal{O}(g, -h)$ each of the same operations does not decrease the expectation (Lemmas 6.2, 6.3), which gives the reversed inequalities.

Theorem 6.11. *Let $g_1, \dots, g_n$ be real 2-forms in $C_0^\infty(\mathbb{R}^d)$, supported at distance greater than 1 from the complement of C_{J_0}. Then for every $b \in \mathbb{R}^d$ and every $z \in \mathbb{C}^n$,*

$$\Big\langle \mathcal{O}\Big(\sum_i z_i T_b g_i, 0\Big)\Big\rangle = \Big\langle \mathcal{O}\Big(\sum_i z_i g_i, 0\Big)\Big\rangle . \tag{6.35}$$

In particular the Schwinger functions of Theorem 6.9 are translation invariant.

Proof. For b as in Lemma 6.10 and z purely imaginary we let $K \to \infty$ and then $J \to \infty$ in (6.34); the three terms converge by Theorem 6.9(ii), the middle one for the sources $T_b g_i$, which are supported at distance greater than 1 from the complement of C_{J_0+1} since $|b|_\infty \leqq M$, and the two ends have the limit (2.18). The difference of the two entire functions $z \mapsto \langle \mathcal{O}(\sum z_i T_b g_i, 0)\rangle$ and $z \mapsto \langle \mathcal{O}(\sum z_i g_i, 0)\rangle$ vanishes on $(i\mathbb{R})^n$, hence everywhere. Composing translations gives all $b \in \bigcup_K \varepsilon_K \mathbb{Z}^d$.

For arbitrary b we choose $b_n \to b$ in this dense set; then $T_{b_n} u \to T_b u$ in L^2 for every real 2-form u, and the bound (6.30) of Theorem 6.9(iv) gives $\langle \mathcal{O}(iT_b u, 0)\rangle = \lim_n \langle \mathcal{O}(iT_{b_n} u, 0)\rangle = \langle \mathcal{O}(iu, 0)\rangle$; the identity theorem extends this from z purely imaginary to all z.

This proves (2.19) for $h = 0$; the case $h \neq 0$ is treated in Sect. 6.6. The uniqueness of the limit (3.54) does not give the invariance by itself, since a translation moves the cubes; the inequalities (6.34) replace it.

Theorem 6.12. *Let $g_1, \dots, g_n$ be real 2-forms in $C_0^\infty(\mathbb{R}^d)$ supported at distance greater than 1 from the complement of C_{J_0}, let P be a signed permutation matrix and $b \in \mathbb{R}^d$, and let Pg be the transform of the 2-form g as in Lemma 6.6. Then for all $z \in \mathbb{C}^n$*

$$\Big\langle \mathcal{O}\Big(\sum_i z_i T_b P g_i, 0\Big)\Big\rangle = \Big\langle \mathcal{O}\Big(\sum_i z_i g_i, 0\Big)\Big\rangle . \tag{6.36}$$

The limit (2.18) at $h = 0$ is therefore invariant under the cubic group about every point, and the Schwinger functions are covariant under it, F transforming as a 2-form.

Proof. For a permutation matrix the lattice expectations of Sect. 2 are exactly invariant (Lemma 6.6), hence so is the limit. Let σ be the reflection $x_3 \to -x_3$. By Lemma 6.6 the expectation of $\mathcal{O}(i\sigma g, 0)$ in C_J equals the expectation of $\mathcal{O}(ig, 0)$ in the reflected cube, whose sites are those of C_J shifted by one layer in the third direction and whose Dirichlet layer in that direction is the bottom layer $x_3 = -MJ + \varepsilon$, the layers in the other two directions being unchanged. For $J \geqq J_0 + 1$ and $K \geqq 1$, so that $2\varepsilon_K \leqq M$, the sites of this cube on which the scalar field is not set to zero contain those of C_{J-1} and are contained in those of C_{J+1}, and the sites of C_{J-1} lie among those of the reflected cube and its sites among those of C_{J+1}; the argument of Lemma 6.10 gives

$$\langle \mathcal{O}(ig, 0)\rangle_{N,D}^{\varepsilon_K, C_{J-1}} \leqq \langle \mathcal{O}(i\sigma g, 0)\rangle_{N,D}^{\varepsilon_K, C_J} \leqq \langle \mathcal{O}(ig, 0)\rangle_{N,D}^{\varepsilon_K, C_{J+1}} . \tag{6.37}$$

Letting $K \to \infty$ and $J \to \infty$ as in the proof of Theorem 6.11 gives the invariance under σ on $(i\mathbb{R})^n$, hence everywhere. The other reflections are conjugates of σ by permutations, every element of the cubic group is a product of permutations and reflections, and the cubic group

about b is the conjugate by T_b of the cubic group about the origin, so Theorem 6.11 completes the proof.

The full rotation group will be treated in Sect. 6.7; nothing in this subsection depends on it.

6.4. Reflection Positivity in the Limit

We first justify the two inequalities (3.52). Let Ω be a box of sites, Ω^* the set of bonds with both ends in Ω and Ω^{**} the set of plaquettes with all corners in Ω, as in (3.38), and let $\langle\cdot\rangle^{\varepsilon,\Omega}_{N,D,*}$ be the expectation (2.10) on Ω with a scalar variable at every site, as in Sect. 6.1; for $\Omega = C_J$ this is the comparison cube. The measure (3.38) is taken throughout this section with the site-independent counterterm δm^2 of (2.10) and the free Wick constant of (2.10) at every site of Ω. We measure lengths in lattice units and write $\mu = \mu_0\varepsilon$ for the mass in these units.

For a positive definite quadratic form M of the vector field on a set of bonds containing Ω^* we write Σ_M for the covariance of $(F(p))_{p\in\Omega^{**}}$ under the Gaussian measure with density $e^{-\langle A,MA\rangle/2}$; it is given by the variational formula

$$\langle v, \Sigma_M v\rangle = \sup_A \Big[2\sum_{p\in\Omega^{**}} v_p F(p) - \langle A, MA\rangle\Big], \tag{6.38}$$

the supremum over the vector fields on the bonds of M. We write Σ_Ω for the form of (3.38) on Ω, $\Sigma_{N,D,*}$ for the form of (2.10) on Ω and Σ_B for the form of (2.10) on a box $B \supset \Omega$.

Lemma 6.13. *(i) For every vector field A on Ω^*,*

$$\sum_{p\in\Omega^{**}} F(p)^2 + \sum_{x\in\Omega}\Big|\sum_{y\in x^*}\Omega(y)A(\langle x,y\rangle)\Big|^2 = \sum_{\mu,\nu}\sum|A_\mu(x+e_\nu)-A_\mu(x)|^2 + \sum|A_b|^2, \tag{6.39}$$

where the first sum on the right runs over the pairs of bonds (x,μ), $(x+e_\nu,\mu)$ both in Ω^* and the second over the bonds $b=(x,\mu)\in\Omega^*$ for which $(x+e_\mu,\mu)$ or $(x-e_\mu,\mu)$ is not in Ω^*, once for each such neighbour. Thus the quadratic form of (3.38) *is the form of* (2.10) *on the bonds of Ω^*, together with a positive semidefinite form in the variables of the bonds of Ω^* with an end on a face of Ω.*

(ii) The covariance of the field strength in the measure (3.38) *is dominated by its covariance in the measure $\langle\cdot\rangle^{\varepsilon,\Omega}_{N,D,*}$ at every e; at $e=0$ these are the Gaussian covariances Σ_Ω, $\Sigma_{N,D,*}$, and the difference is not zero: if x_0 is the corner of Ω with the smallest coordinates, p_0 a plaquette of Ω^{**} with corner x_0, d the exterior derivative on Ω^* and e_{p_0} the indicator of p_0, then for $g=(dd^*+\mu^2)e_{p_0}$ the variance of $F(g)$ in the latter measure is strictly larger than in* (3.38). *Consequently, for g real and $h\geqq 0$,*

$$\langle\mathcal{O}(ig,h)\rangle^{\varepsilon,\Omega}_{N,D,*} \leqq \langle\mathcal{O}(ig,h)\rangle^{\varepsilon}_{\Omega}, \tag{6.40}$$

and the reverse inequality fails at $e=0$ for suitable g.

Proof. (i) Extend A by zero to a vector field a on $\mathbb{Z}^d$. For finitely supported a the lattice Hodge identity gives

$$\sum_p |da(p)|^2 + \sum_x |d^*a(x)|^2 = \sum_{\mu,\nu}\sum_x |a_\mu(x+e_\nu)-a_\mu(x)|^2. \tag{6.41}$$

On the left, $d^*a(x)$ is the divergence in (3.38) for $x\in\Omega$ and 0 outside, and a plaquette outside Ω^{**} which meets Ω^* contains exactly one bond of Ω^*, so it contributes $|A_b|^2$; on the right the pairs with both bonds in Ω^* give the first sum of (6.39), a pair with one bond in Ω^* and $\nu\neq\mu$ corresponds to a plaquette outside Ω^{**} and cancels against it, and the pairs with $\nu=\mu$ are the second sum.

(ii) Write the quadratic form of (2.10) on Ω in blocks, the variables of Ω^* first and the bonds leaving Ω second, and call its first block P. Since Ω is a box, a pair (x,ν), $(x+e_\mu,\nu)$ with $\mu\neq\nu$ and $x, x+e_\mu\in\Omega$ has both bonds in Ω^* or both leaving Ω, so the pairs of (2.10) with exactly one bond leaving Ω are the pairs (x,μ), $(x+e_\mu,\mu)$ with $(x+e_\mu,\mu)$ leaving. By (6.39), P is the first sum of (6.39) plus the squares $|A_b|^2$ with $(x+e_\mu,\mu)\notin\Omega^*$, plus the mass term, so that the form M_Ω of (3.38) is P plus the squares with $(x-e_\mu,\mu)\notin\Omega^*$. The matter does not couple to the bonds leaving Ω, so these variables may be integrated in (2.10); this replaces the form of (2.10) by its Schur complement on Ω^*, which is at most P, hence at most M_Ω, the difference being a positive semidefinite form, that is a sum of squares of linear forms in A.

Adding these squares to the action one at a time does not decrease $\langle\mathcal{O}(ig,h)\rangle$, by Lemma 6.1, which holds for every nondegenerate Gaussian measure of the vector field, and the matter and the counterterms are the same in both measures; this gives (6.40). Since F depends on the variables of Ω^* only, its covariance in (2.10) at $e=0$, the inverse of the Schur complement, dominates that in (3.38): $\Sigma_\Omega \leqq \Sigma_{N,D,*}$. For general e the domination of the covariances of F follows from (6.40): for g real and t real both sides of (6.40) at itg and $h=0$ are even in t by the symmetry $(A,\phi)\to(-A,\bar\phi)$, equal to 1 at $t=0$, and their coefficients of t^2 are $-\frac12$ the variances; hence the variance of $F(g)$ in $\langle\cdot\rangle^{\varepsilon,\Omega}_{N,D,*}$ is at least that in (3.38).

For the strict inequality let x_0 be the corner of Ω with the smallest coordinates and p_0 a plaquette of Ω^{**} with corner x_0; let d and e_{p_0} be as in the statement. Put

$$A = d^* e_{p_0}\,, \qquad v = (dd^*+\mu^2)e_{p_0}\,, \tag{6.42}$$

so that A equals ±1 on the four bonds of p_0 and 0 elsewhere, and has zero divergence. Then $M_\Omega A = d^*v$, so A is the maximiser in (6.38) for $M=M_\Omega$. Using A, extended by zero to the bonds leaving Ω, as a trial field in (6.38) for the form of (2.10) gives

$$\langle v,\Sigma_{N,D,*}v\rangle \geqq 2\langle v,dA\rangle - \langle A,PA\rangle = \langle v,\Sigma_\Omega v\rangle + \sum_{b\in p_0}|A_b|^2\,, \tag{6.43}$$

since every bond of p_0 has its backward neighbour outside Ω^*, x_0 minimizing the coordinates in the two directions of p_0, so that the sum equals 4. At $e=0$ the expectation of $e^{iF(g)}$ is $\exp[-\frac12\operatorname{Var}F(g)]$ in either measure, the matter factor being common, which gives the last statement with $g=v$.

On one and the same box the comparison of (3.38) with the measure (2.10) has, for the class $\mathcal{O}(ig,h)$, the direction of Lemma 6.13(ii), so that the right inequality of (3.52) cannot be obtained from it; this is why (3.52) carries the margin $2n$. The measure (3.38) is nevertheless comparable with the measures (2.10) on cubes of different sizes, which is what (3.52) requires (Fig. 3).

Lemma 6.14. *Let g be real and $h\geqq 0$. (i) If $C\subset\Omega$ are boxes such that C and one layer around it lie in Ω, and g, h vanish outside C (every plaquette of g with its corners in C), then*

$$\langle\mathcal{O}(ig,h)\rangle^{\varepsilon,C}_{N,D,*} \leqq \langle\mathcal{O}(ig,h)\rangle^{\varepsilon}_{\Omega}\,. \tag{6.44}$$

(ii) If $\Omega\subset B$ are boxes such that B contains every site within N lattice steps of Ω, with $N\varepsilon\geqq\pi\sqrt{d}/(2\mu_0)$, and g, h vanish outside Ω, then

$$\langle\mathcal{O}(ig,h)\rangle^{\varepsilon}_{\Omega} \leqq \langle\mathcal{O}(ig,h)\rangle^{\varepsilon,B}_{N,D,*}\,. \tag{6.45}$$

Proof. (i) Starting from the measure (3.38) on Ω we remove the hopping terms across ∂C and outside C (Lemma 6.1); the matter of $\Omega\setminus C$ decouples and cancels, and the matter of C is that of the comparison cube on C, coupled to the vector field through the bonds of C only. By (6.39) the quadratic form of (3.38) contains every gradient term of the form of (2.10) on C, the bonds of C being inside Ω^*, and the remaining terms are non-negative; so it is the form of

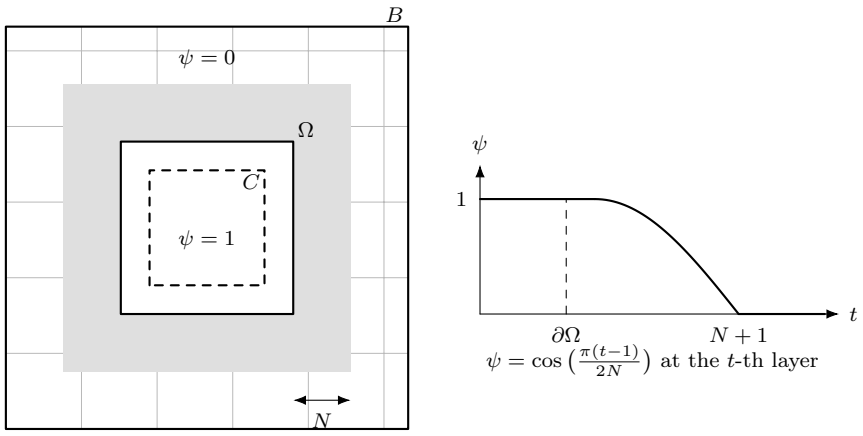


Fig. 3: The box Ω of (3.38) with the cube C of Lemma 6.14(i) inside it and the cube B of Lemma 6.14(ii) around it, the N layers of B around Ω shaded; on the right the cut-off ψ on these layers. In (3.52) the box is Ω_J and the cubes are C_J and C_{J+2n}

the comparison cube on C, plus a Gaussian in the other variables, plus a positive semidefinite form, and adding the latter does not decrease the expectation (Lemma 6.1).

(ii) Let M_B be the quadratic form of the vector field of the measure (2.10) on B and M_Ω that of (3.38) on Ω, with the covariances Σ_B, Σ_Ω of $(F(p))_{p\in\Omega^{**}}$ given by (6.38). We first show that $\Sigma_B \leqq \Sigma_\Omega$.

Given A on the bonds of B, let A' be its restriction to Ω^*, and let ψ be the cut-off which equals 1 on Ω and the layer around it, $\cos(\pi(t-1)/2N)$ at the t-th layer outside for $1 \leqq t \leqq N+1$ and 0 beyond, in each direction, $\psi(x) = \prod_\nu w(x_\nu)$, w the function of one variable just described. By the Hodge identity (6.41), applied to ψA on $\mathbb{Z}^d$,

$$\sum_{\Omega^{**}} F(A')^2 \leqq \sum_{\mu,\nu} \sum_x |(\psi A)_\mu(x+e_\nu) - (\psi A)_\mu(x)|^2 \,, \tag{6.46}$$

and the identity

$$\begin{aligned} \sum_x |\psi(x+e)f(x+e) - \psi(x)f(x)|^2 = {} & \sum_x \psi(x)\psi(x+e)|f(x+e)-f(x)|^2 \\ & + \sum_x \psi(x)\big(2\psi(x) - \psi(x+e) - \psi(x-e)\big)f(x)^2 \end{aligned} \tag{6.47}$$

bounds the right side of (6.46) by the gradient form of B plus $V(x)|A(x)|^2$ summed over $B \setminus \Omega$, with $V \leqq d(\pi/2N)^2 \leqq \mu^2$ by the choice of ψ. Hence

$$\sum_{\Omega^{**}} F(A')^2 + \mu^2 \sum_{\Omega^*} |A'|^2 \leqq \langle A, M_B A\rangle \,. \tag{6.48}$$

Let A'' be the projection of A' onto the divergence-free fields of Ω^*, the orthogonal complement of the gradients of functions on Ω; $F(A'') = F(A')$ on Ω^{**}, the gauge-fixing term vanishes at A'' and $\|A''\| \leqq \|A'\|$, so $\langle A'', M_\Omega A''\rangle \leqq \langle A, M_B A\rangle$ by (6.48). Taking the supremum in (6.38), $\Sigma_B \leqq \Sigma_\Omega$.

Remove now from the measure on B the hopping terms across $\partial\Omega$ and outside Ω; by Lemma 6.1 this does not increase the expectation. The matter of $B \setminus \Omega$ cancels, the matter of Ω is then that of (3.38) on Ω, every bond leaving Ω ending in B since $N \geqq 1$, and the matter of Ω and the sources depend on A only through gauge-invariant functions of A on Ω^*, which on a box are functions of F on Ω^{**}. The expectation is therefore a functional of the law of F, which is Gaussian with covariance Σ_B.

We decompose the 1-forms on Ω^* as $\ker d \oplus \operatorname{ran} d^*$ and the 2-forms on Ω^{**} as $\operatorname{ran} d \oplus \ker d^*$; d maps $\operatorname{ran} d^*$ bijectively onto $\operatorname{ran} d$, and Σ_Ω, Σ_B vanish on $\ker d^*$, the range of $F = dA$ lying in $\operatorname{ran} d = (\ker d^*)^\perp$, and are positive definite on $\operatorname{ran} d$, since $\Sigma_B = dCd^*$ with $C > 0$ the covariance of A'; here and below Σ^{-1} denotes the inverse on $\operatorname{ran} d$. The form M_Ω is block diagonal, with the block $d^*d + \mu^2 = d^*\Sigma_\Omega^{-1}d$ on $\operatorname{ran} d^*$. Let Q have the same block as M_Ω on $\ker d$ and the block $d^*\Sigma_B^{-1}d$ on $\operatorname{ran} d^*$; the covariance of F under the Gaussian with form Q is Σ_B, and

$$Q - M_\Omega = d^*(\Sigma_B^{-1} - \Sigma_\Omega^{-1})d \geqq 0 \tag{6.49}$$

is a sum of squares of linear forms in A, since $\Sigma_B \leqq \Sigma_\Omega$ on their common range $\operatorname{ran} d$ gives $\Sigma_B^{-1} \geqq \Sigma_\Omega^{-1}$ there. Adding these squares to the action (3.38) does not decrease $\langle \mathcal{O}(ig, h)\rangle$, by Lemma 6.1, and the measure so obtained gives F the law of covariance Σ_B, that is the expectation of the measure on B with the hopping terms removed. This proves (6.45).

Both inequalities hold also with the expectation of Sect. 2 in place of the comparison cube: (i) by Lemma 6.4; (ii) because the two measures on B differ only in the scalar field on the top layers of B, at distance at least $N \geqq 1$ from Ω, and the proof removes all hopping terms outside Ω.

The margin in (3.52) is obtained from (ii) with $\Omega = \Omega_J$ and $B = C_{J+2n}$, and from (i) with $C = C_J$: it suffices that C_{J+2n} contain every site within N lattice steps of Ω_J, $N\varepsilon \geqq \pi\sqrt{d}/(2\mu_0)$, and that C_J with one layer around it lie in Ω_J; both hold when $Mn \geqq |c| + R$ with

$$R = 2 + \pi\sqrt{d}/(2\mu_0)\,, \tag{6.50}$$

which is the constant R of Sect. 3.3. Here N is the smallest integer with $N\varepsilon \geqq \pi\sqrt{d}/(2\mu_0)$, so that $N\varepsilon < \pi\sqrt{d}/(2\mu_0) + 1$ for $\varepsilon \leqq 1$, and Ω_J exceeds C_J by at least one layer of sites on every side (Sect. 3.3).

Theorem 6.15. *Let $\nu \in \{1, \dots, d\}$, $c \in \mathbb{R}$, and let $g_{i1}, \dots, g_{il}$, $i = 1, \dots, N$, be real 2-forms supported in $\{x_\nu > c\}$, with complex coefficients z_{il}. Denote by θ the reflection in the plane $x_\nu = c$ acting on 2-forms as in Theorem 2.3, by the pull-back under the reflection, the components $g_{\nu\mu}$ changing sign, so that $F(g)$ at the reflected vector field equals $F(\theta g)$. Then:*

(i) *for all $\gamma \in \mathbb{C}^N$,*

$$\sum_{i,j=1}^{N} \bar\gamma_i \gamma_j \Big\langle \mathcal{O}\Big(\sum_l \bar z_{il}\, \theta g_{il} + \sum_l z_{jl}\, g_{jl},\, 0\Big)\Big\rangle \geqq 0\,; \tag{6.51}$$

(ii) *the Schwinger functions are symmetric multilinear, real for real test functions, and vanish for odd n;*

(iii) *for every polynomial $P = \sum_\alpha c_\alpha \prod_i F(g_{\alpha i})$ in the smeared field strengths with real $g_{\alpha i}$ supported in $\{x_\nu > c\}$, and $\theta P = \sum_\alpha \bar c_\alpha \prod_i F(\theta g_{\alpha i})$, the Schwinger function of $\theta(P)P$ is non-negative.*

Proof. (i) Fix first $c = c_0$ a half-odd multiple of some ε_{K_0}, and the boxes Ω_J of Sect. 3.3, symmetric about $\{x_\nu = c_0\}$; since L is odd, c_0 is a half-odd multiple of every ε_K, $K \geqq K_0$. Let $\delta > 0$ be such that the g_{il} vanish on $\{x_\nu < c_0 + \delta\}$, and let $\varepsilon_K < 2\delta$, so that every sampled plaquette has its corners in Ω_+. By Proposition 3.3 the Hermitian form $(\mathscr{F}, \mathscr{F}') \mapsto \langle \theta(\mathscr{F}')\mathscr{F}\rangle^\varepsilon_{\Omega_J}$ is positive semidefinite on the gauge-invariant functionals supported in Ω_+, and for $\mathscr{F}_i = \mathcal{O}(\sum_l z_{il} g_{il}, 0)$ we have $\theta(\mathscr{F}_i)\mathscr{F}_j = \mathcal{O}(\sum_l \bar z_{il}\theta g_{il} + \sum_l z_{jl} g_{jl}, 0)$, since $F(g)$ composed with the reflection of the fields is $F(\theta g)$; so (6.51) holds for the measure (3.38) on Ω_J at every such ε_K.

By Lemma 6.5(i) and Lemma 6.7 applied to the measure (3.38),

$$|\langle \mathcal{O}(\zeta, 0)\rangle^\varepsilon_{\Omega_J}| \leqq \langle e^{F(g_{\mathrm{Re}\,\zeta})}\rangle^\varepsilon_{\Omega_J} \leqq \exp\Big[\tfrac12 \sum_p \varepsilon^d |g_{\mathrm{Re}\,\zeta}(p)|^2\Big]\,, \tag{6.52}$$

since the quadratic form of (3.38) dominates $\sum_{p\in\Omega^{**}} \varepsilon^d F(p)^2$; so these expectations are bounded uniformly in K and J on compact sets of the coefficients, and on a subsequence they converge, locally uniformly, to an entire function Φ^J of the coefficients (Montel's theorem). Positivity of a Hermitian matrix is preserved under limits, so (6.51) holds for Φ^J.

For purely imaginary coefficients and J so large that the g_{il} and θg_{il} are supported at distance greater than 1 from the complement of C_J, Lemma 6.14 gives

$$\langle \mathcal{O}(\zeta,0)\rangle_{N,D}^{C_J} \leqq \Phi^J(\zeta) \leqq \langle \mathcal{O}(\zeta,0)\rangle_{N,D}^{C_{J+2n}} , \tag{6.53}$$

where ζ stands for any imaginary combination of the g_{il}, θg_{il}, the outer terms converging along the full sequence by Theorem 6.9(ii). As $J\to\infty$ both ends tend to (2.18), so Φ^J converges to (2.18) on the purely imaginary points; since the Φ^J are bounded uniformly in J on compact sets, Montel's theorem and the identity theorem give the locally uniform convergence of Φ^J to (2.18) on all of $\mathbb{C}^m$, in particular at the finitely many points occurring in (6.51), and (6.51) passes to the limit.

For an arbitrary c we choose c_0 as above and $b=(c_0-c)e_\nu$; the sources $T_b g_{il}$ are supported in $\{x_\nu > c_0\}$, $\theta_{c_0} T_b = T_b \theta_c$, and Theorem 6.11 gives (6.51) for c from (6.51) for c_0.

(ii) The limit (2.18) depends on a family of sources only through the total source, so its Taylor coefficients at $z=0$ are symmetric multilinear forms; the lattice expectations are real for real coefficients, so the limit is real on the real points and has real Taylor coefficients; and the symmetry $(A,\phi)\to(-A,\bar\phi)$ of every lattice measure gives $\langle\mathcal{O}(-zg,0)\rangle = \langle\mathcal{O}(zg,0)\rangle$, so the coefficients of odd order vanish.

(iii) For ε_K smaller than the distance of the supports of the $g_{\alpha i}$ from the plane, the sampled P is a gauge-invariant polynomial in the field strengths of Ω_+, square integrable for the measure (3.38), and $\theta(P)P$ is the polynomial obtained by reflection; Proposition 3.3 extends to such functionals by truncation, since the measure has moments of all orders, so $\langle\theta(P)P\rangle_{\Omega_J}^{\varepsilon} \geqq 0$.

This expectation is a sum of mixed moments, each of which is a derivative at 0 of the function of the coefficients considered in (i); the locally uniform convergence established there, together with Weierstrass' theorem, carries the inequality to the derivatives of (2.18), that is to the Schwinger functions, for the planes $x_\nu = c_0$; for an arbitrary c we translate the sources as in (i), the Schwinger functions being translation invariant by Theorem 6.11. This is the positivity condition of [OS 1] in its form for finite sums of products of test functions supported in a half-space.

The reflection positivity of the Schwinger functions with scalar sources, with the amplitude restriction of Theorem 2.3(ii), will be proved in Sect. 6.6 by the same argument.

6.5. The Limit Measure and its Ergodic Components

In this subsection $h=0$, and we write S_n for the Schwinger functions of Theorem 6.9. Let h_α, $\alpha\in\mathbb{N}^d$, be the Hermite functions on $\mathbb{R}^d$, the eigenfunctions of $-\Delta+|x|^2$ with eigenvalues $\lambda_\alpha = 2|\alpha|+d$. For real r we denote by $\mathcal{H}^r$ the completion of the Schwartz 2-forms for the norm

$$\|u\|_r^2 = \sum_\alpha \sum_{\mu<\nu} \lambda_\alpha^r \langle u_{\mu\nu}, h_\alpha\rangle^2 , \tag{6.54}$$

so that $\|u\|_0 = \|u\|_2$, and by $\mathcal{H}^{-r}$ its dual, a separable Hilbert space of tempered distributions with values in 2-forms; we write $\Phi(u)$ for the pairing of $\Phi\in\mathcal{H}^{-r}$ with u. For even r the translations, the reflections and the multiplication by a function of C_0^∞ are bounded on $\mathcal{H}^r$, and hence on $\mathcal{H}^{-r}$; this is why r is taken even below. By Theorem 6.9(iv) the function $u\mapsto \langle\mathcal{O}(iu,0)\rangle$, defined for $u\in C_0^\infty$, extends by continuity to L^2, in particular to the Schwartz 2-forms.

Theorem 6.16. *(i) For every even $r > d$ there is exactly one Borel probability measure μ on $\mathcal{H}^{-r}$ whose moments are the Schwinger functions,*

$$\int \Phi(g_1)\cdots\Phi(g_n)\,d\mu(\Phi) = S_n(g_1,\dots,g_n)\,, \qquad g_i \in C_0^\infty(\mathbb{R}^d)\,; \tag{6.55}$$

its characteristic functional is $\int e^{i\Phi(u)}\,d\mu = \langle \mathcal{O}(iu,0)\rangle$ for every real Schwartz 2-form u, and μ is also the unique Borel probability measure on $\mathcal{H}^{-r}$ with this characteristic functional.

(ii) Let $\chi \in C_0^\infty(\mathbb{R}^d)$ be real and Q a box containing $\operatorname{supp}\chi$ in its interior. The laws of the fields χF on the lattice $\varepsilon_K\mathbb{Z}^d$, as measures on the Sobolev space $H^{-r}(Q)$ of the periodic distributions on Q with values in 2-forms, converge weakly as $K\to\infty$ at every fixed J with $\operatorname{supp}\chi$ at distance greater than 1 from the complement of C_J, and then as $J\to\infty$, to the law of $\chi\Phi$ under μ.

(iii) μ is invariant under all translations and under the cubic group about every point, and, by Theorem 2.5, under every Euclidean motion; and it is reflection positive about every coordinate plane: if θ is the reflection in the plane $x_j = c$, acting on $\mathcal{H}^{-r}$ by $(\theta\Phi)(u) = \Phi(\theta u)$, then $\int \overline{G\circ\theta}\,G\,d\mu \geqq 0$ for every $G \in L^2(\mu)$ which is measurable with respect to the fields $\Phi(u)$, $\operatorname{supp} u \subset \{x_j > c\}$.

(iv) There are a probability space (Ξ,ν) and Borel probability measures μ_ξ on $\mathcal{H}^{-r}$, $\xi\in\Xi$, with $\mu = \int \mu_\xi\,d\nu(\xi)$, such that for ν-almost every ξ: μ_ξ is invariant under the translations and the coordinate reflections, ergodic under the translation group, reflection positive about every coordinate plane, and clusters in the following two senses: for all $G, G' \in L^2(\mu_\xi)$, with $U(a)$ the translation by $a\in\mathbb{R}^d$ and $\Lambda_T = [-T,T]^d$,

$$\begin{gathered}\lim_{T\to\infty}\frac{1}{|\Lambda_T|}\int_{\Lambda_T} E_\xi[G'\,U(a)G]\,da = E_\xi[G']\,E_\xi[G]\,,\\ \lim_{\min_j|a_j|\to\infty} E_\xi[G'\,U(a)G] = E_\xi[G']\,E_\xi[G]\,;\end{gathered} \tag{6.56}$$

in the reconstruction of [OS 1] from μ_ξ in any coordinate direction the vacuum is the only vector invariant under the semigroup e^{-tH}, $t\geqq 0$, and the transverse translations; and the moments of μ_ξ satisfy $|E_\xi[\Phi(u_1)\cdots\Phi(u_N)]| \leqq \sigma_N(\xi)\prod_i \|u_i\|_{r+2}$ with $\sigma_N(\xi)\leqq \alpha_\beta(\xi)(N!)^\beta$ for every $\beta > 1$.

(v) Moreover ν-almost every μ_ξ is invariant under the Euclidean group, reflection positive about every plane, and $E_\xi[G'\,U(a)G] \to E_\xi[G']E_\xi[G]$ as $|a|\to\infty$ in any manner; so almost every component satisfies the cluster property of [OS 1] for every choice of the time direction, and the vacuum is unique in the reconstruction from μ_ξ in every direction.

The moments (6.55) vanish for odd n by the symmetry $(A,\theta)\to(-A,-\theta)$ of the lattice models; the moments of odd order of a component μ_ξ need not vanish. The value $r\geqq 14$ is needed only in Theorem 6.22(vi), for the pair of fields.

Proof. (i) At every ε and J, and for real $u_1,\dots,u_N\in C_0^\infty$ and $c\in\mathbb{C}^N$,

$$\sum_{l,m}\bar c_l c_m \langle \mathcal{O}(i(u_m-u_l),0)\rangle_{N,D}^{\varepsilon,C_J} = \Big\langle \Big|\sum_m c_m e^{iF(u_m)}\Big|^2\Big\rangle_{N,D}^{\varepsilon,C_J} \geqq 0\,, \tag{6.57}$$

since F is linear in the source; the limits (2.17), (2.18) exist at these points, so $u\mapsto\langle\mathcal{O}(iu,0)\rangle$ is positive definite on C_0^∞, equal to 1 at $u=0$, and continuous in the L^2 norm by Theorem 6.9(iv), hence continuous on the Schwartz space. By Minlos' theorem [GJ 1] there is a unique probability measure μ' on the tempered distributions with this characteristic functional.

The function $t\mapsto\langle\mathcal{O}(itu,0)\rangle$ is the restriction to the imaginary axis of the entire function $z\mapsto\langle\mathcal{O}(zu,0)\rangle$, which is bounded by $e^{(\operatorname{Re} z)^2\|u\|_2^2}$ by (6.27); so, by the theorem on analytic characteristic functions, $\int e^{t\Phi(u)}d\mu' = \langle\mathcal{O}(tu,0)\rangle$ for real t, all moments of $\Phi(u)$ exist and are the derivatives of this function at $t=0$, and (6.55) follows by polarization. In particular $\int\Phi(u)^2 d\mu' = S_2(u,u)\leqq 2\|u\|_2^2$ for $u\in C_0^\infty$ by Theorem 6.9(i), and by Fatou's lemma along a

sequence $u_n \to h_\alpha e_{\mu\nu}$ in L^2, for which $\Phi(u_n) \to \Phi(h_\alpha e_{\mu\nu})$ in probability by Theorem 6.9(iv), hence almost surely along a subsequence, the same bound holds for the Hermite functions. Hence

$$\int \sum_{\alpha,\mu<\nu} \lambda_\alpha^{-r} \Phi(h_\alpha e_{\mu\nu})^2 \, d\mu' \leqq 2\binom{d}{2} \sum_\alpha \lambda_\alpha^{-r} < \infty \qquad \text{for } r > d\,, \tag{6.58}$$

so that μ' is carried by $\mathcal{H}^{-r}$, and its restriction μ to the Borel sets of $\mathcal{H}^{-r}$ has the same characteristic functional and moments.

The uniqueness by the characteristic functional is that of Minlos' theorem. The moments satisfy $|\int \Phi(u)^n d\mu| \leqq (\sqrt{2}\, e)^n (n!)^{1/2} \|u\|_2^n$ by (6.29), and the same bound holds for a linear combination $\sum_j t_j \Phi(u_j) = \Phi(u)$, $u = \sum_j t_j u_j$. So $\int e^{i\sum_j t_j \Phi(u_j)} d\mu = \sum_n i^n \int (\sum_j t_j \Phi(u_j))^n d\mu / n!$, the series converging absolutely for all real t_j, and the joint law of $\Phi(u_1), \dots, \Phi(u_N)$ is determined by the moments; so is μ, since the $\Phi(u)$, $u \in C_0^\infty$, generate the Borel sets of $\mathcal{H}^{-r}$.

(ii) On the lattice the field $u \mapsto F(\chi u)$ defines a random element of $H^{-r}(Q)$ whose second moments are bounded by $2\|\chi u\|^2$ uniformly in ε and J, by (6.28); expanding in the Fourier basis of Q this bounds the expectation of its norm in $H^{-r}(Q)$ for $r > d/2$, uniformly, so that the laws are tight in $H^{-r'}(Q)$ for $r' > r$ by the compactness of the embedding. Their characteristic functionals converge at every test function by Theorems 2.1 and 2.2, which identifies every limit point with the law of $\chi\Phi$; hence the laws converge, and since r is arbitrary the statement follows.

(iii) A Borel probability measure on $\mathcal{H}^{-r}$ is determined by its characteristic functional, and the characteristic functional of μ is invariant under the translations and the cubic group by Theorems 6.11 and 6.12, and under all rotations by Theorem 2.5. For $u_1, \dots, u_N$ real and supported in $\{x_j > c\}$ and $\gamma \in \mathbb{C}^N$, Theorem 6.15 with purely imaginary coefficients reads $\int \overline{G \circ \theta}\, G \, d\mu \geqq 0$ for $G = \sum_l \gamma_l e^{i\Phi(u_l)}$; these functions are dense in L^2 of the σ-algebra generated by the fields in $\{x_j > c\}$, and θ preserves μ, so the inequality extends to that space.

(iv) Here $(U(a)\Phi)(u) = \Phi(u(\cdot + a))$ on $\mathcal{H}^{-r}$ and $(U(a)G)(\Phi) = G(U(-a)\Phi)$ on functions, so that $U(a)G$ is measurable with respect to the fields in $K + a$ when G is so with respect to those in K, and $\theta_j U(a) = U(\theta_j a)\theta_j$. Let $\mathcal{I}$ be the σ-algebra of the Borel sets of $\mathcal{H}^{-r}$ which are invariant under all translations, and let μ_ξ, $\xi \in \mathcal{H}^{-r}$, be a regular conditional probability of μ given $\mathcal{I}$, with $(\Xi, \nu) = (\mathcal{H}^{-r}, \mu)$. Since μ is translation invariant and $a \mapsto U(a)$ is strongly continuous on $\mathcal{H}^{-r}$, ν-almost every μ_ξ is invariant under all translations, the invariance for a countable dense set of a extending by continuity. Moreover ν-almost every μ_ξ is ergodic under the translations, in the sense that a function of $L^2(\mu_\xi)$ invariant under every $U(a)$ is almost surely constant.

Let $\{f_m\}$ be the finite linear combinations with coefficients in $\mathbb{Q} + i\mathbb{Q}$ of the functions $e^{i\Phi(u)}$, u in a countable vector space over the rationals of real C_0^∞ 2-forms dense in $\mathcal{H}^r$; the f_m form an algebra closed under conjugation which generates the Borel sets, so they are dense in $L^2(\rho)$ for every Borel probability measure ρ on $\mathcal{H}^{-r}$. For a bounded Borel function G and $T > 0$ put $A_T G(\Phi) = |\Lambda_T|^{-1} \int_{\Lambda_T} G(U(a)\Phi)\, da$, which is defined for every Φ, $(a, \Phi) \mapsto U(a)\Phi$ being continuous. If G is bounded and $U(a)G = G$ μ-almost everywhere for each a, then $A_n G = G$ μ-almost everywhere for every integer n by Fubini's theorem, while $|A_n G(U(b)\Phi) - A_n G(\Phi)| \leqq \|G\|_\infty |\Lambda_n \triangle (\Lambda_n + b)| / |\Lambda_n| \to 0$ for every Φ and b; so $\limsup_n A_n G$ is invariant under every $U(b)$ and equals G almost everywhere.

By truncation the fixed vectors of the unitary group $U(a)$ on $L^2(\mu)$ are the functions of $L^2(\mathcal{I}, \mu)$, and the mean ergodic theorem gives $A_T f_m \to E[f_m \,|\, \mathcal{I}]$ in $L^2(\mu)$, with the version $E[f_m \,|\, \mathcal{I}](\Phi) = \int f_m \, d\mu_\Phi$. We choose $T_n \to \infty$ such that $A_{T_n} f_m \to \int f_m \, d\mu_{(\cdot)}$ μ-almost everywhere for every m, off a μ-null set N_0. For $B \in \mathcal{I}$ we have $\mu_\xi(B) = 1_B(\xi)$ for ν-almost every ξ, since $\int_C \mu_\xi(B)\, d\nu = \mu(B \cap C)$ for all $C \in \mathcal{I}$; applied to the sets $\{\Phi : \int f_m \, d\mu_\Phi \in I\}$, I a rational interval, and combined with $\int \mu_\xi(N_0)\, d\nu = \mu(N_0) = 0$, which gives $\mu_\xi(N_0) = 0$ for ν-almost every ξ, this shows that for ν-almost every ξ, $A_{T_n} f_m \to \int f_m \, d\mu_\xi$ μ_ξ-almost everywhere for

every m.

Let such a ξ be given, with μ_ξ translation invariant, and let G be bounded with $U(a)G = G$ μ_ξ-almost everywhere for each a. By the invariance of μ_ξ and Fubini's theorem $\int G\, A_{T_n} f_m \, d\mu_\xi = \int G f_m \, d\mu_\xi$, and by dominated convergence $\int G f_m \, d\mu_\xi = \int G \, d\mu_\xi \int f_m \, d\mu_\xi$ for every m; by density G is μ_ξ-almost everywhere constant, and so, by truncation, is every invariant function of $L^2(\mu_\xi)$.

We next show that the components are invariant under the reflections. If W is a bounded $\mathcal{I}$-measurable function, then W is, up to a null set, measurable with respect to the fields in any half space $\{x_j > c\}$: approximating W in $L^2(\mu)$ by functions of the fields in a bounded region and translating, which does not change W, moves the region into the half space. The function $W \circ \theta_j$ is again $\mathcal{I}$-measurable, since θ_j normalises the translations, and so is likewise measurable with respect to the fields in that half space.

Hence the reflection positivity of μ applies to $W - W \circ \theta_j$, which is odd under θ_j, and gives $-\int |W - W \circ \theta_j|^2 d\mu \geqq 0$: $W \circ \theta_j = W$ almost surely. Therefore, for $W \geqq 0$ bounded and $\mathcal{I}$-measurable, G bounded and Borel, and G' measurable with respect to the fields in $\{x_j > c\}$,

$$\int W\, G \circ \theta_j \, d\mu = \int WG \, d\mu \,, \qquad \int W\, \overline{G' \circ \theta_j}\, G' d\mu = \int \overline{(\sqrt{W} G') \circ \theta_j}\, (\sqrt{W} G') \, d\mu \geqq 0 \,; \quad (6.59)$$

disintegrating with respect to $\mathcal{I}$ gives the invariance of μ_ξ under θ_j and its reflection positivity for ν-almost every ξ, the exceptional set being made independent of G, G' by countable dense families. It suffices to take $c = 0$, since ν-almost every μ_ξ is invariant under all translations and the reflection in $\{x_j = c\}$ is $U(ce_j)\theta_j U(-ce_j)$.

The first limit in (6.56) is the mean ergodic theorem for the unitary group $U(a)$ on $L^2(\mu_\xi)$, the fixed vectors being the constants by ergodicity.

For the second limit, let G' and G be bounded and measurable with respect to the fields in $\{|x| < R\}$; such functions are dense in $L^2(\mu_\xi)$, and the expression is bounded by $\|G'\|_2 \|G\|_2$, so this case suffices. Fix a direction j, let $\theta_{j,R} = U(Re_j)\theta_j U(-Re_j)$ be the reflection in the plane $x_j = R$, and let $\mathcal{H}_j$ be the Hilbert space of the reconstruction of [OS 1], [GJ 1] from μ_ξ about this plane: the completion of L^2 of the fields in $\{x_j > R\}$ modulo the null vectors of the form $(F, G)_j = E_\xi[\overline{F \circ \theta_{j,R}}\, G]$, which is positive by the reflection positivity of μ_ξ about $\{x_j = R\}$ established above. The translations by te_j, $t \geqq 0$, define on it a self-adjoint contraction semigroup e^{-tH_j}, the transverse translations unitaries $V_j(a_\perp)$ commuting with it, and $\Omega = [1]$ is the vacuum; $\|[K]\|_j \leqq \|K\|_{L^2(\mu_\xi)}$ by Schwarz's inequality.

Put $G_0 = U(2Re_j)G$, which is measurable with respect to the fields in $\{R < x_j < 3R\}$. For $a_j > 2R$ let $F = \overline{G' \circ \theta_{j,R}}$, which is measurable with respect to the fields in $\theta_{j,R}\{|x| < R\} \subset \{x_j > R\}$; since $\overline{F \circ \theta_{j,R}} = G'$,

$$E_\xi[G'\, U(a) G] = (F, e^{-tH_j} V_j(a_\perp) G_0)_j \,, \qquad t = a_j - 2R \,. \quad (6.60)$$

Since $e^{-tH_j/2} \to P_j$ strongly, P_j the projection on the null space of H_j, and $V_j(a_\perp)$ commutes with $e^{-tH_j/2}$,

$$|E_\xi[G'\, U(a) G] - g_j^+(a_\perp)| \leqq \|e^{-tH_j/2} F - P_j F\| \, \|G_0\| + \|F\| \, \|e^{-tH_j/2} G_0 - P_j G_0\| \to 0 \quad (6.61)$$

as $a_j \to +\infty$, uniformly in $a_\perp$, where $g_j^+(a_\perp) = (P_j F, V_j(a_\perp) P_j G_0)_j$. For $a_j \to -\infty$ the same holds with a function $g_j^-(a_\perp)$, since $E_\xi[G'\, U(a) G] = E_\xi[(U(-a)G')\, G]$.

Write $f(a) = E_\xi[G'\, U(a) G]$, fix $\sigma \in \{+, -\}^d$ and let $O_\sigma(R') = \{a : \ \sigma_j a_j > R' \text{ for all } j\}$. Given $\epsilon > 0$ there is R_ϵ such that, if $a, a' \in O_\sigma(R_\epsilon)$ differ in the coordinate a_j only, both $f(a)$ and $f(a')$ are within ϵ of $g_j^{\sigma_j}(a_\perp)$; changing one coordinate at a time, $|f(a) - f(a')| \leqq 2d\epsilon$ on $O_\sigma(R_\epsilon)$, so f has a limit ℓ_σ as $\min_j |a_j| \to \infty$ in O_σ. The mean ergodic theorem holds for the averages over any cube of side T, in particular over $\Lambda_T^\sigma = \{a : \ \sigma_j a_j \in [0, T] \text{ for all } j\}$, so $T^{-d} \int_{\Lambda_T^\sigma} f \, da \to E_\xi[G'] E_\xi[G]$; the set of $a \in \Lambda_T^\sigma$ with some $|a_j| \leqq R'$ has volume at most

$dR'T^{d-1}$ and $|f| \leqq \|G'\|\,\|G\|$, so these averages converge to ℓ_σ as well. Hence $\ell_\sigma = E_\xi[G']E_\xi[G]$ for every σ, which is the second limit in (6.56); the limit along one direction alone need not be the product.

Let now $\Psi \in \mathcal{H}_j$ be invariant under e^{-tH_j} and all $V_j(a_\perp)$. For K bounded and measurable with respect to the fields in a bounded region, choose t so large that $U(te_j)K$ is measurable with respect to the fields in $\{x_j > R\}$, and put $f(K) = (\Psi, [U(te_j)K])_j$; this does not depend on t, since $e^{-sH_j}[U(te_j)K] = [U((t+s)e_j)K]$ and Ψ is invariant, it satisfies $|f(K)| \leqq \|\Psi\|_j \|K\|_{L^2(\mu_\xi)}$, and $f(U(a)K) = f(K)$ for every $a \in \mathbb{R}^d$. So $f(K) = E_\xi[\bar\psi K]$ with $\psi \in L^2(\mu_\xi)$ invariant under all translations, hence a constant by ergodicity; and since the classes $[U(te_j)K]$ are dense in $\mathcal{H}_j$, $\Psi = \psi\Omega$ is a multiple of Ω.

The moments of μ satisfy $\int \Phi(u)^{2n} d\mu \leqq C^{2n}(2n)!\,\|u\|_2^{2n}$ by (6.29); expanding Φ in the Hermite basis and using Jensen's inequality,

$$\int \|\Phi\|_{-r-2}^{2N} d\mu \leqq C^N (N!)^2\,, \tag{6.62}$$

and since $\int E_\xi[\|\Phi\|_{-r-2}^{2N}]\,d\nu(\xi)$ is this integral, for ν-almost every ξ all these expectations are finite, which gives the bound on the moments of μ_ξ with $\sigma_N(\xi) = E_\xi[\|\Phi\|_{-r-2}^{2N}]^{1/2}$, and its growth by Chebyshev's inequality and the Borel–Cantelli lemma. This growth satisfies the linear growth condition of [OS 2] used in the reconstruction.

(v) By Theorem 2.5 the limit (2.18) is invariant under $O(d)$, so by (iii) μ is invariant under $O(d)$ and reflection positive about every plane, the reflection in any plane being conjugate to a coordinate reflection by a rotation. The argument of (iv) applies to the reflection θ in any plane: every bounded $\mathcal{I}$-measurable W is, up to a null set, measurable with respect to the fields in either half space, $W \circ \theta$ is again $\mathcal{I}$-measurable since θ normalises the translations, and so $W \circ \theta = W$ almost surely.

Since the reflections in planes through the origin generate $O(d)$, $W \circ R = W$ almost surely for every $R \in O(d)$, and disintegrating, for R in a countable dense subgroup of $O(d)$ and then for all R by the continuity of $R \mapsto R\Phi$ on $\mathcal{H}^{-r}$, $\int W\,G \circ R\,d\mu = \int WG\,d\mu$ as in (iv) gives the invariance of ν-almost every μ_ξ under $O(d)$, and with (iv) under the Euclidean group; the reflection positivity of μ_ξ about any plane follows from that about a coordinate plane by conjugation.

If $|a| \to \infty$, a rotation R_a carries a to a vector all of whose coordinates are at least $|a|/\sqrt{d}$; by the invariance of μ_ξ,

$$E_\xi[G'\,U(a)G] = E_\xi[(G' \circ R_a^{-1})\,U(R_a a)\,(G \circ R_a^{-1})]\,, \tag{6.63}$$

and the second limit in (6.56), which holds uniformly for G, G' in a compact subset of $L^2(\mu_\xi)$, by the bound $|E_\xi[G'\,U(a)G]| \leqq \|G'\|_2\|G\|_2$ and a finite net, such as $\{G \circ R^{-1} :\ R \in O(d)\}$ for a fixed bounded continuous G, such G being dense, gives the required result. The remaining statements then follow from the reconstruction in an arbitrary direction, as in (iv).

6.6. The Scalar Source

In this subsection we prove Theorems 2.1–2.3 for $h \neq 0$ and the bound (3.53). The expectations with the source h are controlled through the periodic model on the torus

$$T_J = \mathbb{R}^d/(2MJ\mathbb{Z})^d\,, \tag{6.64}$$

whose fundamental cell is C_J;[11] we write $\langle\cdot\rangle_p^{\varepsilon,T_J}$ for the expectation of the action (2.5) on the lattice $\varepsilon\mathbb{Z}^d$ modulo $(2MJ\mathbb{Z})^d$, with the counterterm δm^2 and the subtraction (2.13) of this paper.

[11] This is the cubic torus (2.22) of [I] with $L_\mu = MJ$, to which Theorems 2.1 and 3.4 of [I] apply.

The model of [I] on the same torus differs from it in the counterterm, the Wick constant and the constants E_0, E_1, which are computed there with the periodic propagators; we refer to the two as the free and the torus convention. They give the same expectations up to a shift of the scalar source by a constant and a field-independent factor, and at fixed ε both the shift and the difference of the Wick constants tend to zero as $J \to \infty$, since the periodic propagators converge to those of $\varepsilon\mathbb{Z}^d$.[12]

For a real number u we write

$$\Phi_{\varepsilon,J}(u) = |T_J|^{-1} \ln\langle \mathcal{O}(0,u)\rangle_p^{\varepsilon,T_J} \tag{6.65}$$

for the free energy density of the constant source $h \equiv u$. By Hölder's inequality $\Phi_{\varepsilon,J}$ is convex, and $\Phi_{\varepsilon,J}(0) = 0$.

Lemma 6.17. *(i) The expectation $\langle\cdot\rangle_p^{\varepsilon,T_J}$ is invariant under the lattice translations, and it is reflection positive, in the sense of* (3.39)*, with respect to the reflection of T_J in every pair of planes $x_\nu = c$, $x_\nu = c + MJ$ with $c \in \varepsilon(\mathbb{Z} + \frac{1}{2})$, for gauge-invariant functionals of the fields on one side. (ii) For every real function q on the sites of T_J,*

$$\ln\langle \mathcal{O}(0,q)\rangle_p^{\varepsilon,T_J} \leqq \sum_x \varepsilon^d \, \Phi_{\varepsilon,J}(q(x)) \,. \tag{6.66}$$

Proof. (i) The proof of Proposition 3.3 given in Sect. 3.3 applies with the pair of planes in place of the plane π and without the Dirichlet terms. For gauge-invariant functionals the expectation equals the expectation with the unitary form, by the Hodge decomposition on the torus and a gauge transformation of the scalar field, as there.

We write each crossing variable as $A_b = u_b - v_b$: since $\mu_0 > 0$, the law of A_b under $e^{-\frac{1}{2}\mu_0^2\varepsilon^d A_b^2} dA_b$ is that of the difference of two independent Gaussian variables u_b, v_b of variance $(2\mu_0^2\varepsilon^d)^{-1}$, which we attach to the two sides with $\theta(u_b) = v_b$; the mass term of the crossing bond is thereby absorbed in the factors $e^{-\mu_0^2\varepsilon^d u_b^2} du_b$, $e^{-\mu_0^2\varepsilon^d v_b^2} dv_b$, each a function of the variables on one side. Since $\theta(A_b) = -A_b$, the hopping term across a plane is $\varepsilon^{d-2}\,\mathrm{Re}[F\theta(F)]$ with $F = r(x)e^{i(\theta(x) - e\varepsilon u_b)}$, and a straddling plaquette contributes $e^{-\frac{1}{2}\varepsilon^{d-2}(a-\theta a)^2}$ with a a function of the variables on one side; so every coupling is a sum of terms $G\theta(G)$ with non-negative weights, and (3.39) follows by Schwarz's inequality as in [BFS 1].

(ii) This is the chessboard estimate of [FILS 1],[13] applied to the functions

$$G_x = \exp[\varepsilon^d q(x)(r(x)^2 - 2(-\Delta^\varepsilon + m^2)^{-1}(x,x) - a_\varepsilon)] \,, \tag{6.67}$$

which are gauge invariant and positive with all moments: its hypotheses are the translation invariance and the reflection positivity of (i) with respect to all the planes midway between neighbouring lattice planes, and the number $2MJ/\varepsilon$ of sites on a side is even. The estimate is

$$\Big\langle \prod_x G_x \Big\rangle \leqq \prod_x \Big\langle \prod_y G_x^{(y)} \Big\rangle^{\varepsilon^d/|T_J|} , \tag{6.68}$$

where $G_x^{(y)}$ is the copy of G_x carried to y by reflections; since the counterterm and the constants of (2.13) are the same at every site, $\prod_y G_x^{(y)} = \mathcal{O}(0, q(x))$ for the constant source $q(x)$, and taking logarithms gives (6.66).

We now introduce the condition on the coupling constants under which the results with $h \neq 0$ are proved. Let $\delta m_\varepsilon^2(M)$ be the counterterm of [I] on $\varepsilon\mathbb{Z}^d$ at the scalar mass M,[14] and $\alpha_\varepsilon(M) = M^2 + \delta m_\varepsilon^2(M)$ the bare coefficient of $\frac{1}{2}r^2$ in the action at that mass. Put

$$\sigma_1(M_1) = \liminf_{\varepsilon\to 0} \tfrac{1}{2}\big[\alpha_\varepsilon(m) - \alpha_\varepsilon(M_1)\big] \qquad \text{for } 0 < M_1 < m \,; \tag{6.69}$$

[12] Sect. 5.2 of [I].

[13] Theorem 4.1 of [FILS 1].

[14] (3.51) of [I], as in Sect. 5.2 there.

by Theorem 5.1(i) of [I] the limit exists, so that the lower limit in (6.69) is a limit. The condition is

$$\sigma_1(M_1) > 0 \qquad \text{for some } M_1 \in (0, m)\,, \tag{6.70}$$

that is, the difference of the bare coefficients at the two masses keeps, in the limit, the sign of $m^2 - M_1^2$. It is discussed in Sect. 6.8, where it is shown to hold for e^2 and λ small compared with m. We fix such an M_1 and a number $s_1 \in (0, \sigma_1(M_1))$, and put

$$\sigma_+ = s_1\,, \qquad H = s_1/16\,, \qquad \sigma_W = \sigma_+/8\,. \tag{6.71}$$

Throughout, a_ε is the constant of (2.13); the constant which the renormalization transformation of Sect. 3 of [I] produces differs from it by a sequence which converges as $\varepsilon \to 0$ (the proof of Theorem 6.21(i)), and the results below hold for either, with C_U increased by $s_1 \sup_\varepsilon |a_\varepsilon - a'_\varepsilon|$ for any other choice a'_ε whose difference from a_ε converges, and for no choice whose difference from a_ε diverges, when $\int h \neq 0$ (Sect. 6.10).

Proposition 6.18. *Assume* (6.70). *There is a constant $C_U \geqq 0$, depending on the coupling constants, on M, M_1 and s_1, such that*

$$\sup_K \limsup_{J\to\infty} \Phi_{\varepsilon_K,J}(s_1) \leqq C_U\,. \tag{6.72}$$

Proof. We first work in the torus convention of [I] at fixed $\varepsilon = \varepsilon_K$ and write $\Phi^T_{\varepsilon,J}$ for the corresponding free energy density, which is convex with value 0 at 0 by the proof of Lemma 6.17. Fix $p > 1$ with $ps_1 < \sigma_1(M_1)$. Let $s = s_{\varepsilon,J}$ be half the difference of the bare coefficients of the torus convention at the masses m and M_1; as $J \to \infty$ it converges to half the difference of the free-boundary coefficients, whose lower limit as $\varepsilon \to 0$ is $\sigma_1(M_1) > ps_1$. So there are K_1 and, for $K \geqq K_1$, $J_*(K)$ such that $s > ps_1$ for $K \geqq K_1$ and $J \geqq J_*(K)$. For such K, J the convexity gives

$$\Phi^T(u) \leqq (u/s)\,\Phi^T(s) \leqq [\Phi^T(s)]_+ \qquad \text{for } 0 \leqq u \leqq ps_1\,. \tag{6.73}$$

The source $s\sum_x \varepsilon^d r(x)^2$ shifts the bare coefficient at the mass m exactly to the bare coefficient at the mass M_1, and the action of [I] contains, besides the terms fixed by the mass, only the counterterm and the constants $E_0 + E_1$; hence

$$|T_J|\,\Phi^T(s) = \ln Z^{(M_1)} - \ln Z^{(m)} + B_{\varepsilon,J}\,, \tag{6.74}$$

where $Z^{(M)}$ is the partition function of [I] at the scalar mass M and $B_{\varepsilon,J}$ collects the differences of the constants $E_0 + E_1$ at the two masses and the Wick constants of the source. By the ultraviolet stability bound of [I],[15] applied at the mass M_1 as an upper bound and at the mass m as a lower bound,

$$\ln Z^{(M_1)} - \ln Z^{(m)} \leqq C|T_J| \tag{6.75}$$

with C independent of ε and J.

The constant $B_{\varepsilon,J}$ satisfies $|B_{\varepsilon,J}| \leqq C|T_J|$ uniformly in ε and in $J \geqq J_*(K)$, its terms of order ε^{-1} and $\ln\varepsilon^{-1}$ cancelling.[16] The terms of E_0 and of the Wick constant proportional to $(m^2 - M_1^2)\sum_x \varepsilon^d C_m(x,x)$ cancel exactly, and the remaining terms of E_0 are convergent. The constant E_1 contains, in $d = 3$, terms of order ε^{-2} which depend on the scalar mass, such as the term of order λ, proportional to $\lambda C_M(x,x)^2|T_J|$; their difference at the two masses is of order ε^{-1} and cancels against the part $-(\delta m^2_\varepsilon(m) - \delta m^2_\varepsilon(M_1))\sum_x \varepsilon^d C_m(x,x)$ of the Wick constant, the remainder being $O(|T_J|)$. The same holds for every pair of scalar masses $\mu > \mu_1 \geqq M_1$, with C depending on μ, μ_1. This proves (6.72) in the torus convention for $K \geqq K_1$.

[15] Theorem 2.1(ii) of [I], which is the bound of [Ba 1–2].

[16] E_0 and E_1 are given by (2.6) and (3.53) of [I].

For the finitely many $K < K_1$ the bound follows, at fixed ε, by bounding the hopping terms in absolute value by $d\,\varepsilon^{-2}\sum_x \varepsilon^d r(x)^2$, above in the numerator and below in the denominator, after which the vector field decouples from the scalar field and the ratio is a product of one-site integrals, the torus counterterm and Wick constant being bounded in J at fixed ε; this gives $\Phi^T_{\varepsilon,J}(u) \leqq C_\varepsilon(u)$ for all J.

Finally, the free convention differs from the torus convention by a shift $\sigma_{\varepsilon,J}$ of the source and a change $\Delta c_{\varepsilon,J}$ of the Wick constant, both of which tend to zero as $J \to \infty$ at fixed ε:

$$\Phi_{\varepsilon,J}(u) = \Phi^T_{\varepsilon,J}(u + \sigma_{\varepsilon,J}) - \Phi^T_{\varepsilon,J}(\sigma_{\varepsilon,J}) - 2\Delta c_{\varepsilon,J}\, u\,, \tag{6.76}$$

and by convexity

$$\Phi^T(u+\sigma) \leqq p^{-1}\Phi^T(pu) + (1-p^{-1})\Phi^T(p'\sigma)\,, \qquad (1-p^{-1})p' = 1\,; \tag{6.77}$$

the terms containing $\sigma_{\varepsilon,J}$ are bounded by $C_\varepsilon|\sigma_{\varepsilon,J}|$ and vanish in the limit. Hence

$$\limsup_J \Phi_{\varepsilon,J}(s_1) \leqq p^{-1} \sup_K \limsup_J \Phi^T_{\varepsilon_K,J}(ps_1)\,, \tag{6.78}$$

which is (6.72), with C_U the positive part of the right side. The same holds, with the same proof, for any pair of scalar masses $\mu > \mu_1 \geqq M_1$ and any s_1' with $0 < s_1' < \liminf_{\varepsilon\to 0} \frac{1}{2}[\alpha_\varepsilon(\mu) - \alpha_\varepsilon(\mu_1)]$, with s_1' in place of s_1 and a constant $C_U(\mu, \mu_1, s_1')$, the model at the mass μ playing the role of the model at the mass m; the case of the proposition is $(\mu, \mu_1, s_1') = (m, M_1, s_1)$, and the model at the mass M_2 of Lemma 6.20 with m in the role of M_1 is the case (M_2, m, s_1') with $2\sup q < s_1'$.

Lemma 6.19. *Assume* (6.70)*, and let* $\kappa = C_U/s_1$*. For every box* X *of sites of* $\varepsilon\mathbb{Z}^d$ *and every function* q *on* X *with* $0 \leqq q \leqq s_1$*,*

$$\ln\langle \mathcal{O}(0,q)\rangle_X \leqq \kappa \sum_x \varepsilon^d q(x)\,, \tag{6.79}$$

for the expectation (2.10) *on* $X = C_J$*, for the expectation* (3.38) *on a box, and for the expectation of Sect. 6.1 in which the scalar field vanishes outside* X*; the bound is uniform in* ε *and* X*. In particular* (3.53) *holds with* κ *and* $\sigma_+ = s_1$*.*

Proof. Let J' be such that X and its neighbouring sites lie in $C_{J'}$; for the expectation (3.38) on a box let J' be such that $C_{J'}$ exceeds the box by the margin of $2n$ cubes of (3.52), so that the second inequality of (3.52) bounds the expectation on the box by the expectation (2.10) on $C_{J'}$. The observable $e^{\sum_x \varepsilon^d q(x) r(x)^2}$ is of the class of (3.24), so by Lemmas 6.2 and 6.3 its expectation does not decrease when the scalar field is given back at the sites of $C_{J'}$ outside X and the couplings across $\partial C_{J'}$ are restored, which gives the periodic expectation on $T_{J'}$; the constants of (2.13) are common to all these expectations. The chessboard estimate (6.66) then bounds the left-hand side of (6.79) by $\sum_x \varepsilon^d \Phi_{\varepsilon,J'}(q(x))$, and by the convexity of $\Phi_{\varepsilon,J'}$, as in (6.73), $\Phi_{\varepsilon,J'}(u) \leqq (u/s_1)[\Phi_{\varepsilon,J'}(s_1)]_+$ for $0 \leqq u \leqq s_1$. The left-hand side of (6.79) does not depend on J'; letting $J' \to \infty$ and using (6.72) gives (6.79).

Lemma 6.20. *Assume* (6.70)*. (i) Let* $q \geqq 0$ *be a smooth function supported at distance greater than* 1 *from the complement of* C_J*. Then* $\sup_\varepsilon \langle \mathcal{O}(0,-q)\rangle^{\varepsilon,C_J}_{N,D} < \infty$*. (ii) There is a constant* K_-*, depending on the coupling constants, on* M*,* M_1*,* s_1 *and the auxiliary choices* M_2*,* ℓ_c*,* χ_c *below, such that for every* $R > 0$*, every* J *with* $MJ \geqq R + \ell_c$*, where* ℓ_c *is a fixed multiple of* $2M$*, and every lattice function* q *supported in* $Q_R = \{|x_\mu| \leqq R\}$ *with* $0 \leqq q \leqq H$*,*

$$\ln\langle \mathcal{O}(0,-q)\rangle^{\varepsilon,C_J}_{N,D} \leqq K_- \sup_x (1+|x|^2)^2 q(x)\,, \tag{6.80}$$

uniformly in ε *and* J*.*

Proof. (i) The source $-q$ raises the bare coefficient of $\frac{1}{2}r^2$ by $2q$. Choose $M_2 > m$ so large that

$$\lim_{\varepsilon\to 0} \tfrac{1}{2}[\alpha_\varepsilon(M_2) - \alpha_\varepsilon(m)] > 2\sup q \qquad \text{and} \qquad \sup_\varepsilon |a_\varepsilon(M_2) - a_\varepsilon(m)| < \infty\,. \tag{6.81}$$

Such M_2 exist for every q: by Theorem 5.1 of [I] the limit exists and tends to $+\infty$ with M_2, its leading term being $\frac{1}{2}(M_2^2 - m^2)$, the divergent parts of the difference cancelling, and the constants a_ε at the two masses differ by a bounded sequence. For ε small the coefficient $\alpha_\varepsilon(m) + 2tq(x)$ of the model at the mass m with the source $-tq$ lies below the coefficient $\alpha_\varepsilon(M_2) - 2tq(x)$ of the model at the mass M_2 with the source $+tq$, at every site and for $0 \leqq t \leqq 1$; the finitely many remaining ε contribute finite numbers to the supremum.

By the inequality (3.25), which holds for a site-dependent coefficient of $\frac{1}{2}r^2$, the expectation of $\sum_x \varepsilon^d q(x) r(x)^2$ does not increase when the bare coefficient is increased at any site; integrating this along the two shifts of the coefficient, by $+2tq$ from the mass m and by $-2tq$ from the mass M_2, gives

$$\langle \mathcal{O}(0,-q)\rangle_{N,D}^{\varepsilon,C_J}\, \langle \mathcal{O}(0,q)\rangle_{N,D}^{\varepsilon,C_J,(M_2)} \leqq \exp\Big[C\sum_x \varepsilon^d q(x)\Big]\,, \tag{6.82}$$

where the second expectation is that of the model at the mass M_2 and C bounds the difference, at the two masses, of the constants $2(-\Delta^\varepsilon + M^2)^{-1}(x,x) + a_\varepsilon(M)$ of (2.13), which is bounded uniformly in ε since the divergent parts of these constants do not depend on the mass.

The second expectation is bounded below by e^{-C} uniformly in ε by the lower bound of ultra-violet stability with the source,[17] for the model at the mass M_2 with m in the role of M_1, for which the analogue of (6.70) holds by the choice of M_2, with an amplitude $s_1' \in (2\sup q, \lim_\varepsilon \frac{1}{2}[\alpha_\varepsilon(M_2) - \alpha_\varepsilon(m)])$; this bound is obtained in the proof of Theorem 6.21(i). (i) follows.

(ii) Let $\mathcal{T}_s$, $s \in \{0,1\}^d$, be the 2^d tilings of $\mathbb{R}^d$ by the cubes of side ℓ_c with corners in $\ell_c\mathbb{Z}^d + \frac{1}{2}\ell_c s$, which we call cells, and let $1 = \sum_s \chi_s$ be a partition of unity with $0 \leqq \chi_s \leqq 1$ smooth and χ_s vanishing within distance 2 of the boundaries of the cells of $\mathcal{T}_s$; ℓ_c is chosen so large that this is possible. By Hölder's inequality

$$\langle \mathcal{O}(0,-q)\rangle \leqq \prod_s \langle \mathcal{O}(0,-2^d\chi_s q)\rangle^{2^{-d}}\,. \tag{6.83}$$

For each s, by the inequalities (3.27), (3.29) (Lemma 6.3), the expectation of $e^{-r^2(2^d\chi_s q)}$, with $r^2(h)$ as in (3.23), does not decrease when the terms $\frac{1}{2}\varepsilon^{d-2}|A_\mu(b_+) - A_\mu(b_-)|^2$ coupling bonds in different cells and the terms $-\varepsilon^{d-2}r(x)r(y)\cos(\theta(x) - e\varepsilon A_{\langle x,y\rangle} - \theta(y))$ coupling sites in different cells are removed from the action, the terms $\frac{1}{2}\varepsilon^{d-2}r(x)^2$ being kept. As in the proof of Theorem 3.2, the expectation then factorizes into a product over the cells, the factor of a cell c being the expectation on c in which every site of c carries a scalar variable and the vector field has N b.c. on ∂c, that is the comparison cube $\langle\cdot\rangle_{N,D,*}^{\varepsilon,C_{J_c}}$ of Sect. 6.1 with $2MJ_c = \ell_c$, translated by a vector of $\varepsilon\mathbb{Z}^d$; the cells meeting ∂C_J carry no source since $MJ \geqq R + \ell_c$.

On a cell c the source $\tilde{q} = 2^d\chi_s q$ satisfies $0 \leqq \tilde{q} \leqq 2^d H = H'$ and vanishes within distance 2 of its boundary. Writing $-\tilde{q} = (H'\chi_c - \tilde{q}) - H'\chi_c$ with χ_c a fixed smooth function equal to 1 on the support of $\chi_s q$ in c and vanishing near ∂c, the Cauchy–Schwarz inequality gives

$$\langle \mathcal{O}(0,-\tilde{q})\rangle_c \leqq \langle \mathcal{O}(0,-2H'\chi_c)\rangle_c^{1/2} \langle \mathcal{O}(0,2(H'\chi_c - \tilde{q}))\rangle_c^{1/2}\,; \tag{6.84}$$

the second factor is bounded by Lemma 6.19, since $2H' = s_1$ for $d = 3$, and the first by (i), applied, after translation of c to the cube C_{J_c}, to the cube of Sect. 2, whose expectation of $\mathcal{O}(0,-2H'\chi_c)$ dominates that of the comparison cube by Lemma 6.4, χ_c vanishing near ∂c. So

[17] (3.4) of Theorem 3.1 of [I].

$\ln\langle\mathcal{O}(0,-\tilde{q})\rangle_c \leqq C_{\rm cell}$ for a constant independent of ε and of the cell, and by convexity in the amplitude

$$\ln\langle\mathcal{O}(0,-\tilde{q})\rangle_c \leqq ([C_{\rm cell}]_+/H')\sup_c \tilde{q} \leqq (2^d[C_{\rm cell}]_+/H')\sup_x(1+|x|^2)^2 q(x)\cdot\sup_{x\in c}(1+|x|^2)^{-2}\,. \quad (6.85)$$

Summing over the cells of each tiling, $\sum_c \sup_{x\in c}(1+|x|^2)^{-2} \leqq C_\Sigma$, and over the 2^d tilings gives (6.80) with $K_- = 2^d[C_{\rm cell}]_+ C_\Sigma/H'$.

Theorem 6.21. *Assume* (6.70)*, and let g be a real 2-form and h a real function, both C^∞ and supported at distance greater than* 1 *from the complement of C_J. Put*

$$D_J(h) = \{w\in\mathbb{C}:\ 2\sup_x(\mathrm{Re}\, w\, h(x))_+ < s_1\}\,, \quad (6.86)$$

an open vertical strip containing the imaginary axis, which depends on h only through $\sup h_+$ *and* $\sup h_-$ *and contains the half-plane* $\mathrm{Re}\, w \leqq 0$ *when $h \geqq 0$. (i) If $h \geqq 0$, the limit* (2.17) *exists and is positive for real z and $0 < w < s_1/(2\sup h)$. (ii) The limit* (2.17) *exists locally uniformly for $(z,w)\in\mathbb{C}\times D_J(h)$, is holomorphic there, and is positive at every real point.*

Proof. (i) The argument of Sect. 3.2 of [I] applies to the ratio $Z^{\varepsilon_K}(C_J, zg, wh)/Z^{\varepsilon_K}(C_J,0,0)$, the renormalization transformation being carried out on the cube as in Sect. 3.1. Three points are to be checked: the two-sided ultra-violet stability bound with the source, the convergence of the source-dependent part of the effective action, and the positivity of the limit.

The two-sided bound[18] holds with the source wh: its large field term is controlled, by the Cauchy–Schwarz inequality, through the expectation of $\mathcal{O}(2zg, 2wh)$, which is at most $\exp[2z^2\mu_0^{-2}\|\varepsilon^{-1}d^*g\|^2]\langle\mathcal{O}(0,2wh)\rangle$ by Lemma 6.7, applied to the measure with the weight $\mathcal{O}(0,2wh)$, and is bounded by Lemma 6.19 for $2w\sup h < s_1$.

The convergence of the field-dependent part of the effective action, Theorem 3.4 of [I], extends with the same bound to the part which depends on the source h and on the fields: the graphs containing the source vertex[19] are localized as in Sect. 3.4 of [I], the source entering as a small perturbation, and the extension carries over the bound on the degree of the renormalized classes with one inserted vertex at a point of the block lattice and the inequalities for the mass counterterm units of order four,[20] and with them the arguments of Sects. 3.5 and 3.6 of [I]. On the support of the small field characteristic function the source-dependent part of the effective action, the constant $a_{\varepsilon_K}\sum_x \varepsilon^d wh(x)$ removed, is therefore bounded by $C(h)$ uniformly in K. The part of the effective action which is independent of the fields and linear in h is not bounded uniformly in the number of steps: at order e^2 it grows like the number of steps. It is removed, up to terms which converge, by the constant a_ε of (2.13), the difference $a_{\varepsilon_K} - a_{\varepsilon_{K+1}}$ converging to the growth per step; the constant of (2.13) differs from the constant which the transformation produces by a sequence which converges as $\varepsilon\to 0$.

The ratios then form a Cauchy sequence, the remainder of each step being compared with $Z^{\varepsilon_K}(C_J,0,0)$ rather than with the ratio itself, as in [I].[21] Their limit is positive: the small field integral with the source is at least $e^{-C(h)}$ times the one without it, and the two bounds, the lower one with the source and the upper one without it, applied at a scale $L^k\varepsilon_K$ fixed so small, depending on J, that the large field term of the upper bound is at most one half of the lower bound $e^{-C|C_J|}$ of Sect. 5.2 for $Z^{\varepsilon_K}(C_J,0,0)$, give, with the upper bound $e^{C|C_J|}$ of Sect. 5.2,

$$Z^{\varepsilon_K}(C_J,0,wh)/Z^{\varepsilon_K}(C_J,0,0) \geqq e^{-C} \quad (6.87)$$

uniformly in K, C depending on J and h; for real z the ratio with the source zg is at least this one, by Jensen's inequality in the measure with the source wh, under which $\langle F(g)\rangle = 0$ by the symmetry $(A,\theta)\to(-A,-\theta)$.

[18] (3.4), (3.5) of Theorem 3.1 of [I].
[19] The vertex (3.37) of [I].
[20] (3.98) of [I].
[21] (3.16)–(3.22) of [I].

(ii) We first show that the expectations are bounded on compact sets. Write $h = h^+ - h^-$ with $h^\pm \geqq 0$ smooth, supported at distance greater than 1 from the complement of C_J, with $\sup h^\pm \leqq \sup h_\pm + \eta$; for real x, $y \geqq 0$ with $2y \sup h^+ < s_1$, Hölder's inequality with the exponents $4, 2, 4$ gives

$$\langle \mathcal{O}(xg, yh)\rangle \leqq \langle \mathcal{O}(4xg, 0)\rangle^{1/4} \langle \mathcal{O}(0, 2yh^+)\rangle^{1/2} \langle \mathcal{O}(0, -4yh^-)\rangle^{1/4}\,, \tag{6.88}$$

the constants of (2.13) being distributed accordingly, and symmetrically for $y < 0$. The first factor is bounded by (6.27), the second by Lemma 6.19, since $0 \leqq 2yh^+ \leqq s_1$, and the third by Lemma 6.20(i), uniformly in ε. Since $|\langle \mathcal{O}(zg, wh)\rangle| \leqq \langle \mathcal{O}(\operatorname{Re} z\, g, \operatorname{Re} w\, h)\rangle$ and the logarithms of the expectations at real points are convex functions of (x, y), the expectations are bounded on compact subsets of $\mathbb{C} \times D_J(h)$ uniformly in ε, the sets $\{2y \sup h^+ < s_1\}$ exhausting $D_J(h)$ as $\eta \downarrow 0$.

Next we obtain the convergence on a real open set. Consider the family $(g; h^+, h^-)$ with the coefficients $(z; w^+, w^-)$; its expectations at $(z; w, -w)$ are those of $(g; h)$ at $(z; w)$. By (i), applied to the sources $w^+h^+ + w^-h^-$ with $w^\pm \in (0, \delta)$, and by the argument of (i) for the curvature source, the limits exist on a non-empty open subset of the real points.

The domain $\mathbb{C} \times \{|\operatorname{Re} w^\pm| < \delta'\}$ is connected and the family is locally bounded there by the first step; a holomorphic function vanishing on a non-empty open subset of the real points vanishes identically, so by Montel's theorem the whole sequence converges locally uniformly to a holomorphic limit. Restricting to $(z; w, -w)$ gives the convergence on a neighbourhood of the imaginary w-axis, and the same argument on the connected domain $\mathbb{C} \times D_J(h)$ gives (ii).

Finally, at real points the limit is nonnegative and its logarithm is convex, with value 0 at $(0; 0)$. It is positive at every real point of $D_J(h)$: the limits exist on the whole domain by the third step, so at the real points $(z; w)$ with $|w| < \delta$ the limits at $(z; w)$ and $(-z; -w)$ exist and their product is at least 1, since $\langle e^X\rangle\langle e^{-X}\rangle \geqq 1$ on the lattice by Jensen's inequality; and a convex function on the convex set of real points of $D_J(h)$ which is nowhere $+\infty$ and finite on an open subset is finite everywhere.

We can now prove Theorems 2.2 and 2.3 for $h \neq 0$. For a finite family $g_1, \dots, g_n$ of real 2-forms and $h_1, \dots, h_p$ of real functions, all supported in Q_R, and $\zeta = (z, w) \in \mathbb{C}^{n+p}$, write

$$\begin{gathered} g(\zeta) = \sum_i (\operatorname{Re} z_i) g_i\,, \qquad k(\zeta) = \sum_j (\operatorname{Re} w_j) h_j = k_+ - k_-\,, \\ \langle\langle q\rangle\rangle = \sup_x (1 + |x|^2)^2 |q(x)|\,; \end{gathered} \tag{6.89}$$

the tube $\mathcal{D}$ of Sect. 2 is the set of ζ with $\sup k_+ < \sigma_+/2$ and $\sup k_- < H/2$, an open convex set containing every ζ with w purely imaginary. The joint Schwinger functions are

$$S_{n,p}(g_1, \dots, g_n; h_1, \dots, h_p) = \partial_{z_1} \cdots \partial_{z_n} \partial_{w_1} \cdots \partial_{w_p} \Big\langle \mathcal{O}\Big(\sum_i z_i g_i, \sum_j w_j h_j\Big)\Big\rangle\Big|_{\zeta=0}\,. \tag{6.90}$$

Theorem 6.22. *Assume* (6.70).

(i) For every $\zeta \in \mathcal{D}$, every ε and every J with $MJ \geqq R + \ell_c$,

$$\Big|\Big\langle \mathcal{O}\Big(\sum_i z_i g_i, \sum_j w_j h_j\Big)\Big\rangle_{N,D}^{\varepsilon, C_J}\Big| \leqq \exp\Big[\|g(\zeta)\|^2 + \kappa \sum_x \varepsilon^d k_+(\zeta)(x) + K_- \langle\langle k_-(\zeta)\rangle\rangle\Big]\,. \tag{6.91}$$

(ii) The limits (2.17) *and* (2.18) *exist locally uniformly on $\mathcal{D}$ and are holomorphic there; the limit* (2.18) *satisfies* (6.91) *with the continuum norms, is positive at the real points of $\mathcal{D}$, and at the points with z purely imaginary, w real and $k(\zeta) \geqq 0$ it is the nondecreasing limit of* (2.17) *in J.*

(iii) For $\sigma_0 = H/4$ and $K' = \kappa\pi^2 + K_-$, the joint Schwinger functions (6.90) *satisfy, for real test functions,*

$$|S_{n,p}| \leqq e^{K'\sigma_0}(\sqrt{2}\,e)^n (n!)^{1/2} (e/\sigma_0)^p\, p! \prod_i \|g_i\|_2 \prod_j \langle\langle h_j \rangle\rangle\,, \tag{6.92}$$

so that they extend to the Schwartz space and are tempered distributions.

(iv) The limit (2.18) *is invariant under the translations and the cubic group on $\mathcal{D}$, h transforming as a scalar, which is* (2.19) *for $h \neq 0$; it is symmetric in the g_i and in the h_j, real for real test functions and real ζ, and even in z.*

(v) Reflection positivity (2.20) *holds for finite families of sources supported in $\{x_\nu > c\}$ with complex coefficients of amplitude $\sum_j |\operatorname{Re} w_j| \sup|h_j| < \sigma_W$, in the matrix form* (6.51) *and in the polynomial form; the limit* (2.18) *exists at the points which enter* (6.51)*, which need not lie in $\mathcal{D}$.*

(vi) The $S_{n,p}$ are the moments of a unique probability measure on $\mathcal{H}^{-r}(\mathbb{R}^d; \Lambda^2 \oplus \mathbb{R})$, $r \geqq 14$ even, for which the statements of Theorem 6.16(ii)–(iv) hold, with the Euclidean clause of (iii) there replaced by invariance under the translations and the cubic group.

Proof. (i) Since $|\langle \mathcal{O}(\zeta)\rangle| \leqq \langle \mathcal{O}(\operatorname{Re}\zeta)\rangle$ it suffices to bound the expectation at the real point $(g,k) = (g(\zeta), k(\zeta))$. Integrating over the scalar field first, the expectation with the source k in the exponent is a measure of the class (6.1) in the vector field, with the same Gaussian part, so Lemma 6.7 gives $\langle \mathcal{O}(g,k)\rangle \leqq e^{\|g\|^2} \langle \mathcal{O}(0,k)\rangle$ as in Theorem 6.9(i); the Cauchy–Schwarz inequality gives

$$\langle \mathcal{O}(0, k_+ - k_-)\rangle \leqq \langle \mathcal{O}(0, 2k_+)\rangle^{1/2} \langle \mathcal{O}(0, -2k_-)\rangle^{1/2}\,; \tag{6.93}$$

and $0 \leqq 2k_+ \leqq s_1$, $0 \leqq 2k_- \leqq H$ on $\mathcal{D}$, so Lemmas 6.19 and 6.20(ii) bound the two factors.

(ii) Each lattice expectation is an entire function of ζ. On compact subsets of $\mathcal{D}$ the family is bounded uniformly in ε by (6.91), and it converges at every real point of $\mathcal{D}$ by Theorem 6.21(ii), since there $2 \sup k_+ < \sigma_+ = s_1$ and the negative part is unrestricted; the real points form a non-empty open subset of $\mathbb{R}^{n+p}$, a set of uniqueness in the connected open set $\mathcal{D}$, and Vitali's theorem gives the limit (2.17) locally uniformly on $\mathcal{D}$, with (6.91) in the limit.

For the limit in J, write again $h_j = h_j^+ - h_j^-$, with $h_j^\pm$ supported in Q_{R+1}, so that (6.91) holds for the split family when $MJ \geqq R + 1 + \ell_c$, and consider the family $(g; h^+, h^-)$ on the domain $\mathcal{D}''$ of the points with $|\operatorname{Re} w_j^\pm| < \delta_j$, where $\sum_j \delta_j (\sup h_j^+ + \sup h_j^-) < H/2$, which lies in the tube of that family. At the points of $\mathcal{D}''$ with z purely imaginary and all $w_j^\pm > 0$ the observable is of the class of (3.24), so by Theorem 3.2 the lattice expectations are nondecreasing in J, and so are their limits in ε, which are bounded by (6.91); these points form an open subset of the totally real subspace $(i\mathbb{R})^n \times \mathbb{R}^{2p}$, a set of uniqueness, so Montel's theorem gives the convergence as $J \to \infty$ on $\mathcal{D}''$.

At $w_j^\pm = \pm w_j$ this is the convergence on a neighbourhood $\mathcal{D}'$ of the imaginary w-axis in $\mathcal{D}$, and a second application of Montel's theorem on $\mathcal{D}$, with the uniqueness set $\mathcal{D}'$, gives (2.18) locally uniformly on $\mathcal{D}$. Positivity at real points follows from the convexity of the logarithm as in Theorem 6.21, and the monotonicity in J at the points stated from Theorem 3.2.

(iii) Normalize $\|g_i\|_2 = 1$ and put $\rho_j = \sigma_0/(p\langle\langle h_j\rangle\rangle)$. On the polydisc $|w_j| \leqq \rho_j$ the source $k(\zeta)$ has supremum at most $\sigma_0 < H/2$, so the polydisc lies in $\mathcal{D}$; there (6.91) holds for the limit with the continuum norms, by (ii), and $\int k_+\,dx \leqq \pi^2 \langle\langle k_+\rangle\rangle$ bounds its exponent by

$$\|g(\zeta)\|_2^2 + K'\sigma_0\,. \tag{6.94}$$

Cauchy's estimate for the mixed derivative in the w_j at $w = 0$ gives the factors

$$(p/\sigma_0)^p \prod_j \langle\langle h_j\rangle\rangle\, e^{K'\sigma_0}\,, \qquad p^p \leqq e^p p!\,, \tag{6.95}$$

times an entire function of z bounded by $e^{\|g(z)\|_2^2}$, and Cauchy's estimate in the z_i on the polydisc of radius $(2n)^{-1/2}$ gives the remaining factors as in Theorem 6.9(iii). The extension to the Schwartz space follows as there, with $\langle\langle h\rangle\rangle \leqq \|h\|_4$, the norm (6.25) applied to h.

(iv) Lemma 6.10 holds for h of either sign, after the splitting $h = h^+ - h^-$ of (ii), at the points where z is purely imaginary and both w^+, w^- are nonnegative, by the increasing clause of Theorem 3.2 applied to the source $w^+h^+ + w^-h^- \geqq 0$; at these points the limits of the three terms exist by (ii), so (2.19) holds there for translations by $b \in \varepsilon_K\mathbb{Z}^d$, $|b|_\infty \leqq M$, and by the identity theorem on the connected tube of the split family, for all $\zeta \in \mathcal{D}$.

The logarithm of the limit is a convex function of the real sources (g, h) on the open convex set where $\sup h_+ < \sigma_+/2$ and $\sup h_- < H/2$, the real points of $\mathcal{D}$; it is bounded above there by (6.91), hence continuous in the norm $\|g\|_2 + \langle\langle h\rangle\rangle$. Since $T_b g \to g$ and $T_b h \to h$ in these norms as $b \to 0$ and the translations by $b \in \bigcup_K \varepsilon_K\mathbb{Z}^d$ are dense, this gives the invariance under all translations as in Theorem 6.11, first on the real points of $\mathcal{D}$ and then, by the identity theorem, on $\mathcal{D}$; the invariance under the cubic group follows as in Theorem 6.12. The symmetries are those of the lattice expectations.

(v) The proof of Theorem 6.15 applies: for a family with scalar sources we split each $h_{il} = h_{il}^+ - h_{il}^-$ as above, so that all the scalar members of the enlarged family, which contains the reflected sources, are nonnegative; the positive part of the total source at the points which enter (6.51) is at most twice the amplitude, so for amplitude less than $\sigma_W = \sigma_+/8$ these points lie in the set where $2\sup k_+ < \sigma_+/2$, with no condition on the negative part.

On this set the expectations on the cubes are bounded uniformly in J. All scalar members $h_{il}^\pm$ of the enlarged family being nonnegative, we write the total source at a point entering (6.51) as $k'_+ - k'_-$, where k'_+ (resp. k'_-) is the sum of the members with positive (resp. negative) real coefficient, weighted by the moduli of these coefficients; both are smooth and nonnegative, and $\sup k'_+$ is at most twice the amplitude.

By the Cauchy–Schwarz inequality, the curvature source being removed first as in the proof of (6.91), the expectation is bounded by $\langle\mathcal{O}(0, 2k'_+)\rangle^{1/2}\langle\mathcal{O}(0, -2k'_-)\rangle^{1/2}$; the first factor is bounded by Lemma 6.19, since $2\sup k'_+ \leqq s_1$, and the second is nonincreasing in J by Theorem 3.2 and, on a fixed cube, bounded uniformly in ε by Lemma 6.20(i), $2k'_-$ being smooth and nonnegative. At the points where z is purely imaginary and all scalar coefficients are positive the limits in J exist by monotonicity, and Montel's theorem gives the existence of (2.18) on this set, which is the required result.

The comparison (6.53) holds there by (3.50), (3.52) with the margin of $2n$ cubes between the box and the two cubes which bracket it, and by Proposition 3.3 for the box expectation, which is reflection positive for all amplitudes, and the argument of Theorem 6.15 gives (6.51); the polynomial form follows by differentiation at the origin, where the limit is holomorphic.

(vi) The proof of Theorem 6.16 applies to the pair of fields $(F, :\phi^2:)$ with the bounds (6.91), (6.92) in place of (6.27), (6.29), the characteristic functional being the limit (2.18) at purely imaginary coefficients, which lie in $\mathcal{D}$. The carrier is $\mathcal{H}^{-r}$ for even $r \geqq 14$ because the second moments of $:\phi^2:(h)$ are bounded by $C\langle\langle h\rangle\rangle^2$ and $\langle\langle h\rangle\rangle \leqq C\|h\|_{10}$ in the norm (6.54), so that the sum (6.58) converges for $r > 10 + d$; the reflection positivity needed is that of (v) at purely imaginary coefficients, where the amplitude is zero.

6.7. Rotation Invariance

In Sect. 4 the proof of Theorem 2.4 was reduced to the lower bound (4.9) and to Proposition 4.3, and in Sect. 2 the proof of Theorem 2.5 to Theorem 2.4 and to the equality in (2.28). We give the proofs of (4.9) and of the equality in (2.28) here. The estimates of the renormalization group analysis which enter them, the bound (6.97) on the one-point function of the periodic models, the trace bound (6.106) and the two estimates (6.109) on the models with a heavy shell, are taken from [M] and stated where they are used. For a pair $(D_{J'}, \omega)$ of Sect. 4, a real 2-form

g and a real z the limit $W(izg; D_{J'}, \omega)$ of (2.26) exists by Theorem 2.4 and lies in $[-1, 1]$, the expectation of $\sin(zF(g))$ vanishing by the symmetry $(A, \theta) \to (-A, -\theta)$.

The inequalities (3.24)–(3.26) of Proposition 3.1 hold for every finite lattice model of the form (2.5) in which the vector field has a positive definite Gaussian action and the scalar field is present on an arbitrary set of sites with nonnegative site-dependent masses: this is Lemma 6.1, whose proof uses nothing else. In particular they hold on the torus $D_{J'}$ with the mass term ω, and for the models of Lemma 6.26 below. The remark of Sect. 1 on periodic boundary conditions concerns the monotonicity in the volume, Theorem 3.2, only. Lemmas 6.1, 6.2 and 6.7 hold in the same generality and are the forms used below.

Theorem 6.23. *Let D_J be a torus of the class of Sect. 4, ω a real function on its sites with $|\omega| \leqq \Omega$, and g a real 2-form supported in D_J. Then for real z, on both lattices of Sect. 4 and in the torus convention of [I] (Sect. 6.6), in which the counterterm, the Wick constant and the constants E_0, E_1 are computed with the periodic propagators,*

$$Z(D_J, zg, -\omega) \geqq e^{-C|D_J|\varepsilon^{q'}}, \tag{6.96}$$

where $Z(D_J, zg, -\omega)$ is the generating functional of Sect. 4 with the real source $e^{zF(g)}$ and $h = -\omega$, and C and q' depend on Ω, $|D_J|$ and the coupling constants, but not on g, z, ε or the sign of ω.

The dependence of q' on ω does not matter: in Sect. 4 the lower bound (6.96) divides only the large field term $e^{-cp(L^k\varepsilon)^2 + C|D_J|\varepsilon^{-q}}$ of (4.8), the other error terms being proportional to the partition functions themselves, and $k = K - [\vartheta K]$ there, so that $L^k\varepsilon \leqq \varepsilon^{\vartheta/2}$ for K large and $p(L^k\varepsilon)^2$, a power of $\ln(L^k\varepsilon)^{-1}$ with an exponent greater than one,[22] exceeds $(q + q')\ln\varepsilon^{-1}$ for every fixed ω and D_J. The functional of Sect. 4, in the convention of this paper, is the torus-convention functional at the mass term $\omega + \frac{1}{2}u_\varepsilon$, u_ε a constant bounded uniformly in ε, times a factor independent of the fields and of g, and the large field term of (4.8) carries the same factor (Sect. 4); this is how (6.96) enters (4.8).

Proof. By Jensen's inequality and the symmetry $(A, \theta) \to (-A, -\theta)$, under which $F(g)$ changes sign while the mass term is invariant, $Z(D_J, zg, -\omega) \geqq Z(D_J, 0, -\omega)$ for real z.

Next write $X = -\sum_x \varepsilon^d \omega(x) :r(x)^2:$, where $:r(x)^2: = r(x)^2 - 2C(x, x)$ with the periodic propagator C, so that $Z(D_J, 0, -\omega) = Z(D_J, 0, 0)\langle e^X \rangle$, the expectation being that of the torus model with $g = 0$ and $\omega = 0$. By Jensen's inequality $\langle e^X \rangle \geqq e^{\langle X \rangle}$. By translation invariance $\langle :r(x)^2: \rangle$ does not depend on x, and on both lattices of Sect. 4 the one-point function satisfies

$$|\langle :r(0)^2: \rangle| \leqq C'(1 + \ln\varepsilon^{-1}) \tag{6.97}$$

uniformly in ε; this is the first of the estimates taken from [M]. Hence $\langle X \rangle \geqq -C'\Omega|D_J|(1 + \ln\varepsilon^{-1})$. Since $Z(D_J, 0, 0) \geqq e^{-C|D_J|}$ in the torus convention, by the lower bound of [Ba 1],[23] (6.96) follows with $q' = C'\Omega|D_J|$ and C replaced by $C + C'\Omega$; neither depends on g, z, ε or the sign of ω.

We now turn to the equality in (2.28).

Theorem 6.24. *For every real 2-form $g \in C_0^\infty(\mathbb{R}^d)$ and every real z, equality holds in* (2.28)*:* $\lim_{b\to\infty} I_b(izg) = \langle \mathcal{O}(izg, 0) \rangle$.

Fix a real 2-form g supported in Q_b and a real z. We choose $b' \geqq b+3$ and $D \geqq \max\{1, 3/\mu_0\}$, a cube C_L of the family (2.7) which contains every point within distance $D + 3$ of $Q_{b'}$, and a torus $D_{J'}$ of the class of Sect. 4 containing C_L (Fig. 4).

Let ψ and ψ_U be smooth functions on $D_{J'}$ with values in $[0, 1]$, $\psi = 0$ within distance 1 of $Q_{b'}$ and $\psi = 1$ at distance greater than 2 from $Q_{b'}$, $\psi_U > 0$ exactly at distance greater than 2 from $Q_{b'}$ and $\psi_U = 1$ on the set U of points at distance greater than 3 from $Q_{b'}$, so that

[22] (3.1) of [I].

[23] (1.14) of [Ba 1], which is (4.3) at $g = 0$ and $\omega = 0$.

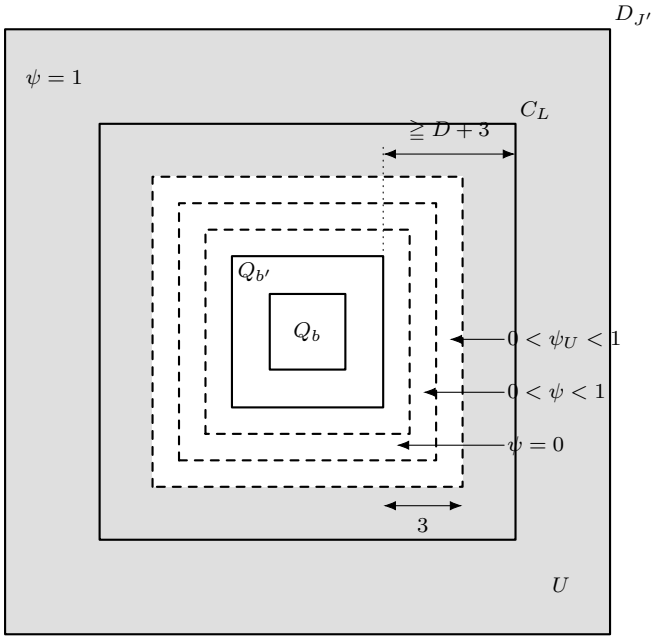


Fig. 4: The regions of Sect. 6.7: the source g in Q_b; the cut-off ψ, which vanishes within distance 1 of $Q_{b'}$ and equals 1 beyond distance 2; the function ψ_U, positive beyond distance 2 and equal to 1 on the shell U (shaded) beyond distance 3, where the mass term has height n; the cube C_L of (2.7), containing every point within distance $D+3$ of $Q_{b'}$, and the torus $D_{J'}$

$U \supset D_{J'} \setminus C_L$ together with the outer layer of C_L; the sets $Q_{b'}$, C_L, U and the torus are unions of blocks of the final scale of the renormalization transformation, as the class requires. For a height $n \geqq 1$ we put

$$\omega_n = \psi + (n-1)\psi_U\,, \qquad \Psi_K(n) = \langle \mathcal{O}(izg,0)\rangle_p^{\varepsilon_K, D_{J'}, \omega_n}\,, \qquad \Psi_K(\infty) = \lim_{n\to\infty} \Psi_K(n)\,, \quad (6.98)$$

where $\Psi_K(\infty)$ is the expectation in the model in which the scalar field vanishes on the set $\{\psi_U > 0\}$, the mass term ψ being kept on the sites within distance 2 of $Q_{b'}$, where $\psi_U = 0$. The limit exists at fixed K: after the change of variables $\phi(x) \to \phi(x)/\sqrt{n-1}$ at the sites of $\{\psi_U > 0\}$, whose Jacobian cancels in the ratio, dominated convergence applies to numerator and denominator for n large at fixed K, the Gaussian factors $e^{-\varepsilon^d \psi_U(x)|\phi(x)|^2}$ being integrable, since ψ_U has a positive minimum on the finitely many sites where it is positive, and dominating the counterterm $\frac{1}{2}\delta m^2|\phi|^2/(n-1)$, and the limit is the expectation of the model in which the scalar field vanishes on $\{\psi_U > 0\}$. By the inequality (3.25) $\Psi_K(n)$ is nonincreasing in n, so that $\Psi_K(n) \geqq \Psi_K(\infty)$.

The mass terms ω_n belong to the class of Sect. 4 once $\sum_x \varepsilon^d \psi_U \geqq c_0 |D_{J'}|$, which holds for J' large, and they vanish near Q_b, so that $(D_{J'}, \omega_n) \in \Omega_b$.

Proposition 6.25. *For every K,*

$$\Psi_K(\infty) \leqq \langle \mathcal{O}(izg,0)\rangle_{N,D,*}^{\varepsilon_K, C_L} + (e^{\tau}-1)^{1/2}\,, \qquad \tau \leqq C_1(1+b')^d D^{-1} e^{-\mu_0 D}\,, \quad (6.99)$$

where the first term is the expectation of Sect. 6.1 in which the scalar field vanishes outside C_L, and C_1 depends on d and μ_0.

The proof uses the following comparison of two Gaussian measures of the vector field.

Lemma 6.26. *Let $\langle\cdot\rangle_\natural$ and $\langle\cdot\rangle_N$ be two expectations with the same scalar field variables, at the sites within distance 2 of $Q_{b'}$, and the same single-site measures and hopping terms, coupled to the vector field as in (2.5); in the first the vector field has the law of the periodic Gaussian*

of $D_{J'}$ conditioned on the vanishing of all its variables except those of the bonds of C_L, in the second the law of the Gaussian with Neumann boundary conditions on C_L. Let $\Theta = e^{iF(g)}$ with g supported in $Q_{b'}$. Then

$$0 \leqq \langle\Theta\rangle_\natural - \langle\Theta\rangle_N \leqq (e^\tau - 1)^{1/2}\,, \tag{6.100}$$

with τ as in (6.99).

Proof. The two Gaussian measures are massive free fields on the bonds of C_L which differ only in their boundary conditions on ∂C_L, at distance at least D from the bonds and sites within distance 2 of $Q_{b'}$, which carry the scalar field variables. The quadratic form of the second is the Neumann form of Lemma 6.8, that of the first is obtained from the periodic form by setting the exterior variables to zero, and it dominates the Neumann form; so the first measure is the second with a positive quadratic form added at the boundary, and the inequality (3.24), in the form of Lemma 6.1, gives $\langle\Theta\rangle_N \leqq \langle\Theta\rangle_\natural$ for $\Theta = e^{iF(g)}$.

For the upper bound let $\gamma_\natural$, γ_N be the laws of the variables of the bonds within distance 2 of $Q_{b'}$ under the two Gaussians, with covariances $C_\natural \leqq C_N$, and

$$\rho = d\gamma_\natural/d\gamma_N = e^c e^{-\frac12\langle A,SA\rangle}\,, \qquad S = C_\natural^{-1} - C_N^{-1} \geqq 0\,, \quad c \geqq 0\,. \tag{6.101}$$

The matter couples to the vector field through these variables only, so $\langle\Theta\rangle_\natural = \langle\Theta\rho\rangle_N/\langle\rho\rangle_N$, where $\langle\cdot\rangle_N$ is the full expectation of the second model, and

$$\langle\Theta\rangle_\natural - \langle\Theta\rangle_N = \frac{\langle(\Theta - \langle\Theta\rangle_N)(\rho - \langle\rho\rangle_N)\rangle_N}{\langle\rho\rangle_N}\,, \qquad |\langle\Theta\rangle_\natural - \langle\Theta\rangle_N| \leqq \Big(\frac{\langle\rho^2\rangle_N}{\langle\rho\rangle_N^2} - 1\Big)^{1/2} \tag{6.102}$$

by the Cauchy–Schwarz inequality and $|\Theta| = 1$. Now

$$e^c(1 - \tfrac12\langle A, SA\rangle) \leqq \rho \leqq e^c\,, \tag{6.103}$$

and by Lemma 6.7, in the form $\langle A(v)^2\rangle_N \leqq \langle v, C_N v\rangle$ which follows from its exponential bound at real argument, applied to the eigenvectors of S,

$$\langle\langle A, SA\rangle\rangle_N \leqq \mathrm{Tr}(SC_N) = \mathrm{Tr}((C_N - C_\natural)C_\natural^{-1}) = \tau'\,. \tag{6.104}$$

Hence, with $\tau = 2\tau'$, for $\tau' \leqq 1$

$$\frac{\langle\rho^2\rangle_N}{\langle\rho\rangle_N^2} \leqq (1 - \tau'/2)^{-2} \leqq e^\tau\,, \tag{6.105}$$

since $-2\ln(1 - \tau'/2) \leqq 2\tau'$ on $[0,1]$, and (6.100) follows from (6.102); if $\tau' > 1$, (6.100) holds since $(e^\tau - 1)^{1/2} > 2 \geqq |\langle\Theta\rangle_\natural - \langle\Theta\rangle_N|$. It remains to bound τ'. By the Markov property of the two Gaussian measures the density ρ depends on the variables of the inner boundary layer Λ of the set of coupled bonds only, so that $\tau' = \mathrm{Tr}_\Lambda((C_N - C_\natural)_\Lambda(C_\natural)_\Lambda^{-1})$, where $(C_N - C_\natural)_\Lambda$ is the restriction of a kernel smooth on the unit scale and of size $e^{-2\mu_0 D}$, by the exponential decay of the massive propagators at the rate μ_0 in the units of $\mathbb{R}^d$, and $(C_\natural)_\Lambda$ is the covariance of the marginal on the layer. The trace is bounded by

$$\tau' \leqq \tfrac12 C_1(1 + b')^d D^{-1} e^{-\mu_0 D} \tag{6.106}$$

uniformly in ε; this is the second of the estimates taken from [M].

Proof of Proposition 6.25. In the model of $\Psi_K(\infty)$ we add $s\,\varepsilon^d A_b^2$ to the action for every bond b of $D_{J'}$ which is not a bond of C_L and let $s \to \infty$. By the inequality (3.24), in the form $\langle\Theta; |A(u)|^2\rangle \leqq 0$ of Lemma 6.1, the expectation of $e^{izF(g)}$ does not decrease, and in the limit the vector field has the conditioned law of Lemma 6.26.

Every site of C_L adjacent to the exterior, and every exterior site, lies in U, where the scalar field vanishes, so no term of the action couples C_L to the exterior; the sites within distance 2

of $Q_{b'}$ carry the mass term ψ and the other sites a vanishing field. Lemma 6.26 replaces the conditioned Gaussian by the Neumann Gaussian of C_L at the cost of $(e^\tau - 1)^{1/2}$. Finally, by Lemma 6.2 the expectation of $e^{izF(g)}$ does not decrease when the scalar field is given back at the sites of $C_L \cap \{\psi_U > 0\}$, and by the inequality (3.25) it does not decrease when the mass term ψ is removed. Here, as in Sect. 3.2, the counterterm δm^2 of every model from (6.98) on, of the tori $(D_{J'}, \omega)$ of Sect. 4 and of the models of Sect. 6.1, is the one defined with free boundary conditions, so that the last expression is the expectation of Sect. 6.1 on C_L, the scalar field vanishing outside C_L; Theorem 6.23 alone is stated in the torus convention, and enters Sect. 4 through the remark which follows it.

Proposition 6.27. *Let $D_{J'}$, C_L, ψ and ψ_U be as above, with the corners of the cube $Q_{b'+3}$, the complement of the shell U, at coordinates congruent to $M'/2$ modulo M' in units of the block of the final scale, where $M = 5M'$, M' an even integer, at least 10; this holds for every M admitted in Sect. 2. For every $\eta > 0$ there is n_1 such that*

$$\limsup_{K\to\infty} [\Psi_K(n) - \Psi_K(\infty)] \leqq \eta \qquad \text{for } n \geqq n_1 \,. \tag{6.107}$$

Proof. Write $\Psi_K(n; z)$ and $\Psi_K(\infty; z)$ for the expectations (6.98) with $e^{izF(g)}$ replaced by $e^{zF(g)}$, $z \in \mathbb{C}$, the real number z of (6.98) being fixed at 1 by absorbing it into g, so that $\Psi_K(n) = \Psi_K(n; i)$; at fixed K these are entire functions of z.

(i) By Lemma 6.7, in every one of these models the vector field Gaussian is the periodic massive Gaussian of $D_{J'}$, whose quadratic form is at least μ_0^2 times the ℓ^2 norm, so that

$$|\Psi_K(n; z)| \leqq \exp[2\mu_0^{-2} \|\varepsilon^{-1} d^* g\|^2] = M_0 \qquad \text{for } |z| \leqq 2 \text{ and all } n \leqq \infty \,, \tag{6.108}$$

uniformly in K, where $\|\varepsilon^{-1} d^* g\|^2 = \sum_x \varepsilon^d |\varepsilon^{-1} d^* g(x)|^2$ is the norm of (6.21), bounded uniformly in ε by $C\|g\|_4$.

(ii) For real $z \in [-1, 1]$ we use two estimates, obtained by carrying the renormalization transformation of [I] through the shell U, on the torus $D_{J'}$ with the mass term ω_n as a scalar source, the scalar field on the shell being integrated out to the scale where the mass n becomes large, so that the small field integral at the final scale is an integral $I_n(z)$ over the fields of a fixed test set with a reduced action. For every $\delta > 0$ there are a stopping scale and numbers $a(\delta)$, $b(\delta)$, which tend to zero with δ and satisfy $b \leqq \frac{1}{2}$, a height $n_0(\delta)$, a function $r(n; \delta)$ with $r(n; \delta) \to 0$ as $n \to \infty$, and $K_0(\delta)$, independent of n and n', such that for $K \geqq K_0(\delta)$, $n_0 \leqq n \leqq n' < \infty$ and $z \in [-1, 1]$:

$$e^{-a} I_n(z) \leqq Z_n(zg) \leqq e^a I_n(z) + b\, Z_n(0) \,, \tag{6.109}$$

where Z_n is the partition function with the mass term ω_n; and the reduced actions at the heights n and n' differ by $c_{n,n'} + \rho_z$ with $c_{n,n'}$ independent of z and $|\rho_z| \leqq r(n; \delta)$ on the test set. These are the two estimates taken from [M]; they are proved there for the mass term $\psi + (n-1)\mathbf{1}_U$, U being a union of blocks of the final scale on a neighbourhood of which $\psi \equiv 1$, as the two-sided bound requires, and extend to ω_n, which differs from it only on the layer $\{\psi_U > 0\} \setminus U$ of points at distance between 2 and 3 from $Q_{b'}$.[24]

The first estimate is the two-sided bound of Theorem 3.1 of [I] for the model in which the shell mass is kept inside the scalar covariance at every scale, the small field integral $I_n(z)$ and the large field remainder being those of this model, with a, b and K_0 independent of the height n; it is not a consequence of Theorem 3.1 of [I] and Theorem 6.23, whose lower bound carries

[24] The proof follows the renormalization transformation of [I] through the shell; its new elements are the heavy insertions on the shell (the local vertex of the two-loop subgraphs with a vertex other than their two ends, the insertions of vertex type, the single-scale lines deep in the shell), the block part of the one-loop term of order z^2, the case of a light scale at the first step, and the two lower bounds on the stopping scale under which the reduction applies.

the exponent q' growing with $\int \omega_n$. The second expresses that, once the scalar field on the shell has been integrated out, the dependence of the reduced action on the height n enters only through terms which become negligible as $n \to \infty$, the test set being the same for all heights.

(iii) From (ii), with $J_n(z) = I_n(z)/I_n(0)$,

$$e^{-2a}(1-b)J_n(z) \leqq \Psi_K(n;z) \leqq e^{2a}J_n(z) + b \qquad \text{for } z \in [-1,1]\,, \tag{6.110}$$

so

$$|\Psi_K(n;z) - J_n(z)| \leqq m_1(\delta)\,, \qquad J_n(z)/J_{n'}(z) \in [e^{-2r}, e^{2r}]\,, \quad r = r(n;\delta)\,, \tag{6.111}$$

with $m_1(\delta) \to 0$ as $\delta \to 0$, the second since the integrands of the quotient differ pointwise by a factor between e^{-2r} and e^{2r}, the constant $c_{n,n'}$ cancelling. Hence, with $J_{n'}(z) \leqq 2e^{2a}M_0$,

$$|\Psi_K(n;z) - \Psi_K(n';z)| \leqq m(n,\delta) = 2m_1(\delta) + 2e^{2a}M_0\big(e^{2\sup_{n''\geqq n} r(n'';\delta)} - 1\big) \tag{6.112}$$

for $z \in [-1,1]$, $n' \geqq n \geqq n_0$ and $K \geqq K_0(\delta)$, and letting $n' \to \infty$ at fixed K, $|\Psi_K(n;z) - \Psi_K(\infty;z)| \leqq m(n,\delta)$ on $[-1,1]$.

(iv) The difference $X(z) = \Psi_K(n;z) - \Psi_K(\infty;z)$ is entire, bounded by $2M_0$ on $|z| \leqq 2$ and by $m(n,\delta)$ on $[-1,1]$; by the two constants theorem for the subharmonic function $\ln|X|$ on the disc of radius 2 slit along $[-1,1]$,

$$|X(i)| \leqq (2M_0)^{1-\vartheta} m(n,\delta)^{\vartheta}\,, \tag{6.113}$$

where $\vartheta \in (0,1)$ is the harmonic measure of the slit at the point i, a numerical constant. Since $\Psi_K(n;i)$ and $\Psi_K(\infty;i)$ are real, $X(i) = \Psi_K(n) - \Psi_K(\infty)$. Given η, we choose δ so small and then $n_1 \geqq n_0(\delta)$ so large that $(2M_0)^{1-\vartheta} m(n_1,\delta)^{\vartheta} \leqq \eta$; then (6.107) holds for $n \geqq n_1$.

The condition on the corners of $Q_{b'+3}$, the complement of the shell, together with the requirements that $Q_{b'+3}$ be a union of blocks of the final scale of side at least two such blocks, at distance at least three such blocks from its periodic images, and that the sides of the torus be multiples of that block, is the one under which the blocks of all scales of the renormalization transformation are compatible with the shell; every interval of M' blocks of the final scale contains an admissible value, so the condition does not restrict the choice of b, $b' \geqq b+3$ being otherwise free.

Proof of Theorem 6.24. Let g be supported in Q_b and let $\eta > 0$. Choose b', D, C_L, $D_{J'}$, ψ and ψ_U as above with $\tau \leqq \eta$, and $n \geqq n_1(\eta)$ as in Proposition 6.27. Then $(D_{J'}, \omega_n) \in \Omega_b$, and by Theorem 2.4, Propositions 6.27 and 6.25, Lemma 6.4 and Theorem 3.2,

$$\begin{aligned} W(izg; D_{J'}, \omega_n) &= \lim_{K\to\infty} \Psi_K(n) \leqq \eta + \limsup_{K\to\infty} \Psi_K(\infty) \\ &\leqq \eta + \langle \mathcal{O}(izg,0)\rangle_{N,D}^{C_{L+1}} + (e^{\eta}-1)^{1/2} \leqq \eta + \langle \mathcal{O}(izg,0)\rangle + (e^{\eta}-1)^{1/2}\,, \end{aligned} \tag{6.114}$$

the expectation of Sect. 6.1 on C_L being at most that of (2.10) on C_{L+1} in the limit $K \to \infty$, and the latter nondecreasing in L for the observable $e^{izF(g)}$. Hence $I_b(izg) \leqq \langle \mathcal{O}(izg,0)\rangle$ for every b, and with (2.28) the theorem follows.

Proof of Theorem 2.5. It remains to justify two steps of the argument of Sect. 2. First, the infimum of the limits (2.26) over the pairs $(D_{J'}, \omega)$ with ω vanishing near the rotated cube $R^{-1}Q_b$ lies between $I_{7b/5}(izg)$ and $I_{5b/7}(izg)$, since $Q_{5b/7} \subset R^{-1}Q_b \subset Q_{7b/5}$ and the set of admissible pairs shrinks as the cube grows; so it converges to $\langle \mathcal{O}(izg,0)\rangle$ as $b \to \infty$ by Theorem 6.24, and (2.30) follows for z purely imaginary. Second, by Theorem 2.4 the limit (2.26) is the same on the two lattices, so the chain of Sect. 2 from the rotated cube on the rotated lattice to the tori $(D_{J'}, \omega)$ is valid.

The passage from the rotations $R^{\pm 1}$ in the coordinate planes to all of $O(d)$ and from purely imaginary to all z is the one given after (2.30): the rotations generated by R in a coordinate plane

are dense in the rotations of that plane, since $\theta_0/2\pi$ is irrational (Sect. 4); $R \mapsto \langle \mathcal{O}(iRg, 0)\rangle$ is continuous by Theorem 6.9(iv), since $R \mapsto Rg$ is continuous into L^2; the reflections are covered by Theorem 6.12; and both sides of (2.31) are entire in z and agree for z purely imaginary. This proves Theorem 2.5, and with it the Euclidean clauses of Theorem 6.16.

6.8. The Condition on the Coupling Constants

The results with scalar sources were proved under the condition (6.70): the difference $\sigma_1(M_1)$ of (6.69) between the renormalized bare coefficients at the masses m and M_1, a condition on the limit $\varepsilon \to 0$, is positive for some $M_1 \in (0, m)$. No other condition is placed on e, λ, m, μ_0 in this paper: Theorems 2.1–2.3 at $h = 0$, Theorems 2.4–2.6 for the field strength and the results of Sects. 6.1–6.5 and 6.7 hold for all $e > 0$ and $\lambda, m, \mu_0 > 0$. The smallness conditions of [Ba 1–4] are conditions on the couplings $e^2 L^k \varepsilon$, $\lambda L^k \varepsilon$ and $m^2 (L^k \varepsilon)^2$ at the scales where they are used, and so are satisfied for all e, λ and m when $L^K \varepsilon$ is small (Sect. 5.3 of [I]); no condition is placed on $\mu_0 L^k \varepsilon$, and no relation between μ_0 and m is needed. The constants of Sect. 6.6 depend on μ_0 and are not uniform in it.

By Theorem 5.1 of [I], $\sigma_1(M_1)$ is a polynomial in e^2 and λ of degree two whose coefficients are convergent continuum Feynman integrals depending on m, M_1 and μ_0, and its one-loop part is[25]

$$\begin{aligned}\sigma_1(M_1) = {} & \tfrac{1}{2}(m^2 - M_1^2) + \frac{2\lambda(m - M_1)}{\pi} \\ & - e^2 \, \frac{(m - M_1)(m\mu_0 + mM_1 + M_1\mu_0)}{8\pi(\mu_0 + M_1)(m + \mu_0)} + \sigma_1^{(2)}(M_1)\,,\end{aligned} \tag{6.115}$$

where $\sigma_1^{(2)}$ collects the terms of orders λ^2, $e^2\lambda$ and e^4. Since $\sigma_1^{(2)}(M_1)$ is bounded by $C(\lambda^2 + e^2\lambda + e^4)$ with C depending on m, M_1 and μ_0, the condition (6.70) holds, with $M_1 = \frac{7}{10}m$, whenever e^2 and λ are small compared with m, the threshold depending on μ_0/m only, since $\sigma_1(M_1) = m^2 F(\lambda/m, e^2/m, M_1/m, \mu_0/m)$ by dimensional analysis.

Lemma 6.28. *To one-loop order, that is with $\sigma_1^{(2)}$ omitted in* (6.115)*, the condition* (6.70) *holds if and only if*

$$e^2 < e_c^2 = \max\Big\{ \frac{8\pi(m + \mu_0)^2}{m + 2\mu_0}\Big(1 + \frac{2\lambda}{\pi m}\Big),\ 4\pi(m + \mu_0)\Big(1 + \frac{4\lambda}{\pi m}\Big)\Big\}\,, \tag{6.116}$$

the first entry being not smaller than the second if and only if $4\lambda\mu_0 \leqq \pi m^2$.

Proof. The one-loop part of (6.115) is $(m - M_1)\,\beta(M_1)$ with

$$\beta(M_1) = \tfrac{1}{2}(m + M_1) + 2\lambda/\pi - e^2(8\pi(m + \mu_0))^{-1}[(m + \mu_0) - \mu_0^2(\mu_0 + M_1)^{-1}]\,, \tag{6.117}$$

since $m\mu_0 + mM_1 + M_1\mu_0 = (m + \mu_0)(\mu_0 + M_1) - \mu_0^2$. The function β is continuous and convex on $[0, m]$, being the sum of a linear and a convex function, so $\beta(M_1) > 0$ for some $M_1 \in (0, m)$ if and only if $\beta(0) > 0$ or $\beta(m) > 0$; these two inequalities are the two entries of (6.116), as required.

We now consider the condition (6.70) for small couplings and at $e = 0$.

Proposition 6.29. *(i) For every $\mu_0 > 0$ there is $\kappa_0 > 0$, depending on μ_0/m, such that* (6.70) *holds with $M_1 = \frac{7}{10}m$ whenever $\lambda, e^2 \leqq \kappa_0 m$. (ii) Let $e = 0$. Then*

$$\sigma_1(M_1) = \tfrac{1}{2}(m^2 - M_1^2) + 2\lambda(m - M_1)/\pi - (4\lambda^2/\pi^2)\ln(m/M_1)\,, \tag{6.118}$$

so that, for every μ_0, (6.70) *holds with $M_1 = \frac{7}{10}m$ if and only if $\lambda < \lambda_c m$, where*

$$\lambda_c = \frac{\pi}{40\ln(10/7)}\Big(3 + \sqrt{9 + 102\ln(10/7)}\Big) \tag{6.119}$$

[25] The polynomial is (5.11) of [I] and its one-loop part (5.13) there, read off from (1.23) of [Ba 3] with $N = 2$.

is the positive root of

$$\frac{4\ln(10/7)}{\pi^2}\lambda_c^2 - \frac{3}{5\pi}\lambda_c - \frac{51}{200} = 0\,; \tag{6.120}$$

and (6.70) *holds for some* $M_1 \in (0, m)$ *if and only if* $\lambda < \frac{\pi}{4}(1+\sqrt{5})\,m$.

Proof. (i) is the remark following (6.115). (ii) At $e = 0$ the terms of order λ^2 in δm^2 are the sunset graph, the tadpole with a tadpole insertion and the tadpole with the insertion of the counterterm of order λ; the last two cancel by the zero-momentum condition defining δm^2, the inserted quantities being $16\lambda C^\varepsilon(0)$ and $-16\lambda C^\varepsilon(0)$ for the two real components of ϕ. For the sunset graph the vertex of $\lambda|\phi|^4$ is $V_{abcd} = 8\lambda(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})$, so that

$$\tfrac{1}{6}\sum_{ijk} V_{1ijk}^2 = \tfrac{1}{6}(24^2 + 3\cdot 8^2)\lambda^2 = 128\lambda^2\,, \tag{6.121}$$

and the term of order λ^2 in $\delta m_\varepsilon^2(M)$ is $128\lambda^2 \sum_x \varepsilon^3 C_M^\varepsilon(x)^3$, where $C_M^\varepsilon = (-\Delta^\varepsilon + M^2)^{-1}$ is the free propagator on $\varepsilon\mathbb{Z}^3$, the counterterm being defined with free propagators, as in Sect. 2. As $\varepsilon \to 0$,

$$\begin{aligned}\sum_x \varepsilon^3 \big[C_m^\varepsilon(x)^3 - C_{M_1}^\varepsilon(x)^3\big] &\to (4\pi)^{-2}\int_0^\infty \big(e^{-3mr} - e^{-3M_1 r}\big)\frac{dr}{r} \\ &= -(16\pi^2)^{-1}\ln(m/M_1)\,.\end{aligned} \tag{6.122}$$

The sum converges to the integral: $C_M^\varepsilon(x)$ is bounded by $Ce^{-\delta|x|}(|x|+\varepsilon)^{-1}$ and $C_m^\varepsilon - C_{M_1}^\varepsilon$ by $Ce^{-\delta|x|}$, uniformly in ε, by the heat kernel representation of the lattice propagator, and $C_M^\varepsilon(x)$ converges to $e^{-M|x|}/4\pi|x|$ for $x \neq 0$; the sites with $|x| \leqq \varepsilon$ contribute $O(\varepsilon)$. With the factor $\frac{1}{2}$ of (6.69) this gives the last term of (6.118).

At $M_1 = \frac{7}{10}m$ the right side of (6.118) is a concave quadratic in λ, positive at $\lambda = 0$, with the positive root $\lambda_c m$. Finally put $x = M_1/m$, $\ell = \lambda/m$ and

$$f(x) = \tfrac{1}{2}(1-x^2) + (2\ell/\pi)(1-x) + (4\ell^2/\pi^2)\ln x\,, \qquad xf'(x) = -x^2 - (2\ell/\pi)x + 4\ell^2/\pi^2\,; \tag{6.123}$$

then $f(1) = 0$, and $xf'(x)$ is positive for $x < x_+ = (\ell/\pi)(\sqrt{5}-1)$ and negative for $x > x_+$. If $x_+ < 1$, that is $\ell < \frac{\pi}{4}(1+\sqrt{5})$, f decreases on $(x_+, 1)$ to $f(1) = 0$, so $f > 0$ there; if $x_+ \geqq 1$, f increases on $(0,1)$ and $f \leqq 0$, as required.

6.9. The Main Theorem

We collect the results. Let $d = 3$, let L, M and the cubes C_J be as in Sect. 2, let $e > 0$ and $\lambda, m, \mu_0 > 0$, and let $\langle\cdot\rangle$ be the limit (2.18) of the expectations (2.10) on the cubes (2.7).

Theorem 6.30. *(A) For every finite family* $g_1, \ldots, g_n$ *of real 2-forms in* $C_0^\infty(\mathbb{R}^d)$ *the limit*

$$\Big\langle \mathcal{O}\Big(\sum_i z_i g_i, 0\Big)\Big\rangle = \lim_{J\to\infty}\lim_{K\to\infty}\Big\langle \mathcal{O}\Big(\sum_i z_i g_i, 0\Big)\Big\rangle_{N,D}^{\varepsilon_K, C_J} \tag{6.124}$$

exists locally uniformly in $z \in \mathbb{C}^n$, *is entire, depends on the family only through* $\sum_i z_i g_i$, *is real and at least* 1 *for real* z, *lies in* $[0,1]$ *for purely imaginary* z, *and satisfies*

$$\Big|\Big\langle \mathcal{O}\Big(\sum_i z_i g_i, 0\Big)\Big\rangle\Big| \leqq \exp\Big[\Big\|\sum_i (\mathrm{Re}\, z_i) g_i\Big\|_2^2\Big]\,, \tag{6.125}$$

with no dependence on e, λ, m, μ_0 *or* M.

(B) The Schwinger functions $S_n(g_1, \ldots, g_n)$ *of the field strength satisfy:*

(i) the bounds (6.27), (6.29), *the extension to the Schwartz space, the exponential bound of the regularity axiom of* [GJ 1] *and the linear growth condition (E0′) of* [OS 2] *(Sect. 6.2);*

(ii) invariance under the Euclidean group, F transforming as a 2-form;

(iii) reflection positivity about every plane, in the matrix form (6.51) *and in the polynomial form;*

(iv) symmetry, reality for real test functions, and vanishing for odd n;

(v) they are the moments of a unique probability measure μ on the tempered distributions with values in 2-forms, carried by the space $\mathcal{H}^{-r}$ of Sect. 6.5 for every even $r > d$, with the characteristic functional $\langle \mathcal{O}(iu, 0)\rangle$, to which the laws of the localized lattice fields converge;

(vi) μ is an integral of components which are invariant under the translations and the coordinate reflections, ergodic under the translations, reflection positive about every coordinate plane and satisfy the cluster property (6.56)*, the vacuum of the reconstruction from each of them in every coordinate direction being the only vector invariant under the time semigroup and the transverse translations; the components are invariant under the Euclidean group (Theorem 6.16(v)), reflection positive about every plane, and satisfy the cluster property of* [OS 1] *for every time direction.*

(C) Let the coupling constants satisfy (6.70)*, and let σ_+, H, κ, K_- be the constants of Sect. 6.6 and $\sigma_W = \sigma_+/8$. For every finite family of real 2-forms g_i and real functions h_j in $C_0^\infty(\mathbb{R}^d)$ the limit $\langle \mathcal{O}(\sum_i z_i g_i, \sum_j w_j h_j)\rangle$ exists locally uniformly on the tube $\mathcal{D}$ of Sect. 2 and is holomorphic there, and the joint Schwinger functions $S_{n,p}$ of the field strength and of the renormalized square* (2.13)*, with the free subtraction and the constant a_ε, satisfy:*

(i) the bound (6.91) *on $\mathcal{D}$, the bound* (6.92) *and the extension to the Schwartz space;*

(ii) invariance under the translations and the cubic group about every point, h transforming as a scalar;

(iii) reflection positivity about every coordinate plane, in the matrix form for families of amplitude less than σ_W and in the polynomial form;

(iv) symmetry, reality and vanishing for odd n;

(v), (vi) the statements of (B)(v), (vi) for the pair of fields on $\mathcal{H}^{-r}(\mathbb{R}^d; \Lambda^2 \oplus \mathbb{R})$, $r \geqq 14$ even, with invariance under the translations and the cubic group in place of the Euclidean group, reflection positivity about the coordinate planes, and the cluster property of the components in the forms (6.56)*.*

Part (A) follows from Theorems 2.1, 2.2 and 6.9; (B) from Theorems 6.9, 6.11, 6.12, 6.15 and 6.16 and from Theorem 2.5, which rests on Theorems 2.4 and 6.24 (Sects. 4 and 6.7); (C) from Theorems 6.21, 6.22 and 6.16 applied to the pair of fields, the constant a_ε of (2.13) being needed for the existence of the limit (Sect. 6.10). The bound in (A) and the constants in (B) are independent of e, λ, m, μ_0 and M; those in (C) depend on the coupling constants, on M, on the auxiliary mass M_1 and the amplitude s_1 of Sect. 6.6, and on the cells and cut-offs of Lemma 6.20, and they are not uniform in μ_0.

The limit objects do not depend on the choice of M among the values admitted in Sect. 2. If M and $\tilde{M}$ are two such values, their least common multiple $\check{M}$ is a third, the admitted values being the multiples of 50 exceeding a fixed bound (Sect. 2); the lattice expectations (2.10) do not depend on M, which enters only through the cubes (2.7), the mass counterterm δm^2 and the constant a_ε of (2.13) being defined by the free lattice propagators alone, without reference to M, J or the cubes, and the constants E_0, E_1 cancelling in the normalized expectations (2.10); and every cube $[-\check{M}J, \check{M}J)^3$ is a cube of M and of $\tilde{M}$, so that each of the two iterated limits (2.18) equals the limit along the cubes of $\check{M}$, and the two coincide. For (C), whose tube depends on M through its constants, the two limits coincide on the intersection of the two tubes.

6.10. Open Problems

We list the questions left open here.

The cluster property. Theorem 6.16 gives the cluster property for almost every ergodic component of the limit measure, not for the measure itself. Whether $\langle\cdot\rangle$ is ergodic under the

translations, that is whether the decomposition of Theorem 6.16(iv) is trivial, is not known. A cluster expansion with a mass gap would give this property, but none is available for $e > 0$ in the continuum limit.

Periodic boundary conditions. We do not know whether the limits of the periodic models on the tori D_J, without a mass term, exist as $J \to \infty$, nor whether they coincide with (2.18). The correlation inequalities give no monotonicity in the volume for periodic boundary conditions, and the comparison of Sect. 6.7 holds only in the presence of the mass term. This is also the reason why Theorem 2.4 is stated with the mass term.

Non-triviality. Nothing is proved here about the non-triviality of the limit: whether the field strength is not a Gaussian field, or whether it is not independent of the renormalized square. On every lattice model to which Proposition 3.1 applies we have

$$\langle :\phi^2:(h); F(g)^2\rangle \leqq 0 \qquad \text{for } h \geqq 0\,. \tag{6.126}$$

To see this, we use (3.24) at $s = 1$, whose derivative there is the truncated expectation, with the function g of (3.24) equal to zero, h replaced by th, $t > 0$, and $u = \varepsilon^{-1} d^* g$, where $(dA)(p) = \sum_{b \in \partial p} A(b)$ and d^* is its adjoint with respect to the counting measure on bonds and plaquettes, so that $A(u) = F(g)$. The hypothesis of Proposition 3.1, that the quadratic form of the vector field exceeds $|A(u)|^2$, is satisfied by cu for $c > 0$ small, since $\mu_0 > 0$, and (6.126) is homogeneous of degree two in g; so we may replace g by cg. The truncated expectation $\langle \exp(t\, r^2(h)); F(g)^2\rangle$ is then $\leqq 0$ for $t > 0$ and vanishes at $t = 0$. The lattice model has all exponential moments of $r^2(h)$ by the term λr^4 of (2.5), so its derivative at $t = 0$, which is $\langle r^2(h); F(g)^2\rangle = \langle :\phi^2:(h); F(g)^2\rangle$ since the Wick ordering (2.13) adds a constant to $r^2(h)$, is $\leqq 0$.

Under the condition (6.70) the inequality holds in the limit (2.18). By the symmetry $(A, \theta) \to (-A, -\theta)$ of (2.5), $\langle F(g) \exp(w{:}\phi^2{:}(h))\rangle = 0$ for all w, so the truncated expectation is the third cumulant $\partial_w \partial_z^2 \ln\langle \mathcal{O}(zg, wh)\rangle$ at $z = w = 0$. Now $\mathcal{D}$ contains a neighbourhood of the origin of $\mathbb{C}^2$, on which the limits of Theorems 2.1 and 2.2 are uniform by Vitali's theorem (Sect. 6.6) and holomorphic in (z, w) by Theorem 2.3. Since $\langle \mathcal{O}(0,0)\rangle = 1$, the logarithm is defined near the origin for all large K, J, and the derivatives converge by Weierstrass' theorem.

If the inequality were strict for some g and some $h \geqq 0$, the field strength and the renormalized square would be neither independent nor jointly Gaussian; for a jointly Gaussian pair $\langle X; Y^2\rangle = 2\langle X; Y\rangle\langle Y\rangle$, and $\langle F(g)\rangle = 0$ as above. The lowest order of perturbation theory in which the truncated expectation can be non-zero is e^2; whether the coefficient is strictly negative is not known, and the bounds of this paper are not uniform in the coupling constants.

The limit $\mu_0 \to 0$. Every constant of Sect. 6.6 depends on the vector field mass μ_0 and is not uniform in it, while the bounds (6.27), (6.29) on the field strength are. The limit of vanishing vector field mass is not treated here. For the field strength it would require the existence of the limit (2.18) uniformly in μ_0 along a family of models with the counterterm δm^2 and the constant a_ε fixed at their values for a fixed mass $\mu_1 > 0$. At fixed counterterms the expectations are monotone in μ_0 for g real and $h \geqq 0$, $\langle \mathcal{O}(ig, h)\rangle$ increasing and $\langle \mathcal{O}(g, -h)\rangle$ decreasing; this follows from (3.24), (3.27) applied, one bond at a time, to the difference of the terms $\frac{1}{2}\mu_0^2 \varepsilon^d |A(b)|^2$ at two values of μ_0, the remaining form being positive definite, as in the proof of Theorem 3.2. For the scalar source it would require in addition a bound of the kind of Lemma 6.20 uniform in μ_0.

The scalar source at large amplitudes. The results with the scalar source are stated on the tube $\mathcal{D}$ and under the condition (6.70). Whether the limit (2.18) exists for all $w \in \mathbb{C}$, so that (2.21) is entire in w as well, whether the bound $\sup_J \langle \mathcal{O}(0,q)\rangle_{N,D}^{C_J} < \infty$ holds for every $q \geqq 0$ and not only for $\sup q \leqq s_1$, and whether reflection positivity (2.20) holds for arbitrary complex w, are open. The proofs of Sect. 6.6 would give these statements from the two bounds just named, the extensive bound of Proposition 6.18 at every amplitude and the bound of Lemma 6.20(ii) for every $q \geqq 0$. The rotation invariance of the Schwinger functions with scalar sources is also open: the comparison of Sect. 6.7 is made at $h = 0$, and with $h \neq 0$ a comparison of the

two lattices at equal spacing, with the same constant a_ε, would be needed, together with the equality of Theorem 6.24 in the presence of the scalar source.

The condition on the coupling constants. The condition (6.70) is a condition on the renormalized mass difference at two values of the scalar mass. Lemma 6.28 gives its one-loop form and Proposition 6.29 the small-coupling and the $e = 0$ cases; whether it holds on the whole range $e > 0$, $\lambda, m, \mu_0 > 0$, or fails for some values, is not known. Its failure would not affect the results for the field strength.

Relation to the original papers. This paper is a corrected edition of the second of the two papers [K 1], [K 2], of which the paper [I] is the corrected edition of the first. The statements below concern [K 2] in three dimensions. The text of [K 2] is kept wherever it is correct; its misprints are corrected without comment, and its results are restated, and their proofs changed, at the following places.

Theorems 2.1 and 2.2 are stated with the source h on the set $\mathcal{D}$, the function of Theorem 2.3(iii) is holomorphic in w on the tube only, and the Wick ordering (2.13) carries the free subtraction and the constant a_ε. Without the constant the limit (2.17) is 0 or $+\infty$ according to the sign of $w \int h$, for $\int h \neq 0$, real z and real $w \neq 0$ in the set of Theorem 2.1. For the two Wick orderings differ by the constant $a_{\varepsilon_K} \int h$ and by a term bounded uniformly in K, the covariances with Dirichlet and with free boundary conditions differing at coinciding points by at most Ce^{-mR}, R the distance from the support of h to ∂C_J (Sect. 3.2); $a_{\varepsilon_K} \to +\infty$, growing like $\ln \varepsilon_K^{-1}$ (Sect. 2 of [I]); and for real z the limit with the Wick ordering (2.13) exists by Theorem 2.1 and is positive, the expectation being at least $\exp(w\langle{:}\phi^2{:}(h)\rangle)$ by Jensen's inequality and $\langle F(g)\rangle = 0$, and $\langle{:}\phi^2{:}(h)\rangle$ is bounded uniformly in K by Jensen's inequality applied to $\mathcal{O}(0, \pm ch)$, $c > 0$ small, whose expectations converge by Theorem 2.1 and are therefore bounded in K.

The inequalities (3.26), (3.29) carry the factor ε^{d-2}, and Proposition 3.1 the hypothesis on u; the blocks of Sect. 3.1 are placed with the Dirichlet planes on their last layers, where (3.9) holds, and not at their corners.

In Sect. 3.3 the proof of Proposition 3.3 is given directly, the printed derivation from the translation invariant case not covering the box; the comparison (3.52) carries the margin $2n$, since on one and the same box the quadratic form of (3.38) exceeds the form of (2.5) by a positive form on the bonds next to the boundary, so that the second inequality of the printed chain, with C_{J+1}, is reversed, and fails strictly for a field strength source already at $e = 0$ (Lemma 6.13). The bound (3.53) is proved in Sect. 6.6 rather than referred to a standard argument, and the sign-changing sources and the translation invariance are treated in Sects. 3.3 and 6.3.

The passage from the tori to the infinite volume limit, (2.24)–(2.25) of [K 2], which rested on a conjectured identity, is replaced by the comparison of Sect. 6.7, and Theorem 2.4 is stated at equal spacing for 2-form sources with the mass term ω of Sect. 4, since the form of [K 2] fails already at $e = 0$ for a 2-form source equal to the same function g on every plaquette.

At $e = 0$ both lattice models converge to the free vector field of mass μ_0 on D_J. The rotation in the 12-plane leaves F_{12} unchanged and takes $F_{13} + F_{23}$ to $(\cos\theta_0 - \sin\theta_0)F_{13} + (\cos\theta_0 + \sin\theta_0)F_{23}$. We take g even in x_3, so that the cross terms with F_{12}, which are odd in p_3, vanish, and symmetric under $x_1 \leftrightarrow x_2$, so that $\langle F_{13}(g)^2\rangle = \langle F_{23}(g)^2\rangle$. Then the two variances of $F(g)$ differ by $-4\sin^2\theta_0 \langle F_{13}(g)F_{23}(g)\rangle$, where

$$\langle F_{13}(g)F_{23}(g)\rangle = |D_J|^{-1} \sum_p |\hat{g}(p)|^2 p_1 p_2 \hat{G}_{\mu_0}(p)\,, \qquad \hat{g}(p) = \int_{D_J} e^{-ip\cdot x} g(x)\,dx\,, \tag{6.127}$$

the sum over the dual lattice of D_J, and this is not zero when $\hat{g}$ is concentrated near the directions $\pm(1, 1, 0)$.

For $\int h \neq 0$ the form of [K 2] fails at $e > 0$ as well. By Jensen's inequality, applied to the sources $\pm ch$ with $c > 0$ so small that (4.3) holds for them, and by the linearity of $\langle{:}\phi^2{:}(h)\rangle$

in h, $|\langle{:}\phi^2{:}(h)\rangle_P^{\varepsilon_K,D_J}| \leqq C$ uniformly in K with the Wick ordering (2.13). The Wick ordering of [K 2] exceeds (2.13) by the constant $a_{\varepsilon_K}\int h$, $a_{\varepsilon_K}\to+\infty$ (Sect. 2 of [I]), and by twice the sum of $\varepsilon_K^d h(x)$ times the free less the periodic covariance at x, which is bounded uniformly in K by the exponential decay of the free lattice covariance. Hence with the Wick ordering of [K 2] the expectations tend to $+\infty$ for real w with $w\int h>0$, the expectation with the Wick ordering (2.13) being at least $e^{-|w|C}$ by Jensen's inequality, and to 0 for every w with $\operatorname{Re} w\int h<0$ for which (4.3) holds with the source $\operatorname{Re} w\, h$, in particular for all such w near 0, since $|\langle\mathcal{O}(0,wh)\rangle|\leqq\langle\mathcal{O}(0,\operatorname{Re} w\, h)\rangle$; no function holomorphic near $w=0$ and equal to 1 there is their limit.

In Sect. 4 the transformation (4.4)–(4.5), Proposition 4.1, Theorem 4.2 and the passage (4.8)–(4.11) are restated at equal spacing. In Sect. 5 the bound (5.28) is extended to every vector field, the derivative bounds of Proposition 5.1 are stated for rectangular regions and away from the re-entrant edges, the corridor $r(L^k\varepsilon)$ replaces $p(L^k\varepsilon)$ in (5.56)–(5.57), the lower bound (5.45)–(5.48) and the bound (5.77) are corrected, and the change of variables of (5.37) is made without the mass term. The bound (A.4) is stated for $f=1$ only, the proof of (A.6) is given, and the comparison of Appendix B is organized by the common first-step lattice. Section 6 is new. The misprints of [K 1] are treated in [I].

Appendix A

Using the representations (3.7), (3.9), it is sufficient to consider the operators G_k, $G_k(0)$ with free boundary conditions. The only difference from the case studied in [I] is the averaging function $f(x)$, given by (3.8). The Fourier transform of f is equal to

$$\tilde f_\eta(p)=L^{-d}\sum_{x\in B(y)}f(x)e^{-ip(x-y)}=\prod_{\mu=1}^{d}\left[\frac{e^{-ip_\mu\eta(L-1)}-1}{(L-1)(e^{-ip_\mu\eta}-1)}\right]. \tag{A.1}$$

We can easily deduce the explicit representation corresponding to (4.2) of [I]:

$$\begin{aligned}(a_kG_kQ_k^*)(x,y)=(2\pi)^{-d}\int_{|p'|\leqq\pi}dp'\,\Delta^{(k)}(p')\sum_l e^{i(p'+l)(x-y)}\\ \cdot\frac{\tilde f_\eta(p'+l)u_{k-1}^{L\eta}(p'+l)}{\Delta^\eta(p'+l)}\,.\end{aligned} \tag{A.2}$$

The terms in (A.2) have the same meaning as the corresponding terms in (4.2) of [I], with the following change:

$$\Delta^{(k)}(p')=\Big(a_k^{-1}+\sum_l|\tilde f_\eta(p'+l)|^2|u_{k-1}^{L\eta}(p'+l)|^2(\Delta^\eta(p'+l))^{-1}\Big)^{-1}. \tag{A.3}$$

Lemma 5.3 follows by exactly the same considerations used in [Ba 4] with $f=1$. The integrand in (A.2) may be analytically continued in a strip around the real p'_μ-axis, giving exponential decay. The sum over l is controlled, when $f=1$, by the estimates

$$\begin{aligned}|\tilde f_\eta(p'+l)u_{k-1}^{L\eta}(p'+l)|\approx\prod_\mu|(p'+l)_\mu|^{-1}\,,\\ \Delta^\eta(p'+l)^{-1}\approx|p'+l|^{-2}\,.\end{aligned} \tag{A.4}$$

For the function f of (3.8) the first estimate in (A.4) does not hold. At $p'=0$, $l=2\pi L^{k-1}(1,\dots,1)$ we have $u_{k-1}^{L\eta}(l)=1$, and each factor of $\tilde f_\eta(l)$ in (A.1) has modulus $(L-1)^{-1}$, so that $|\tilde f_\eta u_{k-1}^{L\eta}|=(L-1)^{-d}$, independent of k, whereas the right side of (A.4) is $(2\pi L^{k-1})^{-d}$.

However, $|\tilde{f}_\eta| \leqq 1$ for real argument, and $u_{k-1}^{L\eta}(p'+l)$ depends on l only modulo $2\pi L^{k-1}\mathbb{Z}^d$. We therefore write $l = l_1 + l_2$, where $-\pi(L\eta)^{-1} < (p'+l_1)_\mu \leqq \pi(L\eta)^{-1}$ for each μ and $l_2 \in 2\pi L^{k-1}\mathbb{Z}^d$; for each l_1 there are L^d values of l_2. Then $u_{k-1}^{L\eta}(p'+l) = u_{k-1}^{L\eta}(p'+l_1)$, and this is bounded by $C\prod_\mu \min\{1, |(p'+l_1)_\mu|^{-1}\}$.

For $l_2 = 0$ the sum over $l_1 \neq 0$ of this bound times $\Delta^\eta(p'+l_1)^{-1}$ is bounded uniformly in k and p', and the term $l = 0$ is controlled together with $\Delta^{(k)}(p')$, as in [Ba 4]. For the $L^d - 1$ values $l_2 \neq 0$ we have $|p'+l| \geqq \pi(L\eta)^{-1}$, and

$$\Delta^\eta(p'+l)^{-1} \leqq CL^{-2k}\,, \qquad \sum_{l_1} |u_{k-1}^{L\eta}(p'+l_1)| \leqq C(1+k)^d\,. \tag{A.5}$$

Here C depends on L. These bounds persist, with $|\operatorname{Re}(p'+l)_\mu|$ in place of $|(p'+l)_\mu|$ and with $|\tilde{f}_\eta| \leqq C$ in place of $|\tilde{f}_\eta| \leqq 1$, in a strip around the real p'_μ-axis whose width depends on a, L and d, as in [Ba 4], and Lemma 5.3 follows as before.

Furthermore, suppose that we have D b.c. on the hyperplane $x_\mu = 0$. We wish to prove that

$$|\eta^{-1}(G_k(x,y) - G_k(x, y-2y_\mu e_\mu))| \leqq C(\eta + |x-y|)^{1-d}\exp[-\delta|x-y|]\,, \tag{A.6}$$

when $y_\mu = \eta$ (this shows that the propagator with D b.c. goes to zero near the boundary; G_k in (A.6) has free boundary conditions).

We write $H_k = -\Delta^\eta + m^2(L^k\varepsilon)^2 + a_k Q_k^* Q_k$, so that $G_k = H_k^{-1}$ on $\ell^2(\eta\mathbb{Z}^d)$ with the inner product $\langle\phi,\psi\rangle = \sum_x \eta^d \bar{\phi}(x)\psi(x)$, and $Q_k^*Q_k = \sum_{y\in\mathbb{Z}^d} |f_y\rangle\langle f_y|$ with $f_y = f\chi_{B^k(y)}$, $\langle f_y, 1\rangle = 1$ and $\sup f = (L/(L-1))^d$ by (3.8).

First, $H_k \geqq c_0$ with $c_0 = \min\{2,\ a_k/4,\ \frac{1}{2}(\sup f)^{-2}\}$, which is bounded below uniformly in k since $a(1-L^{-2}) \leqq a_k \leqq a$ by (2.15) of [Ba 1]. Dropping the bonds between unit blocks gives $-\Delta^\eta \geqq \bigoplus_y (-\Delta^{\eta,N}_{B^k(y)})$, and $Q_k^*Q_k$ is block diagonal, so it suffices to consider one block. Write $\phi = c + \phi_\perp$, c the mean of ϕ on the block; then $\langle\phi, -\Delta^{\eta,N}\phi\rangle \geqq 4\|\phi_\perp\|^2$ and $\langle f_y,\phi\rangle = c + \langle f_y,\phi_\perp\rangle$ with $|\langle f_y,\phi_\perp\rangle| \leqq \sup f\,\|\phi_\perp\|$. If $\|\phi_\perp\| \geqq |c|/(2\sup f)$ the form is at least $2\|\phi_\perp\|^2 + \frac{1}{2}(\sup f)^{-2}|c|^2$; otherwise $|\langle f_y,\phi\rangle| \geqq |c|/2$ and it is at least $4\|\phi_\perp\|^2 + a_k|c|^2/4$. Since $\|\phi\|^2 = |c|^2 + \|\phi_\perp\|^2$ on the block, the bound follows.

Second, let $w(x) = \exp[\theta(1+|x-y|^2)^{1/2}]$ with $0 \leqq \theta \leqq \theta_0$. Then $wH_kw^{-1} = H_k + B$, where B collects the terms of $-\Delta^\eta$ and of $a_kQ_k^*Q_k$ in which w does not cancel. The ratios $w(x)/w(x')$ over a bond, or within a unit block, are $e^{\theta s}$ with $|s| \leqq \eta$, respectively $|s| \leqq \sqrt{d}$, since $(1+|u|^2)^{1/2}$ has first and second derivatives bounded by 1; hence the part of B coming from $Q_k^*Q_k$ has norm at most $a_k(\sup f)^2(e^{\theta\sqrt{d}} - 1) \leqq C\theta$, and the part B_Δ coming from $-\Delta^\eta$ satisfies

$$|\langle\phi, B_\Delta\phi\rangle| \leqq C(\theta+\theta^2)\|\phi\|^2 + 2\theta\|\phi\|\,\langle\phi,-\Delta^\eta\phi\rangle^{1/2} \leqq \tfrac{1}{2}\langle\phi,H_k\phi\rangle + C(\theta+\theta^2)\|\phi\|^2\,. \tag{A.7}$$

Thus $\operatorname{Re}\langle\phi,(H_k+B)\phi\rangle \geqq \frac{1}{4}c_0\|\phi\|^2$ for $\theta \leqq \theta_0$, θ_0 depending on a, L and d only, and the same holds for the adjoint; so $wG_kw^{-1} = (H_k+B)^{-1}$ has norm at most $4/c_0$.

Third, let $C' = (-\Delta^\eta + m^2(L^k\varepsilon)^2 + 1)^{-1}$ and $W = 1 - a_kQ_k^*Q_k$, so that $H_k = C'^{-1} - W$ and

$$G_k = C' + C'WG_k = C' + C'WC' + C'WG_kWC'\,. \tag{A.8}$$

We take $d = 3$, as throughout this paper (Sect. 2), and let $\kappa(u) = (\eta + |u|)^{-1}$. The kernel of C' satisfies $|C'(x,z)| \leqq C\kappa(x-z)e^{-c|x-z|}$ and $|\eta^{-1}(C'(x,z) - C'(x-2\eta e_\mu, z))| \leqq C(\eta + |x-z|)^{1-d}e^{-c|x-z|}$, with C, c independent of k; these are the standard bounds for the massive lattice Green's function with mass at least 1 and spacing η, from its Fourier representation. Put $\psi_y = WC'(\cdot,y)$ and $\phi_x = WC'(\cdot,x)$. A block average of $\kappa(\cdot - y)e^{-c|\cdot-y|}$ is at most $Ce^{-c(|z-y|-\sqrt{d})}$, so $|\psi_y(z)| \leqq C\kappa(z-y)e^{-c|z-y|/2}$ and $\|w\psi_y\| \leqq C$ for $\theta \leqq c/4$, uniformly in k. Hence

$$|G_k(x,y)| \leqq |C'(x,y)| + |(C'WC')(x,y)| + \|w^{-1}\phi_x\|\,\|wG_kw^{-1}\|\,\|w\psi_y\| \leqq C\kappa(x-y)e^{-\delta|x-y|} \tag{A.9}$$

with $\delta = \min\{c, \theta_0\}/8$, using $e^{-\theta|z-y|} \leqq e^{-\theta|x-y|}e^{\theta|z-x|}$ in the first factor, the middle term being bounded by the convolution of two singularities κ, which is at most $C\kappa(x-y)e^{-c|x-y|/4}$. Consequently $|(WG_k(\cdot, y))(z)| \leqq C\kappa(z-y)e^{-\delta|z-y|/2}$, and differencing $G_k = C' + C'WG_k$ in the first argument,

$$\begin{aligned} |\eta^{-1}(G_k(x,y) - G_k(x-2\eta e_\mu, y))| &\leqq C(\eta + |x-y|)^{1-d}e^{-c|x-y|} \\ &\quad + C\sum_z \eta^d(\eta+|x-z|)^{1-d}e^{-c|x-z|}\kappa(z-y)e^{-\delta|z-y|/2} \\ &\leqq C(\eta+|x-y|)^{1-d}e^{-\delta'|x-y|}\,, \end{aligned} \tag{A.10}$$

$\delta' = \min\{c, \delta/2\}/2$, since $e^{-c|x-z|-\delta|z-y|/2} \leqq e^{-\delta'|x-y|}e^{-\delta'(|x-z|+|z-y|)/2}e^{-\delta'|x-y|/2}$, the sum over z of the product of the two singularities times $e^{-\delta'(|x-z|+|z-y|)/2}$ is at most $C(1+|\ln(\eta+|x-y|)|)$, and $(1+|\ln r|)r^{d-1}e^{-\delta' r/2} \leqq C$ for $r = \eta + |x-y|$. Since $G_k(x,y) = G_k(y,x)$, this is (A.6), with δ' in place of δ. The kernel (A.2) also satisfies

$$|\eta^{-1}((a_kG_kQ_k^*)(x,y) - (a_kG_kQ_k^*)(x-2\eta e_\mu, y))| \leqq C\,, \tag{A.11}$$

$x \in \eta\mathbb{Z}^d$, $y \in \mathbb{Z}^d$; we indicate how the sum over l in (A.2) is controlled.

From (A.2), we get the factor

$$|\eta^{-1}(e^{-i(p'+l)_\mu\eta} - e^{i(p'+l)_\mu\eta})| \leqq C|p'+l|\,. \tag{A.12}$$

There are sufficient negative powers of $|p'+l|$ in the sum over l in (A.2) to control (A.12); for the function f of (3.8) this uses the decomposition $l = l_1 + l_2$ given after (A.4) in place of (A.4): the terms with $l_2 \neq 0$ are summable because of the bound on Δ^η and the periodicity of $u_{k-1}^{L\eta}$, the terms with $l_2 = 0$, $l_1 \neq 0$ because of the bound on $u_{k-1}^{L\eta}$, and the term $l = 0$ gives $|p'|^{-1}$, which is integrable over $|p'| \leqq \pi$. Therefore, (A.11) holds.

In order to prove the necessary convergence estimates for (A.2) which were proved with $f = 1$ in Sect. 4 of [I], we specify below the alterations needed to include f. We leave unchanged the representation (4.6) of [I] and the conditions (4.7), (4.8) of [I]. In order to reprove Lemma 4.1 of [I], we consider

$$\begin{aligned} &|\tilde f_\eta(p)|^2|u_{k-1}^{L\eta}(p)|^2 - |u_k^\eta(p)|^2 \\ &\quad = |u_{k-1}^{L\eta}(p)|^2 \sum_{x,\omega\in B(y)} L^{-2d}[f(x)f(\omega)-1]e^{-ip(x-\omega)} \\ &\quad = 2|u_{k-1}^{L\eta}(p)|^2 \sum_{x,\omega\in B(y)} L^{-2d}[1-f(x)f(\omega)]\sin^2[\tfrac12 p(x-\omega)]\,, \end{aligned} \tag{A.13}$$

since $\tilde f_\eta(0) = 1$. Hence

$$\begin{aligned} &\big||\tilde f_\eta(p)|^2|u_{k-1}^{L\eta}(p)|^2 - |u_k^\eta(p)|^2\big| \\ &\quad \leqq C|u_{k-1}^{L\eta}(p)|^2 \sum_{\mu,\nu=1}^d p_\mu p_\nu \Big\{ \sum_{x,\omega\in B(y)} L^{-2d}(x-\omega)_\mu(x-\omega)_\nu \Big\} \\ &\quad \leqq CL^{-2k}|p|^2|u_{k-1}^{L\eta}(p)|^2\,. \end{aligned} \tag{A.14}$$

Using (A.14) and the estimates (4.10), (4.11) of [I], and also $\sum_l |u_{k-1}^{L\eta}(p'+l)|^2 = L^d$, we deduce Lemma 4.1 of [I]. Next, we again notice that $\Delta^{(k+n)}(p')$ can be written in the form of $\bar\Delta^{(k)}(p')$ with

$$D^{-1}(p) = a_n^{-1}L^{-2k} + \sum_{l'} |\tilde f_{\eta'}(p+l')|^2|u_{n-1}^{L\eta'}(p+l')|^2\Delta^{\eta'}(p+l')^{-1}\,. \tag{A.15}$$

Also, we have

$$|\tilde{f}_{\eta'}(p)|^2 |u_{n-1}^{L\eta'}(p)|^2 = \sum_{x,\omega \in B^n(y)} L^{-2nd} f(x) f(\omega) e^{-ip(x-\omega)} \\ = 1 + \sum_{x,\omega \in B^n(y)} L^{-2nd} f(x) f(\omega) [e^{-ip(x-\omega)} - 1] \,. \tag{A.16}$$

Using (A.15) and (A.16), we can follow the proof of Lemma 4.2 in [I], and deduce convergence of $\Delta^{(k)}(p')$. The rest of Sect. 4 of [I] goes through with one change: the bound (4.21) of [I] on $u_{k+n}^{\eta'}$ holds for $f = 1$ only, since for the function f of (3.8) the weighted average does not vanish at the aliases $2\pi L^{k+n-1}(1, \dots, 1)$ of the coarse lattice, and in the sums over the aliases in (4.26), (4.27) of [I] we use instead the decomposition $l+m = l_1+l_2$, $l_2 \in 2\pi L^{k+n-1}\mathbb{Z}^d$, given after (A.4), with $|\tilde{f}_{\eta'}| \leqq 1$, the periodicity of $u_{k+n-1}^{L\eta'}$ and the bound $|u_{k+n-1}^{L\eta'}(p'+l_1)| \leqq C \prod_\mu |p'_\mu|\,|(p'+l_1)_\mu|^{-1}$, which follows from $|\sin((p'+l_1)_\mu/2)| = |\sin(p'_\mu/2)|$ and $|(p'+l_1)_\mu| L\eta' \leqq \pi$; the terms with $l_2 \neq 0$ give a convergence factor $L^{-\gamma k}$, since $|p'+l+m| \geqq \pi(L\eta')^{-1}$ there, while the sum over l_1 is at most $C(1+k+n)^d$. So we deduce the convergence lemmas needed for the proof of the continuum limit.

Appendix B

We denote by $\hat{G}_k$ the propagator generated by the averaging operators (4.4). Consider a region Ω given by

$$\Omega = \{x \in \mathbb{R}^d : \; 0 < x_\mu \leqq 25 L_\mu, \; \mu = 1, \dots, d\} \,, \tag{B.1}$$

where $\{L_\mu\}$ are integers. We denote by T_η the torus with lattice spacing $\eta = 5^{-k}$ which fits inside Ω. The orientation of this lattice is the same as that of $\partial\Omega$. The lattice operator $\hat{G}_k(\Omega)$ on the torus T_η is then defined by the equation

$$(-\Delta^\eta + m^2 (L^k \varepsilon)^2 + a_k \hat{Q}^* Q_{k-1}^* Q_{k-1} \hat{Q}) \hat{G}_k(\Omega) f = f \,, \tag{B.2}$$

for any function f on T_η. We can now state the multiple reflection representation for $\hat{G}_k(\Omega)$ (the operator $\hat{G}_k$ is defined with free boundary conditions):

$$\hat{G}_k(\Omega)(x,y) = \sum_{n \in \mathbb{Z}^d} \hat{G}_k\Big(x, y + \sum_{\mu=1}^d 25 n_\mu L_\mu e_\mu\Big). \tag{B.3}$$

Therefore, it is sufficient to prove regularity and decay estimates and convergence properties for $\hat{G}_k$. We will denote by $\eta\mathbb{Z}^d$ the original lattice on which $\hat{G}_k$ is defined, and by $L^j\eta\hat{\mathbb{Z}}^d$ the rotated lattices defined by the renormalization transformation, for $j = 1, \dots, k$. The dual of $\eta\mathbb{Z}^d$ is P_η; the dual of $L^j\eta\hat{\mathbb{Z}}^d$ is $\hat{P}_{L^j\eta}$ (of course, $L = 5$ here). We introduce the function

$$w_\eta(p) = 5^{-d} \sum_{x \in \hat{B}(y)} e^{-ip\cdot(x-y)} \,, \tag{B.4}$$

for $p \in P_\eta$. Here $\hat{B}(y)$, $y \in L\eta\hat{\mathbb{Z}}^d$, is the set of the 5^d points of $\eta\mathbb{Z}^d$ in the cube of side $L\eta$ centered at y with sides parallel to the axes of $L\eta\hat{\mathbb{Z}}^d$, as in Sect. 4; these sets are translates of one another and partition $\eta\mathbb{Z}^d$, so that w_η does not depend on y, and $|x-y| \leqq \frac{1}{2}\sqrt{d}\, L\eta$ for $x \in \hat{B}(y)$. By introducing a Fourier representation on $\eta\mathbb{Z}^d$, we deduce the following explicit formula:

$$(a_k \hat{G}_k \hat{Q}^* Q_{k-1}^*)(x,y) = (2\pi)^{-d} \int_{p' \in \hat{P}_1} dp' \, \hat{\Delta}^{(k)}(p') \\ \cdot \sum_l e^{i(p'+l)\cdot(x-y)} \frac{w_\eta(p'+l) u_{k-1}^{L\eta}(p'+l)}{\Delta^\eta(p'+l)} \,, \tag{B.5}$$

where

$$\hat{\Delta}^{(k)}(p') = \Big(a_k^{-1} + \sum_{l'} |w_\eta(p'+l')|^2 |u_{k-1}^{L\eta}(p'+l')|^2 \Delta^\eta(p'+l')^{-1}\Big)^{-1}, \tag{B.6}$$

and we sum l, l' over $2\pi\hat{\mathbb{Z}}^d$, such that $p'+l$ or $p'+l'$ belongs to P_η. Using the representation (B.5), we can deduce the regularity and decay estimates for $\hat{G}_k$ in the same way as in Appendix A. Similarly, the convergence proofs go through in almost the same way. We will give the proof of convergence for $\hat{\Delta}^{(k)}(p') - \Delta^{(k)}(p')$; the other estimates then follow straightforwardly (see Sect. 4 of [I]). First, we have

$$\Delta^{(k)}(p') = \Big(a_k^{-1} + \sum_{l'} |u_k^\eta(p'+l')|^2 \hat{\Delta}^\eta(p'+l')^{-1}\Big)^{-1}, \tag{B.7}$$

where $\hat{\Delta}^\eta(p)$ is the Laplacian on the lattice $\eta\hat{\mathbb{Z}}^d$, and l' is summed over $2\pi\hat{\mathbb{Z}}^d$ such that $p'+l' \in \hat{P}_\eta$. It is clear that

$$\Delta^{(k)}(p'),\ \hat{\Delta}^{(k)}(p') \leqq a\,. \tag{B.8}$$

So we need only prove convergence of $(\Delta^{(k)}(p')^{-1} - \hat{\Delta}^{(k)}(p')^{-1})$. This is done by comparing the two sums. In each of (B.6) and (B.7) we first sum l' over $2\pi\hat{\mathbb{Z}}^d$ such that $p'+l' \in \hat{P}_{L\eta}$, the dual of the lattice $L\eta\hat{\mathbb{Z}}^d$ of the first step, which is the same for both transformations. We then sum over the L^d points of P_η, respectively $\hat{P}_\eta$, which differ from such a $p'+l'$ by an element m of the reciprocal lattice $2\pi(L\eta)^{-1}\hat{\mathbb{Z}}^d$ of $L\eta\hat{\mathbb{Z}}^d$; this is possible since $L\eta\hat{\mathbb{Z}}^d \subset \eta\mathbb{Z}^d$ and $2\pi(L\eta)^{-1}\hat{\mathbb{Z}}^d \subset 2\pi\hat{\mathbb{Z}}^d$, $(L\eta)^{-1}$ being an integer, and every point of $\hat{P}_{L\eta}$ lies in P_η and in $\hat{P}_\eta$, its components in either frame being at most $\sqrt{d}\,\pi(L\eta)^{-1} < \pi\eta^{-1}$, since $d < L^2$. Since $u_k^\eta = u_{k-1}^{L\eta} u_1^\eta$, where u_1^η is the transform of one step of the transformation on $\eta\hat{\mathbb{Z}}^d$, and $|u_{k-1}^{L\eta}|$ is constant on each inner set, the outer sums are identical, and $\sum_{l'} |u_{k-1}^{L\eta}(p'+l')|^2 = 1$.

For $m \neq 0$ some component of m in the frame of $L\eta\hat{\mathbb{Z}}^d$ has modulus at least $2\pi(L\eta)^{-1}$, while the components of $p'+l'$ in that frame are at most $\pi(L\eta)^{-1}$; so each such term is bounded by $CL^{-2k}|u_{k-1}^{L\eta}(p'+l')|^2$, since $|w_\eta|, |u_1^\eta| \leqq 1$ and $\Delta^\eta, \hat{\Delta}^\eta \geqq C^{-1}|p|^2$ on P_η, $\hat{P}_\eta$. For $m = 0$ the point $p'+l'$ lies in both P_η and $\hat{P}_\eta$. The two terms are equal at $p'+l' = 0$, where both equal $(m^2(L^k\varepsilon)^2)^{-1}$ since $w_\eta(0) = u_{k-1}^{L\eta}(0) = 1$, and for $p'+l' \neq 0$ their difference is

$$\begin{aligned} |u_{k-1}^{L\eta}(p'+l')|^2 \Big[&|w_\eta(p'+l')|^2 \big(\Delta^\eta(p'+l')^{-1} - \hat{\Delta}^\eta(p'+l')^{-1}\big) \\ &+ \big(|w_\eta(p'+l')|^2 - |u_1^\eta(p'+l')|^2\big)\hat{\Delta}^\eta(p'+l')^{-1}\Big]\,, \end{aligned} \tag{B.9}$$

with $\hat{\Delta}^\eta(p'+l')^{-1} \leqq C|p'+l'|^{-2}$; we bound the two brackets with the following estimates, together with the bound $|1 - |u_1^\eta(p)|^2| \leqq CL^{-2k}|p|^2$, proved as in (A.13), (A.14). We will use μ, α to denote components taken with respect to $\eta\hat{\mathbb{Z}}^d$, $\eta\mathbb{Z}^d$. Then

$$\begin{aligned} \hat{\Delta}^\eta(p'+l') &= 4\eta^{-2}\sum_\mu \sin^2[\tfrac{1}{2}\eta(p'+l')_\mu] + m^2(L^k\varepsilon)^2\,, \\ \Delta^\eta(p'+l') &= 4\eta^{-2}\sum_\alpha \sin^2[\tfrac{1}{2}\eta(p'+l')_\alpha] + m^2(L^k\varepsilon)^2\,. \end{aligned} \tag{B.10}$$

Using $x^2 \geqq \sin^2 x \geqq x^2 - \frac{1}{3}x^4$, we get

$$|\hat{\Delta}^\eta(p'+l') - \Delta^\eta(p'+l')| \leqq CL^{-2k}[(p'+l')^2]^2\,. \tag{B.11}$$

Hence we deduce

$$|\hat{\Delta}^\eta(p'+l')^{-1} - \Delta^\eta(p'+l')^{-1}| \leqq CL^{-2k}\,. \tag{B.12}$$

It is also straightforward to show that

$$|1 - |w_\eta(p)|^2| \leqq CL^{-2k}|p|^2 \,. \tag{B.13}$$

Combining these estimates with the results obtained in Sect. 4 of [I], that is summing the bounds on (B.9) and on the terms $m \neq 0$ with the weights $|u_{k-1}^{L\eta}(p'+l')|^2$, whose sum is 1, and multiplying by $\Delta^{(k)}(p')\hat{\Delta}^{(k)}(p') \leqq a\Delta^{(k)}(p')$ by (B.8), we deduce

$$|\hat{\Delta}^{(k)}(p') - \Delta^{(k)}(p')| \leqq CL^{-2k}\Delta^{(k)}(p') \,, \tag{B.14}$$

which is the required result.

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